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(12) **United States Patent**
Hensler et al.

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(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **COLLECTING AND HARVESTING CUT
BONE FROM RONGEUR**

(71) Applicant: **H & M INNOVATIONS, LLC,**
Wilmington, NC (US)

(72) Inventors: **Robert Sean Hensler**, Wilmington, NC
(US); **Thomas Eric Melin**, Wilmington,
NC (US); **Ryan Shane Gorman**,
Charlotte, NC (US); **Raeshon Lamont
Mcneil**, Charlotte, NC (US); **Michael
Morgan Starkey**, Kent, OH (US);
Thomas James Philpott, Charlotte, NC
(US)

(73) Assignee: **H & M Innovations, LLC,**
Wilmington, NC (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 234 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/093,814**

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(65) **Prior Publication Data**
US 2023/0149121 A1 May 18, 2023

Related U.S. Application Data
(63) Continuation of application No. 16/443,798, filed on
Jun. 17, 2019, now Pat. No. 11,547,524, which is a
(Continued)

(51) **Int. Cl.**
A61B 90/70 (2016.01)
A46B 9/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A61B 90/70** (2016.02); **A61B 10/025**
(2013.01); **A61B 17/16** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61B 17/1604; A61B 90/70; A61B 17/16;
A61B 17/1608; A61B 2017/00969;
(Continued)

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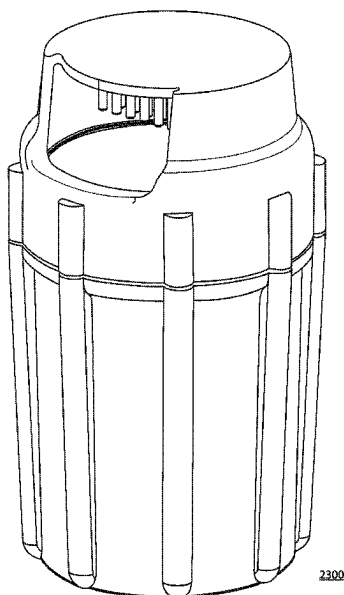
Primary Examiner — Andrew J Mensh

(74) *Attorney, Agent, or Firm* — TILLMAN WRIGHT,
PLLC; Chad D. Tillman

(57) **ABSTRACT**

A collector used to collect bone includes: a container body
defining an interior containment space for receiving and
retaining collected bone, and having at least one open end
for access and removal of collected bone from the interior
containment space; and a cap in covering relation to the open
end of the container body such that access to the interior
containment space for removal of collected bone is inhib-
ited. The collector includes an intake port defining an
opening for receiving therein a distal end of a kerrison
rongeur for collecting cut bone from a cutting area thereof,
and the cap includes at least one scraper for engaging and
dislodging cut bone from the cutting area of a distal end of
a kerrison rongeur when received within the intake port. The
collector preferably is used with a kerrison rongeur for
collecting cut bone therefrom.

12 Claims, 45 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/831,276, filed on Dec. 4, 2017, now Pat. No. 10,321,971, which is a continuation of application No. 15/151,732, filed on May 11, 2016, now Pat. No. 9,833,297, which is a continuation-in-part of application No. 14/679,903, filed on Apr. 6, 2015, now Pat. No. 9,833,246, which is a continuation of application No. PCT/US2015/024402, filed on Apr. 4, 2015.

- (60) Provisional application No. 61/975,698, filed on Apr. 4, 2014.

(51) **Int. Cl.**

A61B 10/02 (2006.01)
A61B 17/00 (2006.01)
A61B 17/16 (2006.01)
A61B 17/32 (2006.01)
A61F 2/46 (2006.01)
A63B 57/60 (2015.01)

(52) **U.S. Cl.**

CPC *A61B 17/1608* (2013.01); *A46B 9/00* (2013.01); *A61B 2017/00004* (2013.01); *A61B 2017/00969* (2013.01); *A61B 17/1604*

(2013.01); *A61B 17/1611* (2013.01); *A61B 2017/320064* (2013.01); *A61B 2217/002* (2013.01); *A61F 2/4644* (2013.01); *A63B 57/60* (2015.10)

(58) **Field of Classification Search**

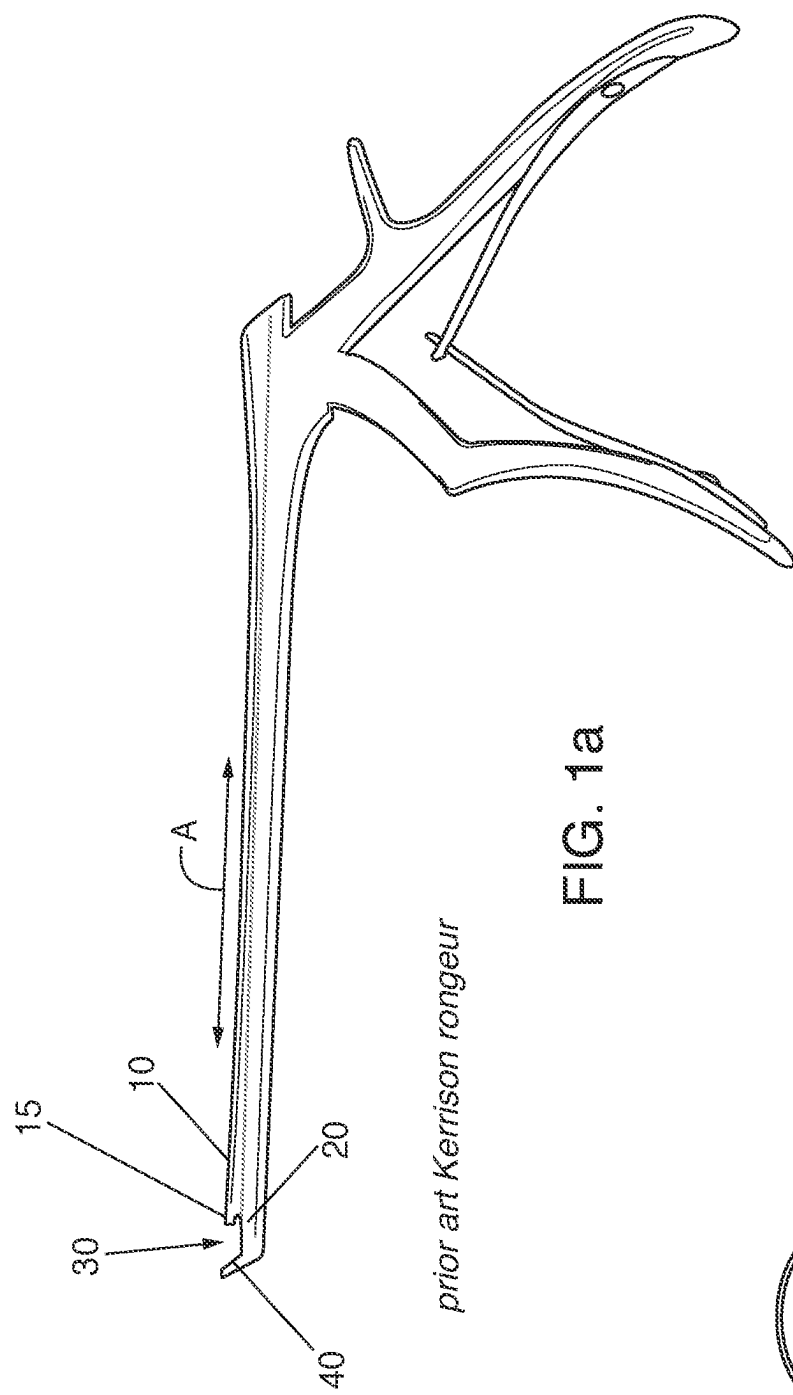
CPC . A61B 2217/002; A61B 10/0096; A61L 2/10; A61L 2/26; A61L 2202/182; A61L 2202/24
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 See application file for complete search history.

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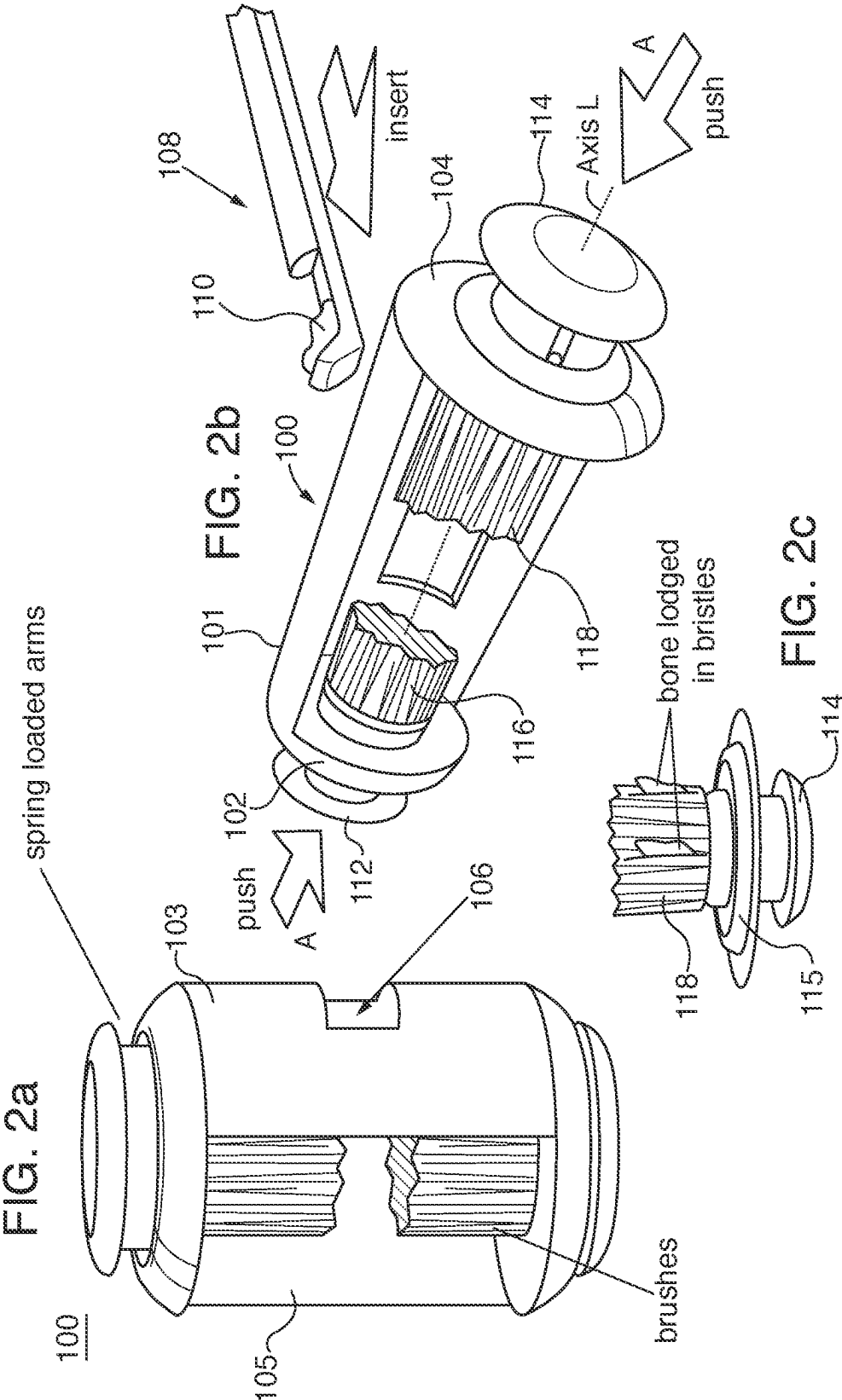


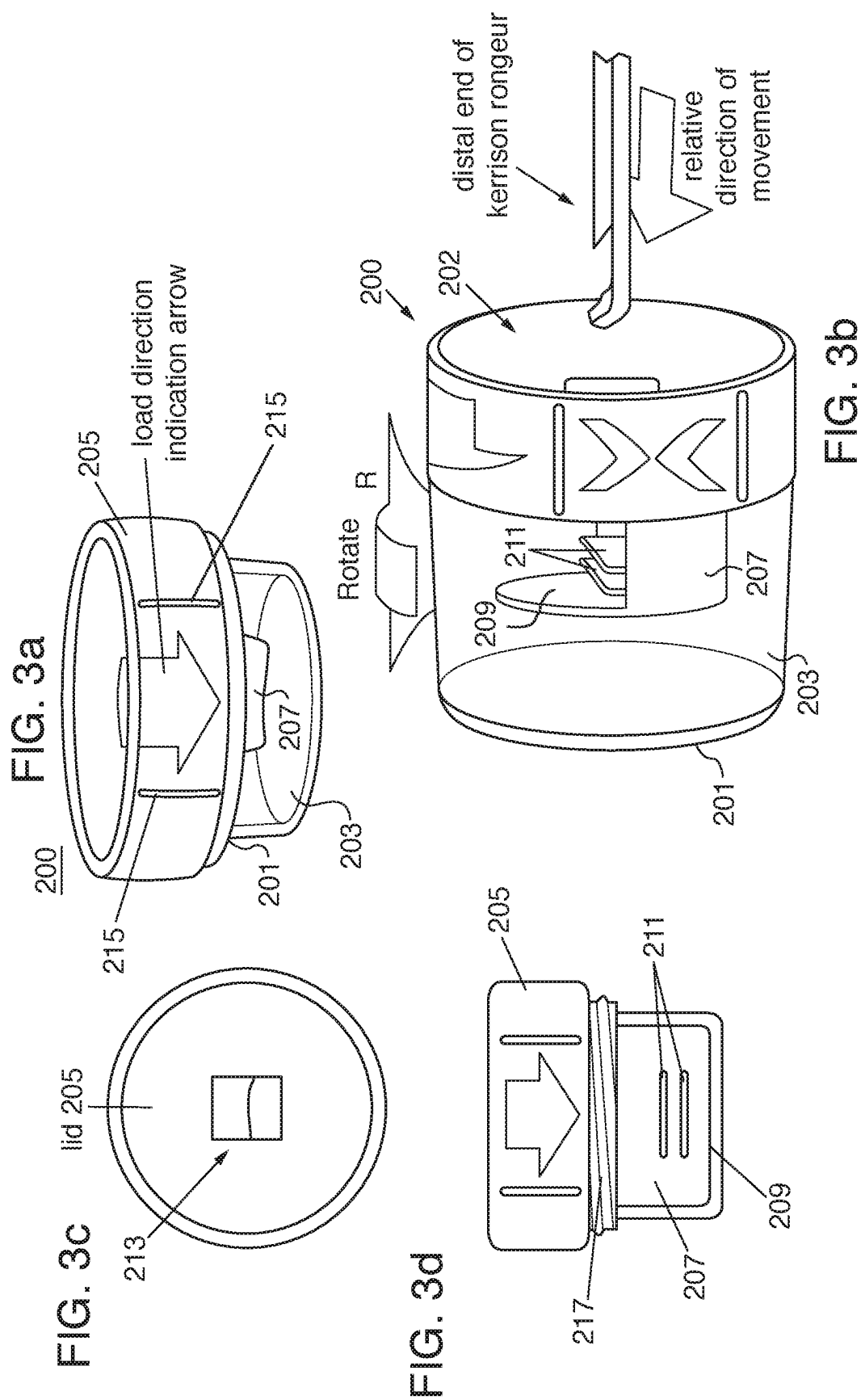
prior art Kerrison rongeur

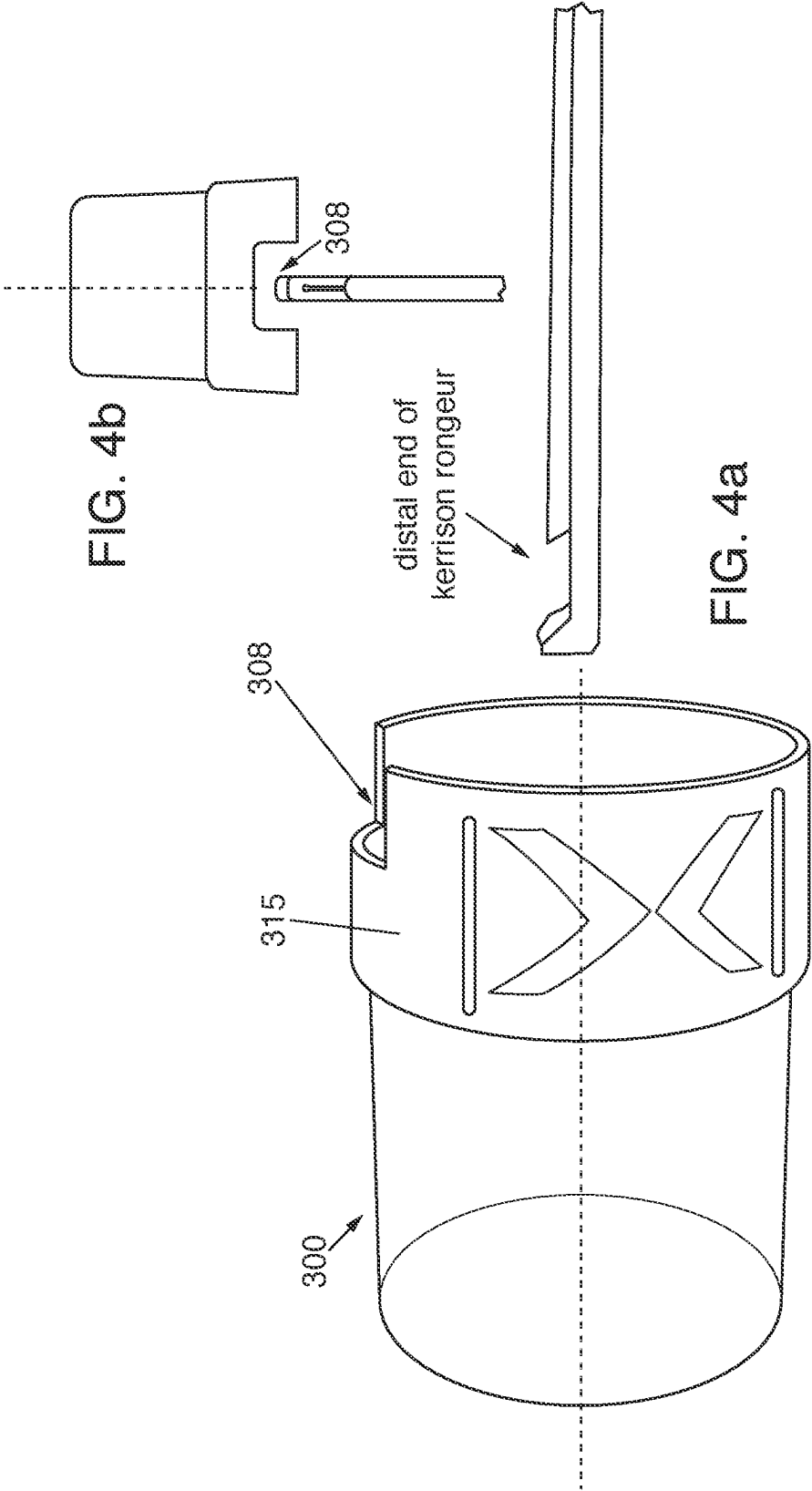
FIG. 1a

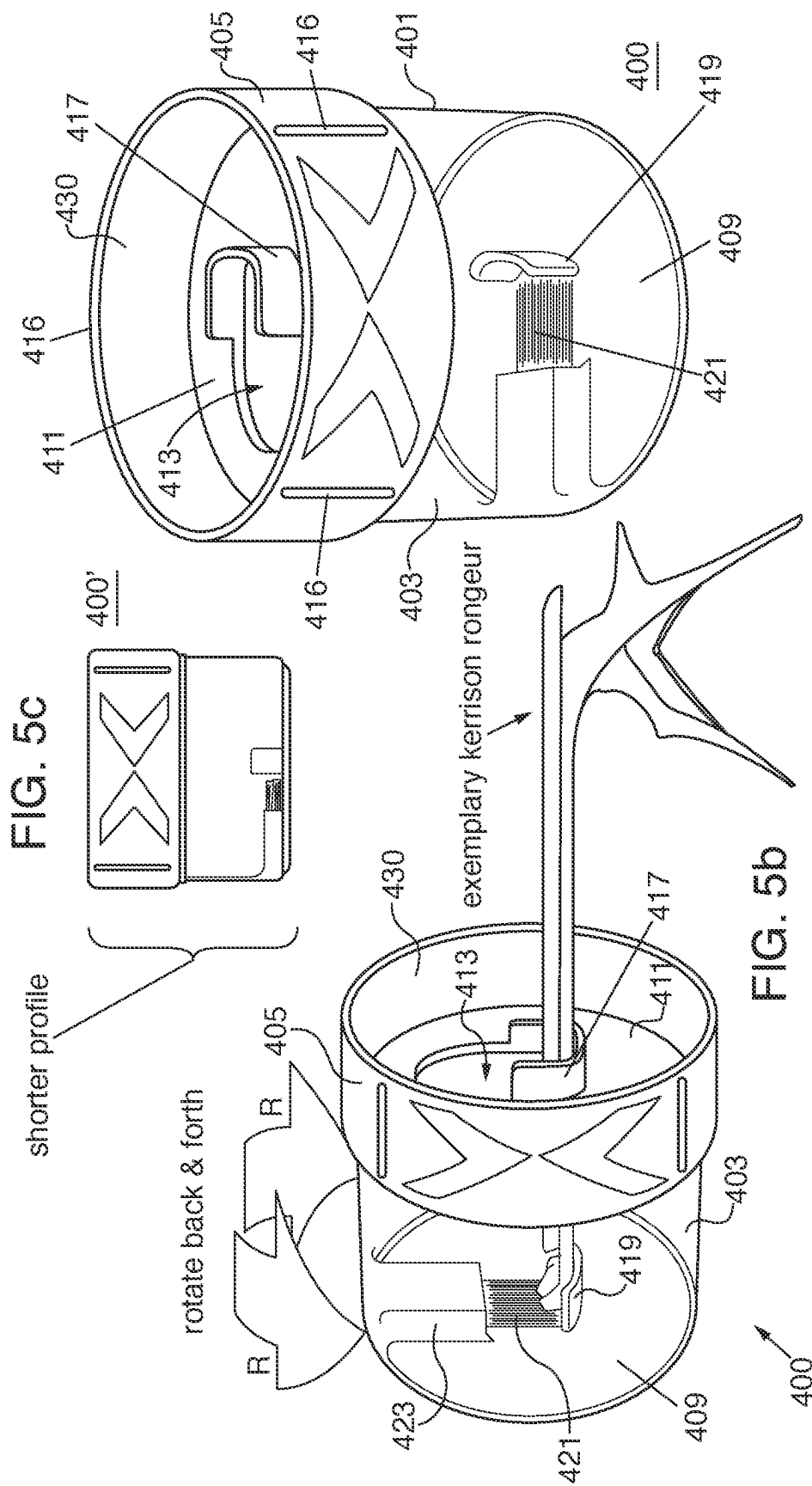


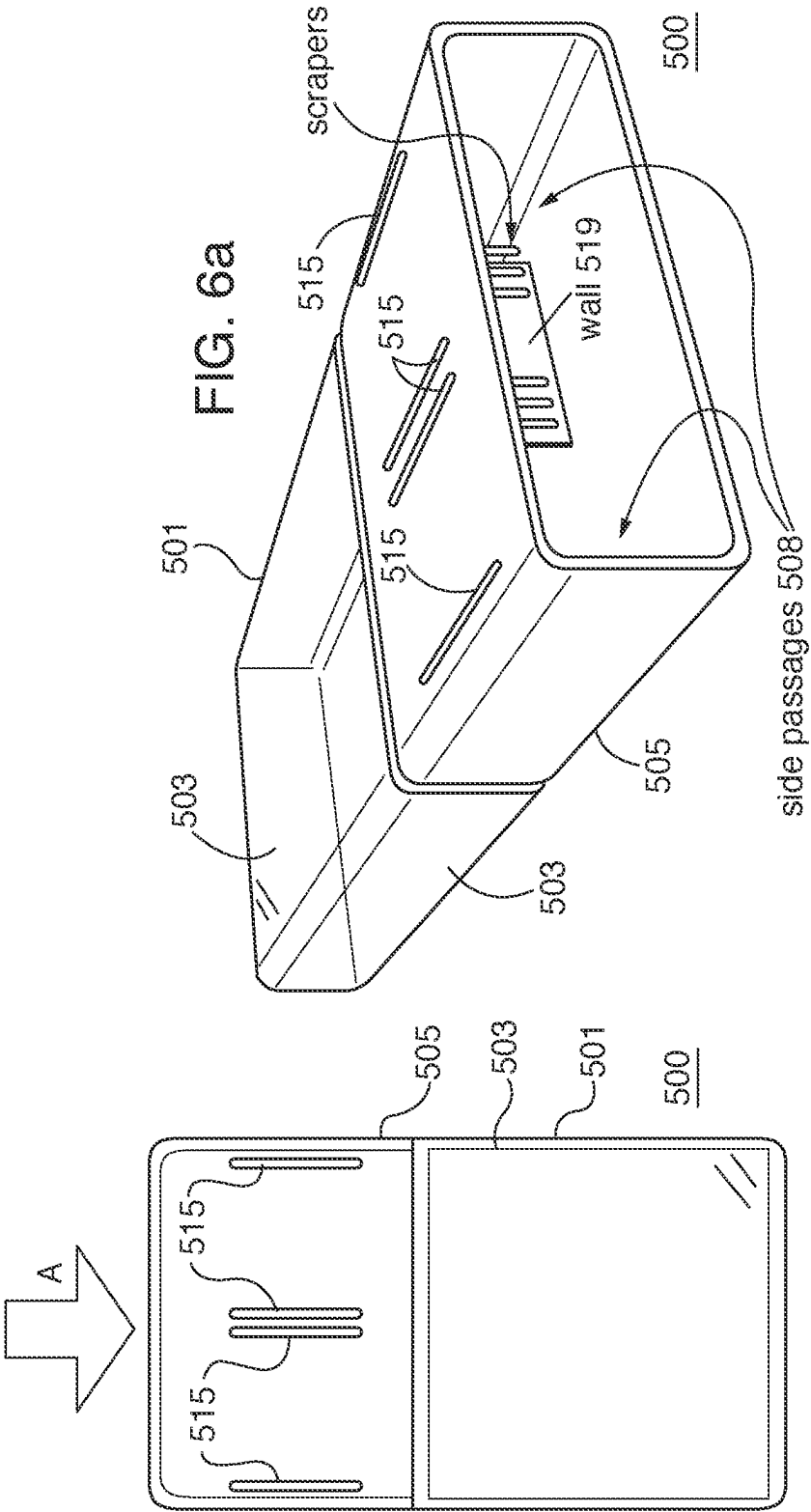
FIG. 1b

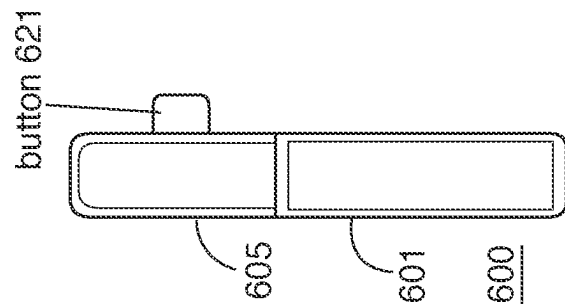




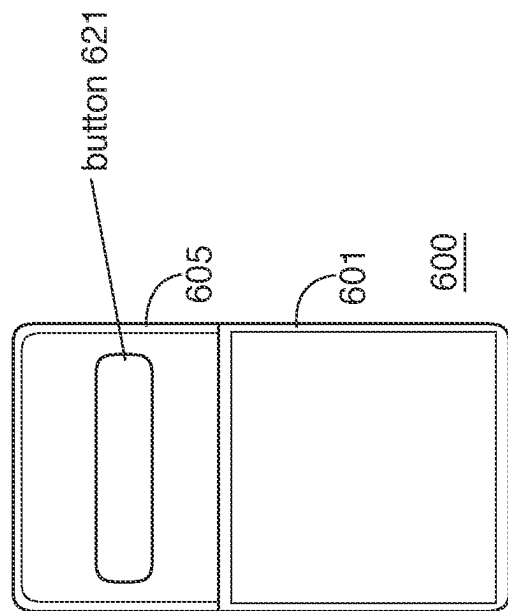




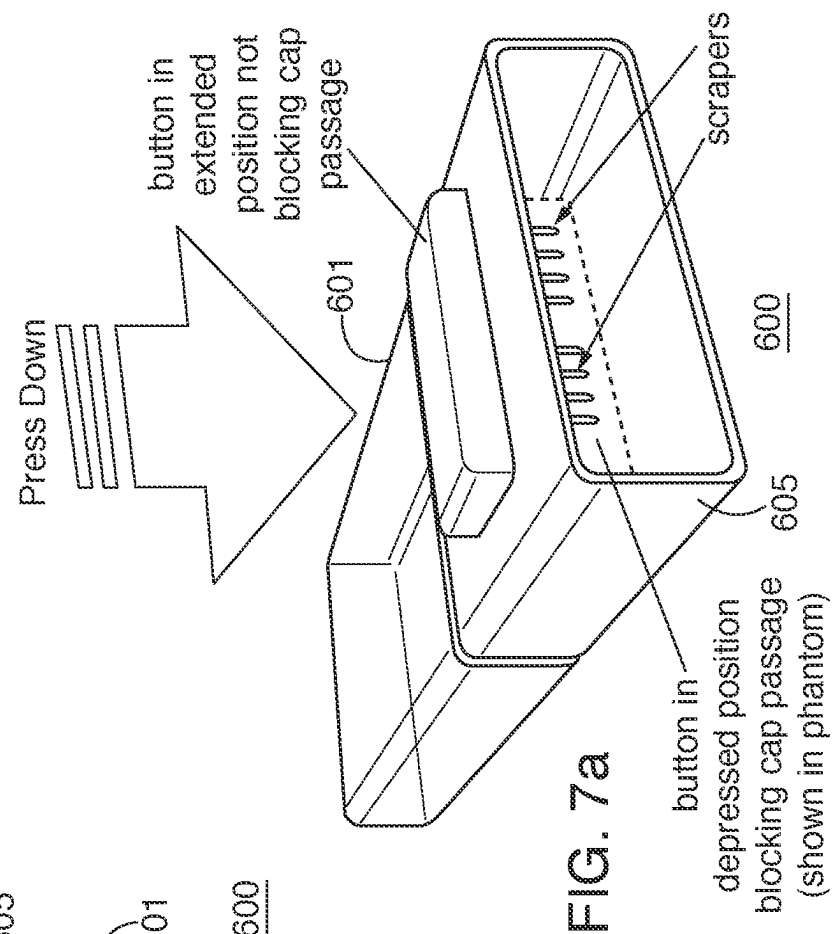




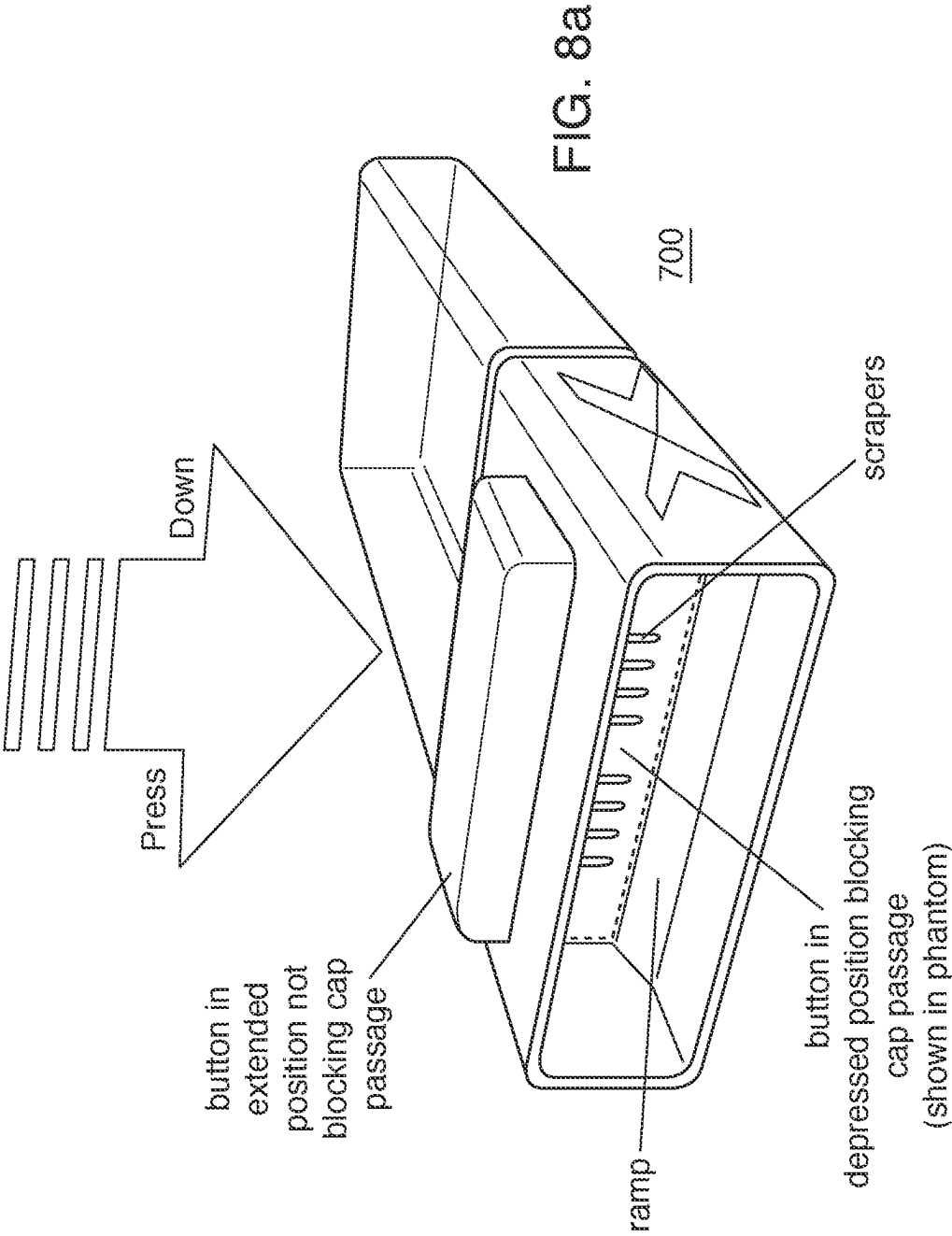
79



70



79



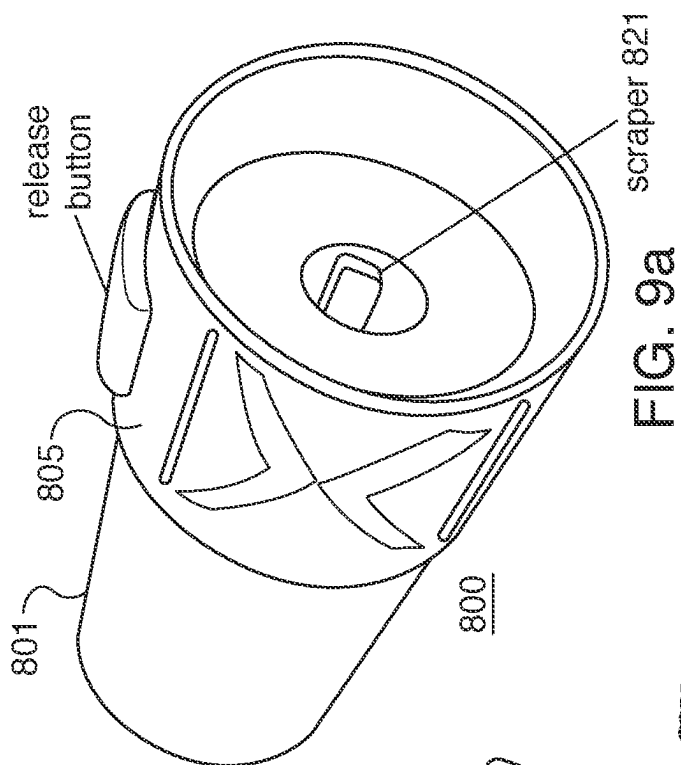


FIG. 9a

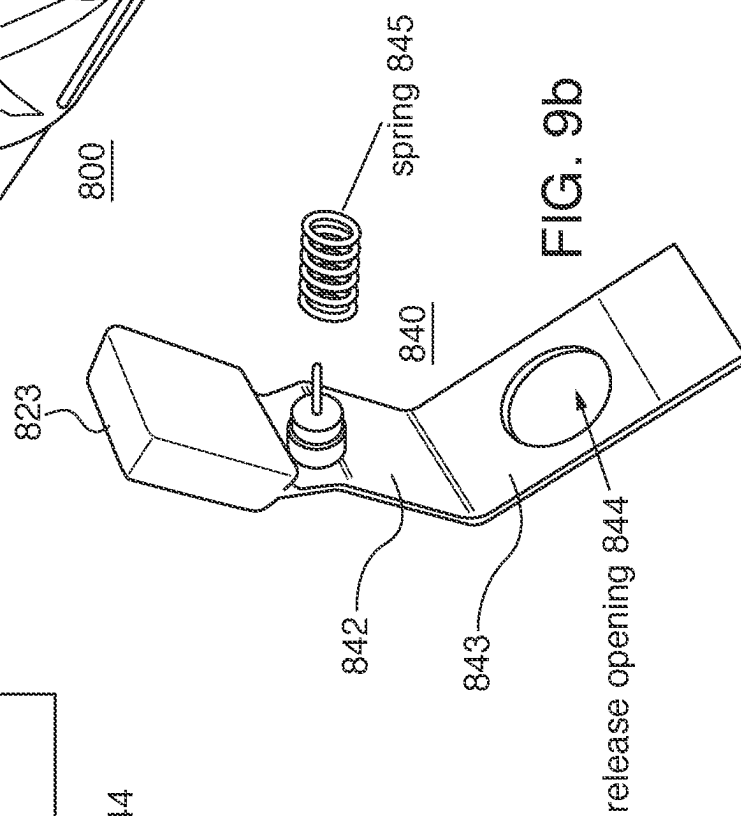


FIG. 9b

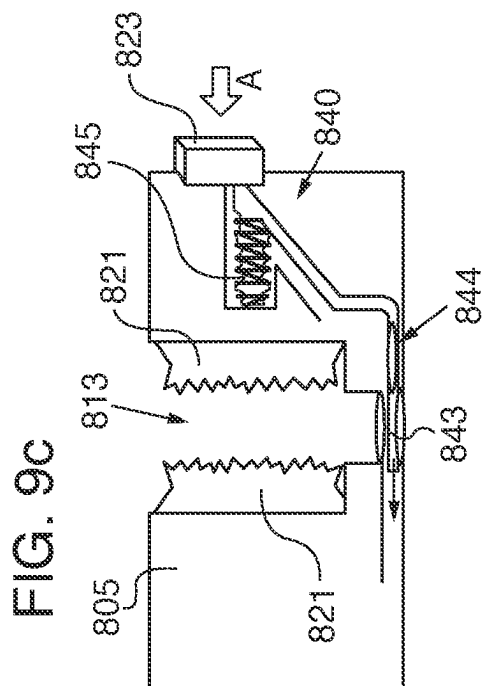
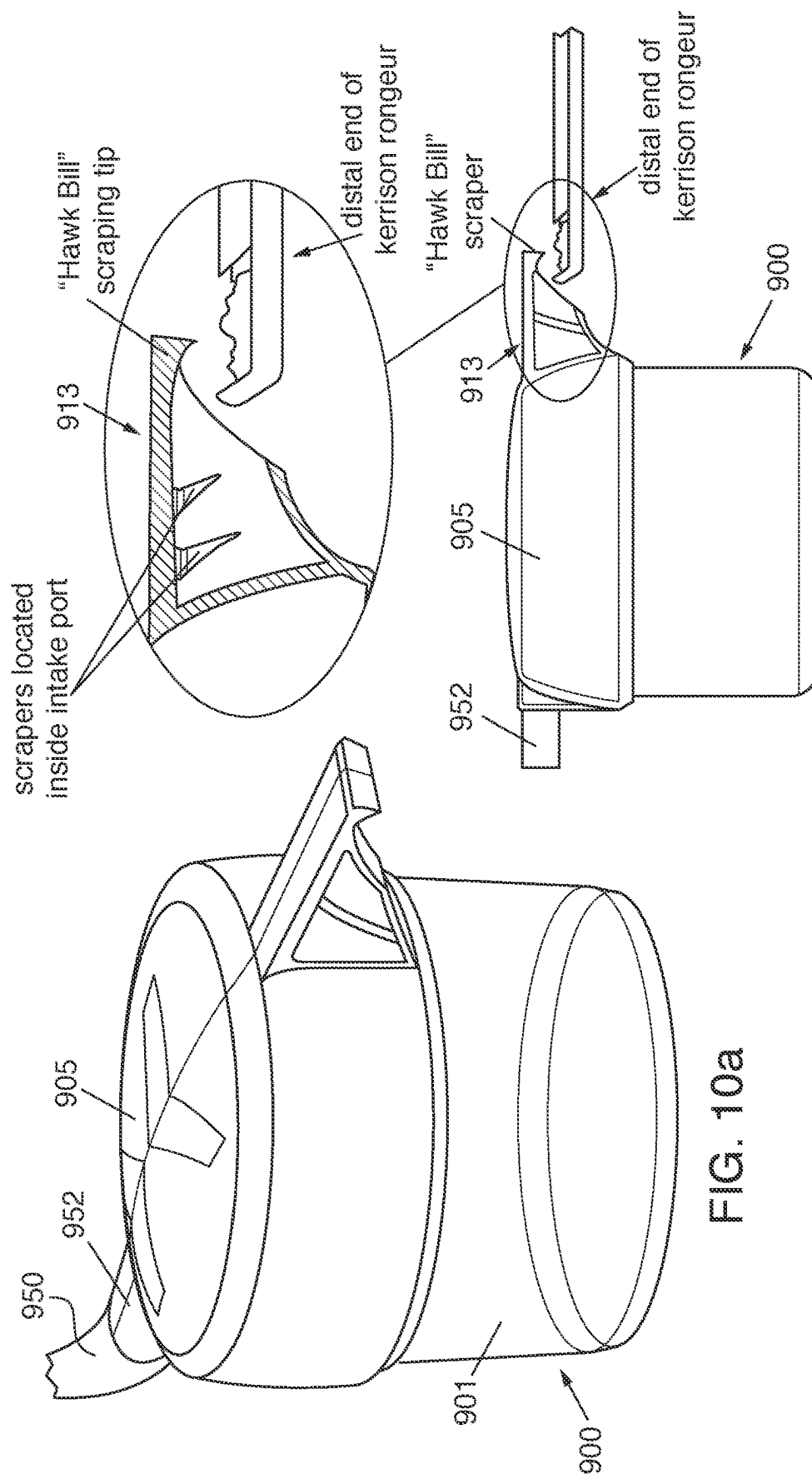
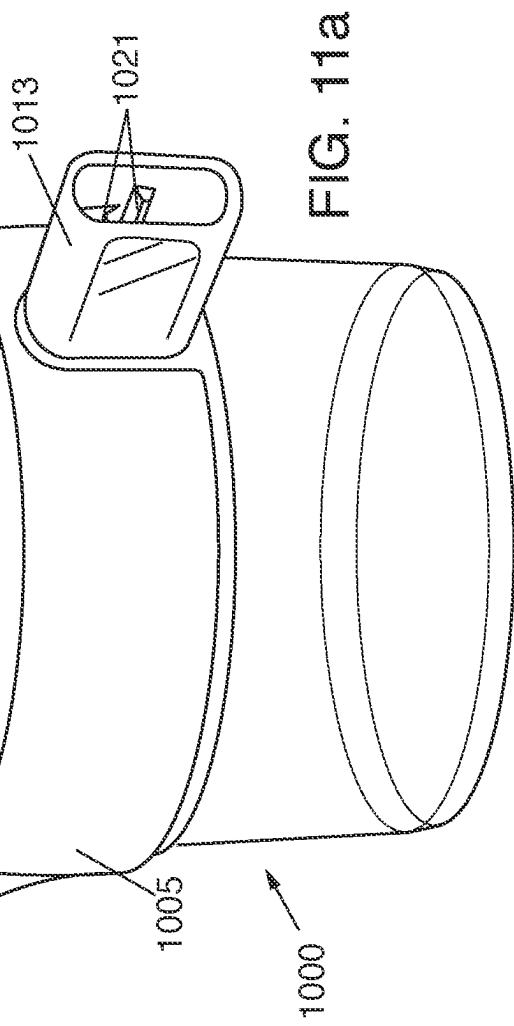
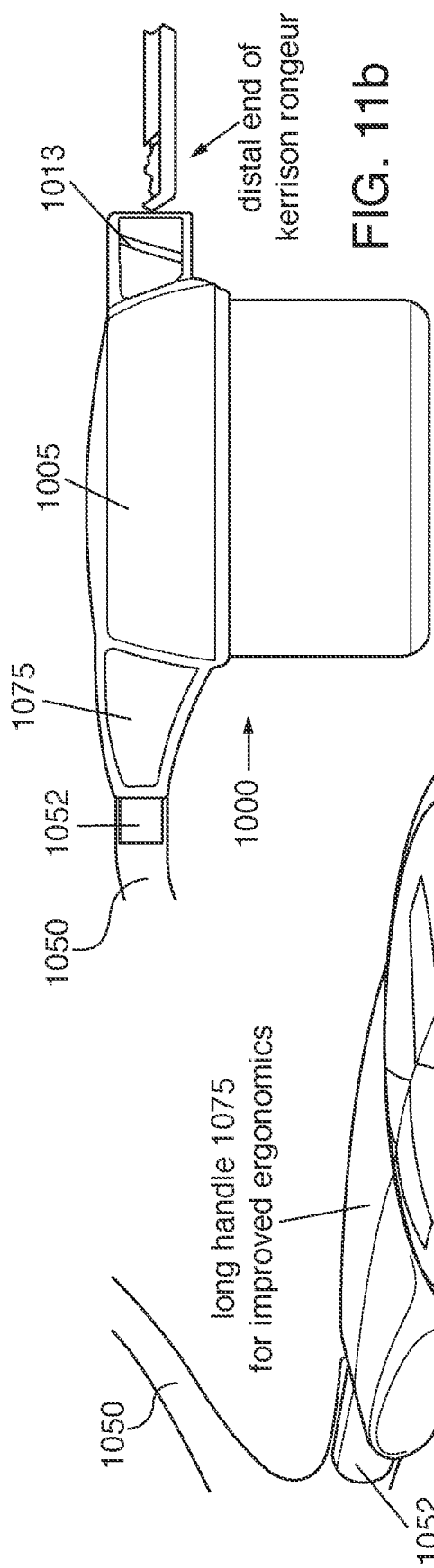


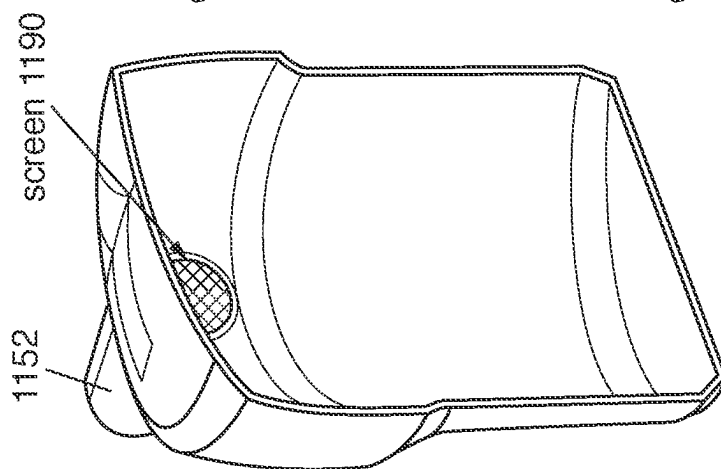
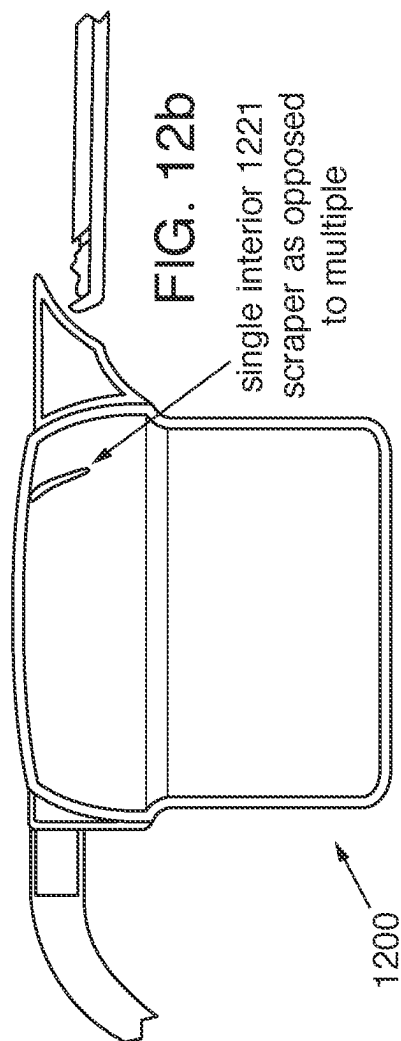
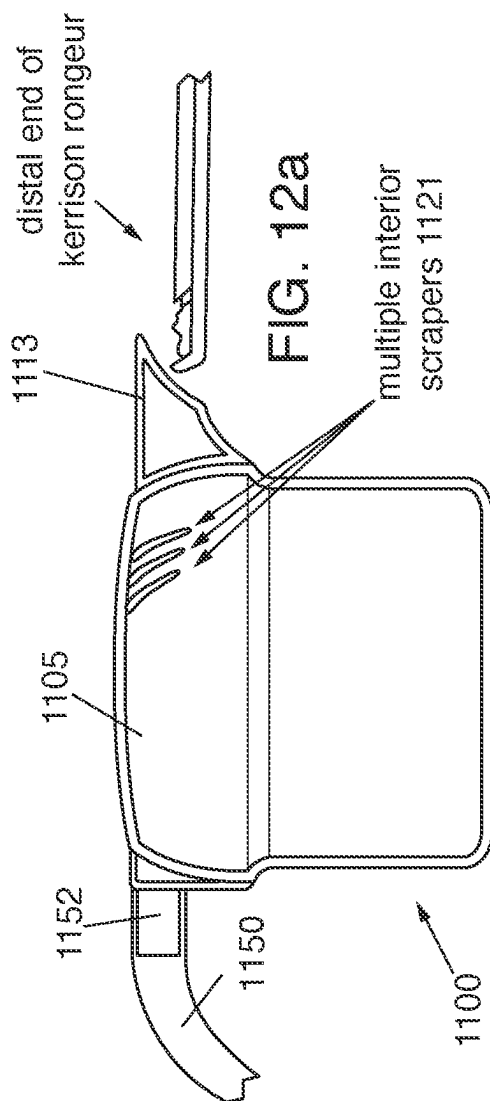
FIG. 9c

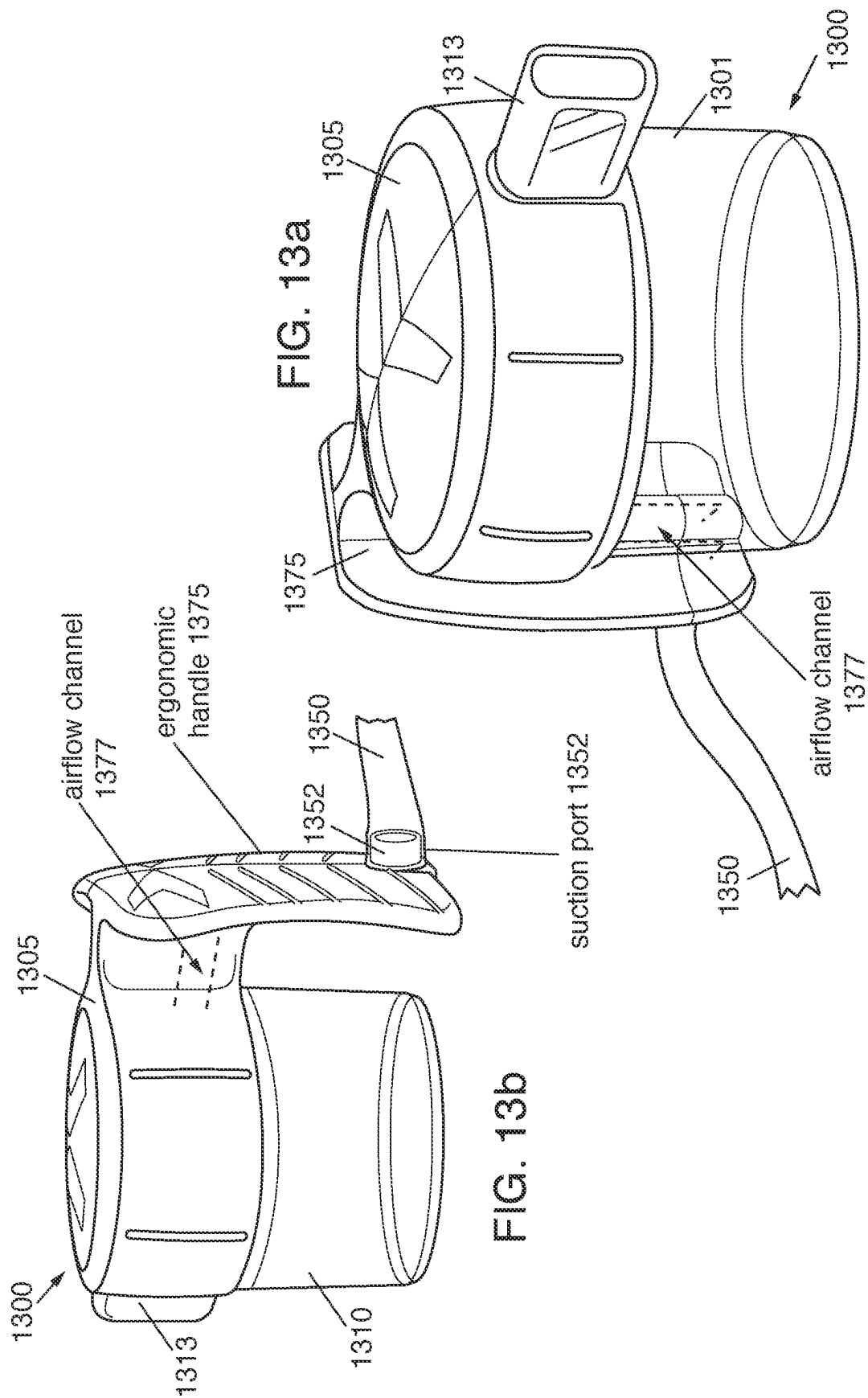


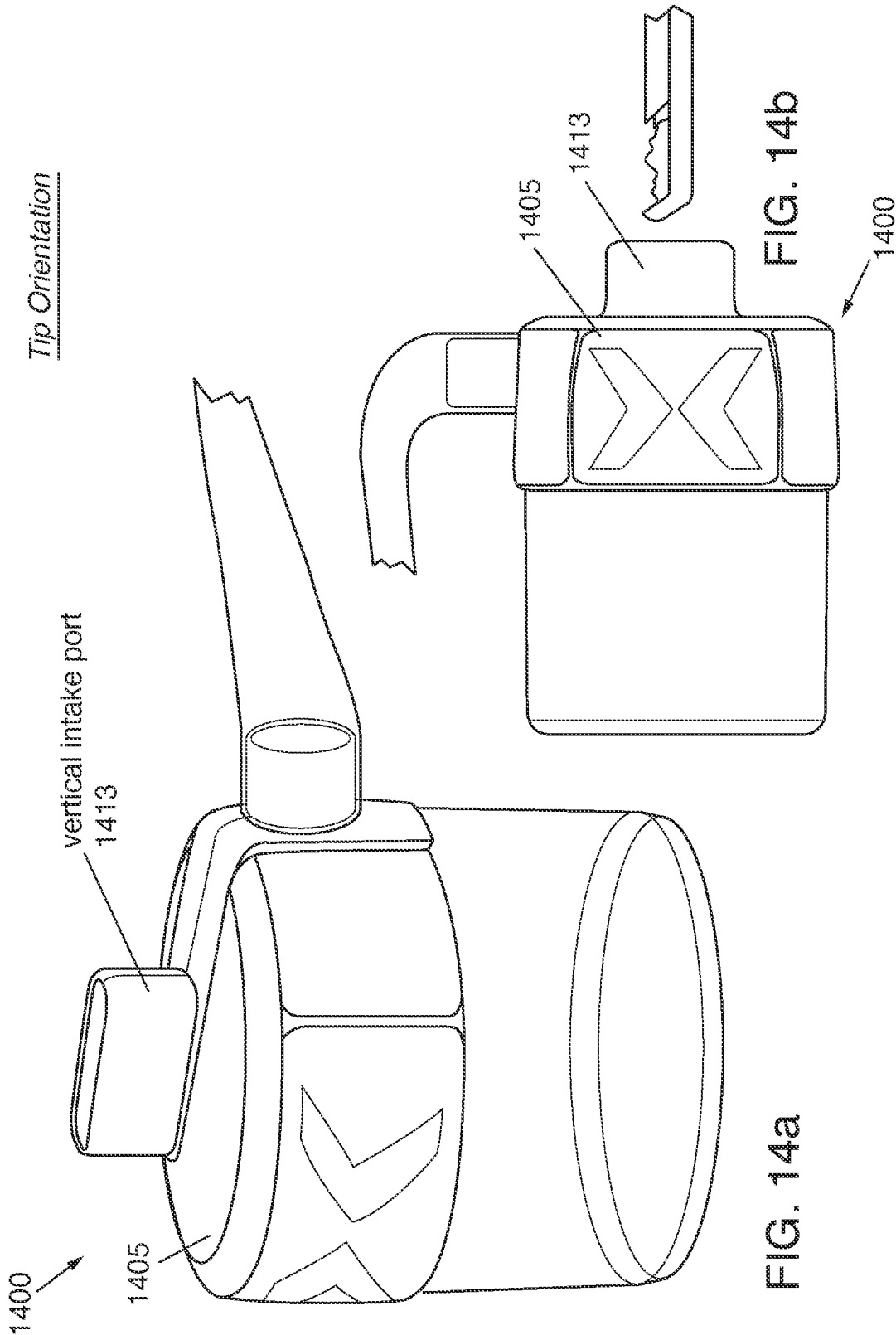
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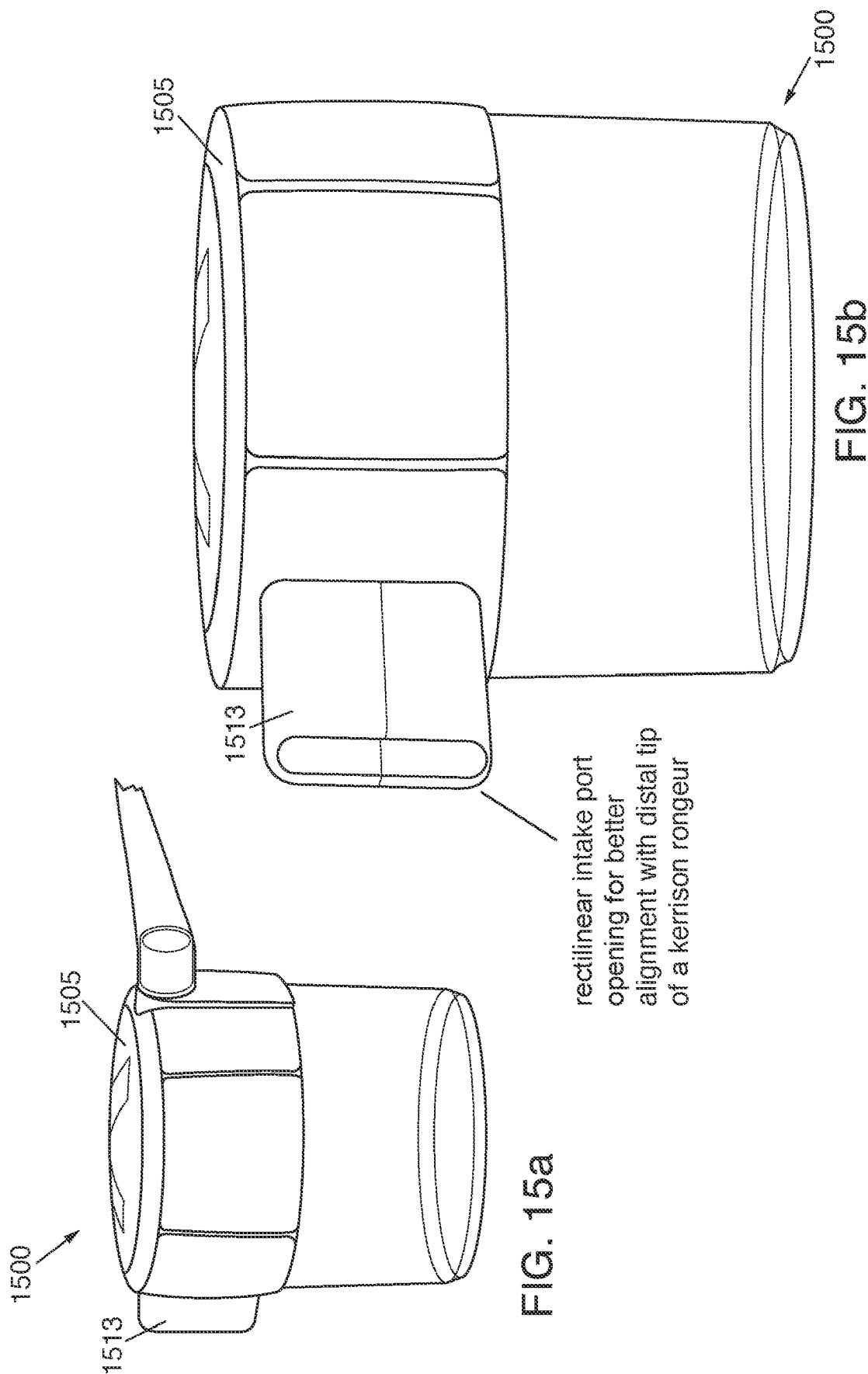










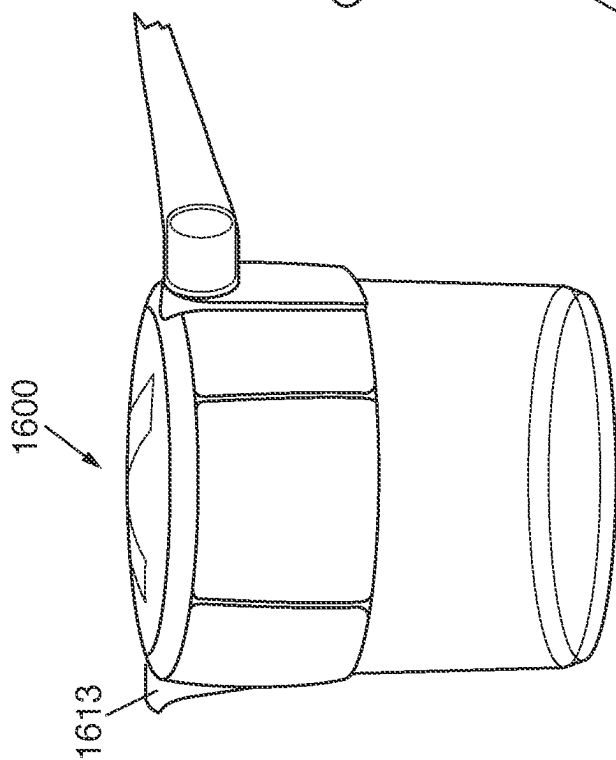


FIG. 16a

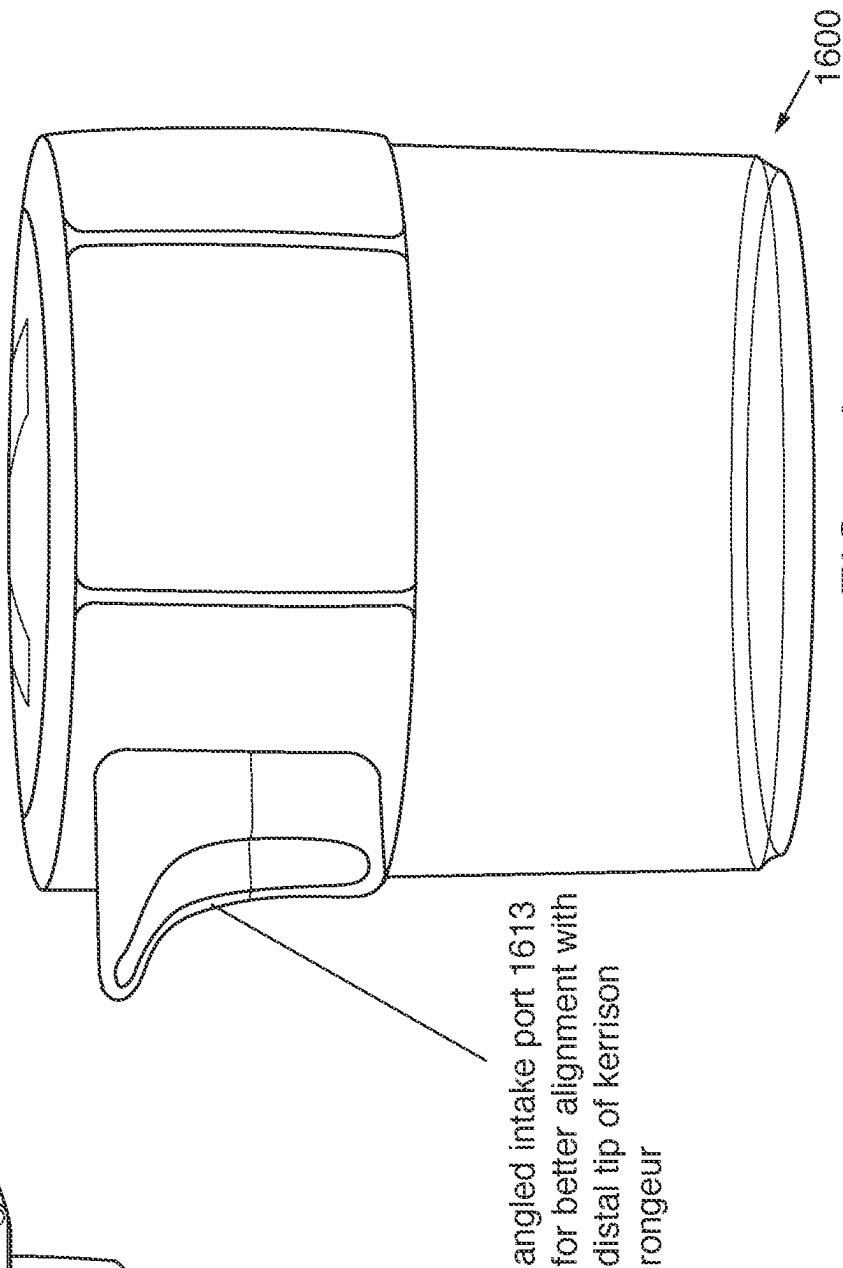


FIG. 16b

FIG. 17a

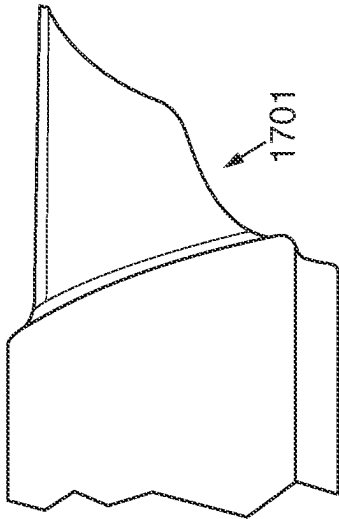


FIG. 17b

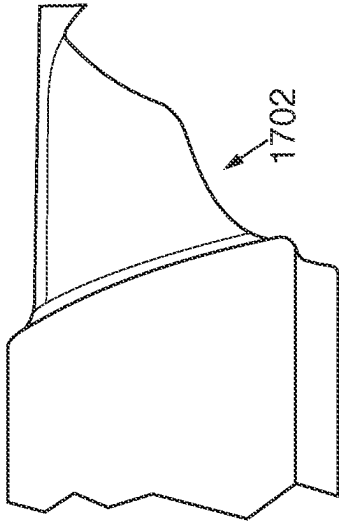


FIG. 17c

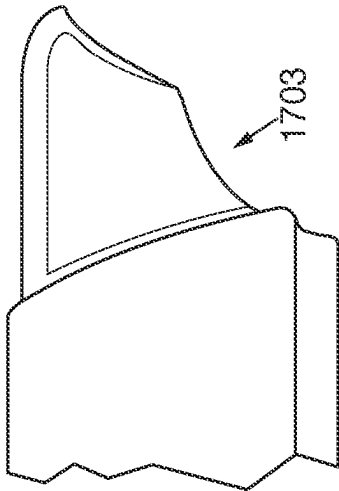


FIG. 17d

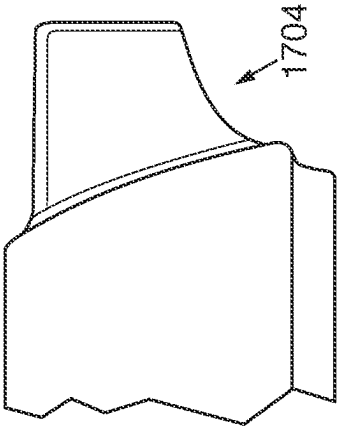


FIG. 17e

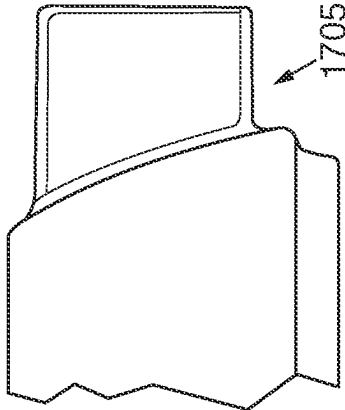


FIG. 17f

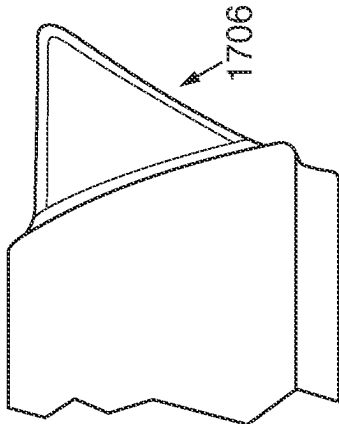


FIG. 18a

brushes

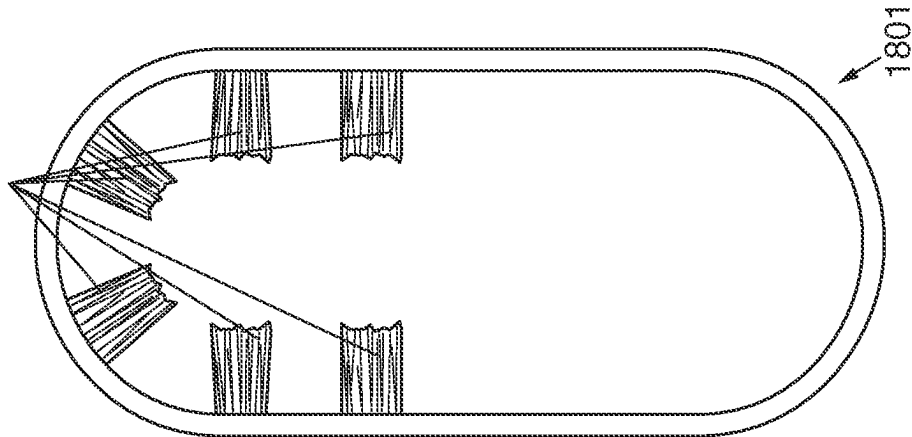


FIG. 18b

top scraper

Side
scraper

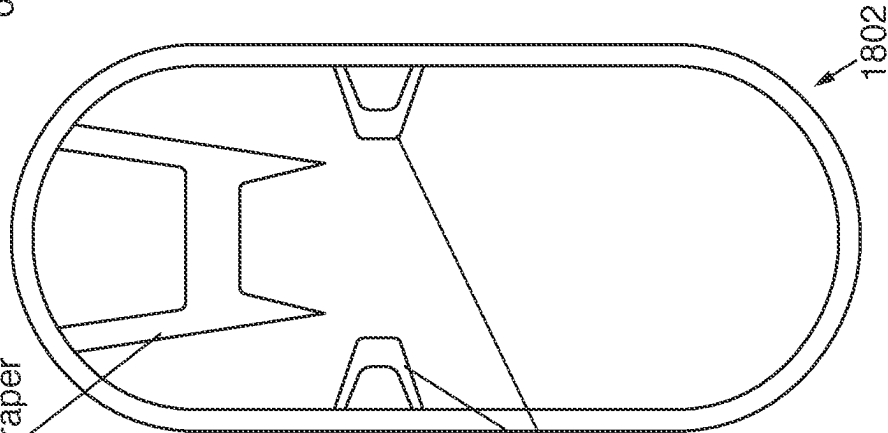
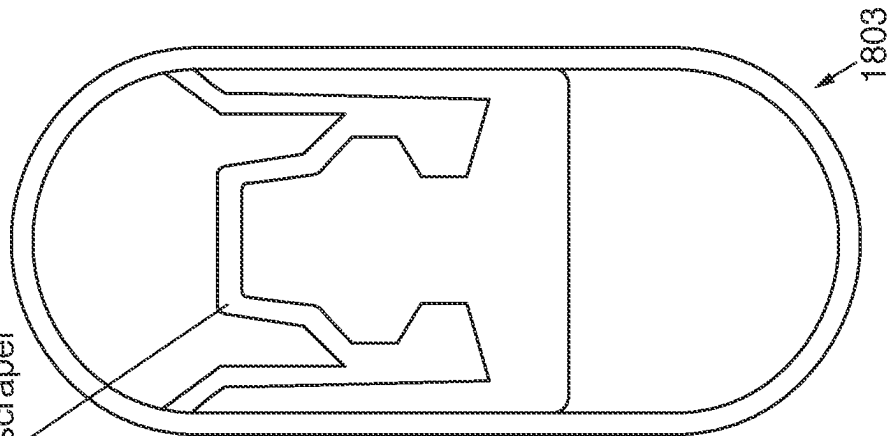


FIG. 18c

one piece scraper



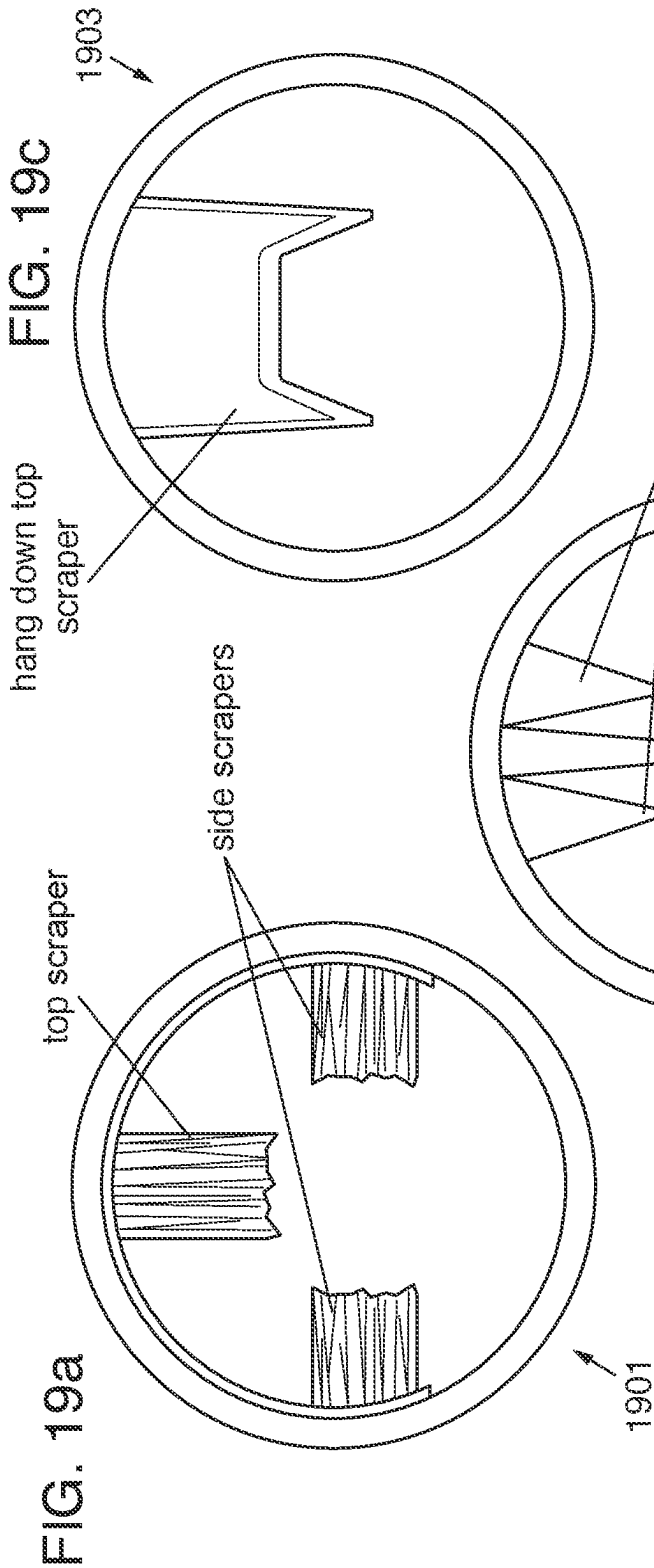


FIG. 19c

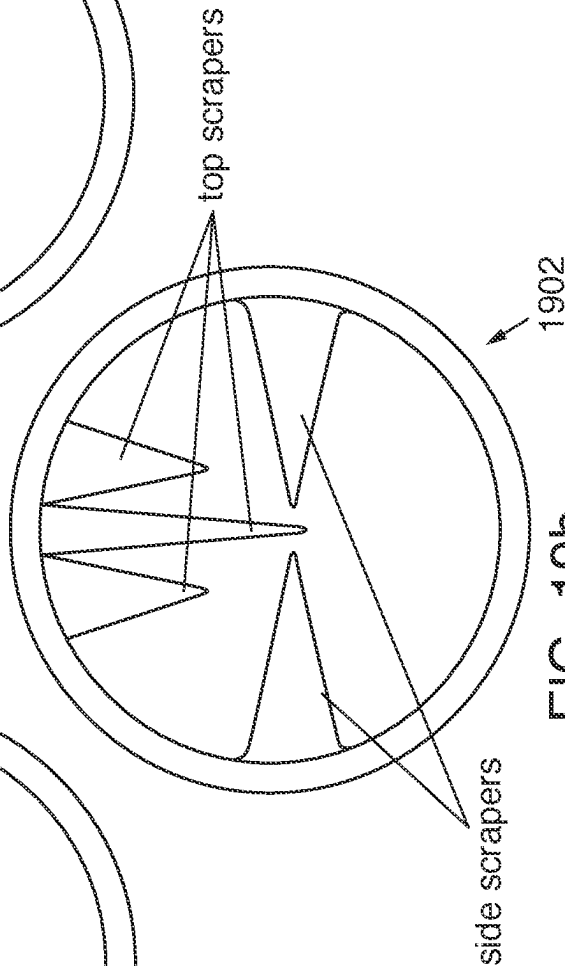
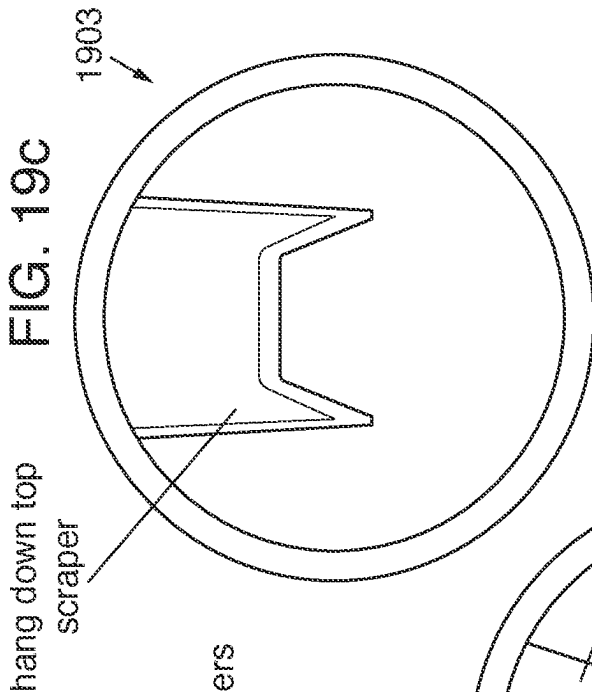


FIG. 19b

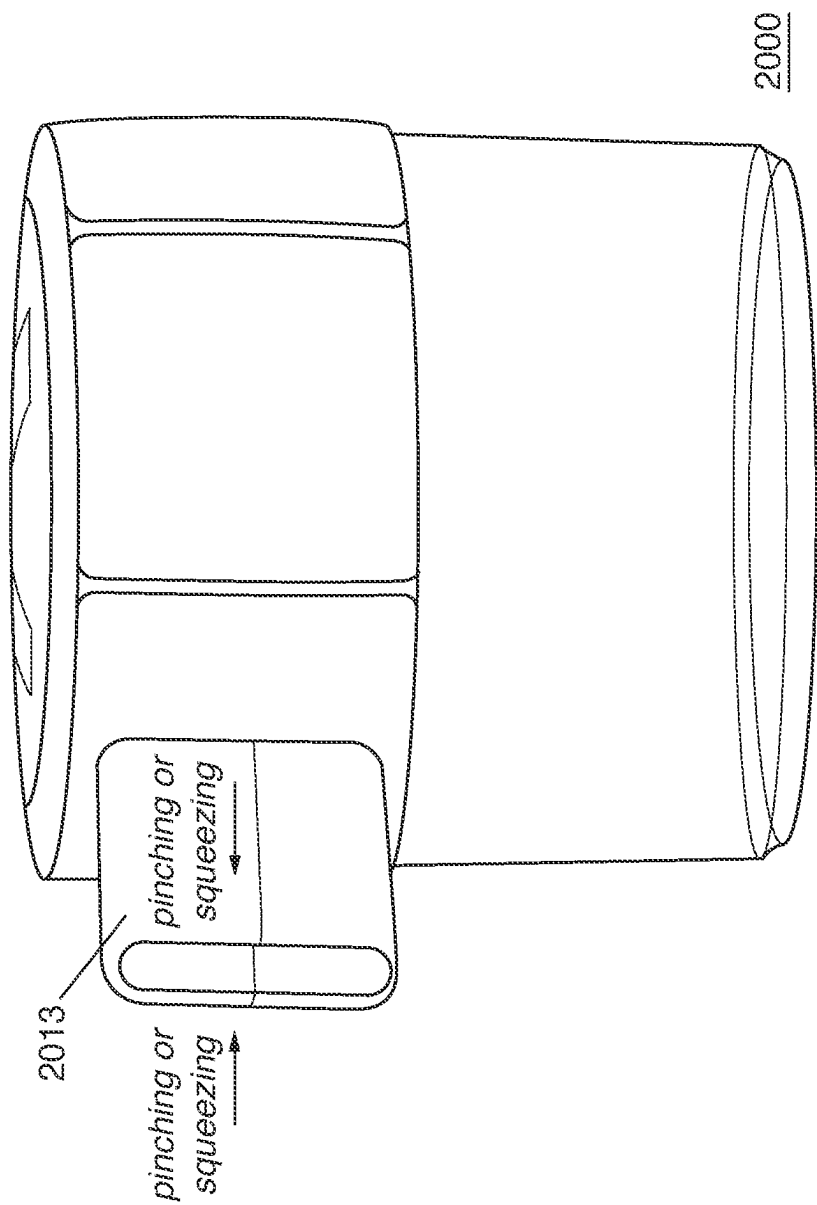


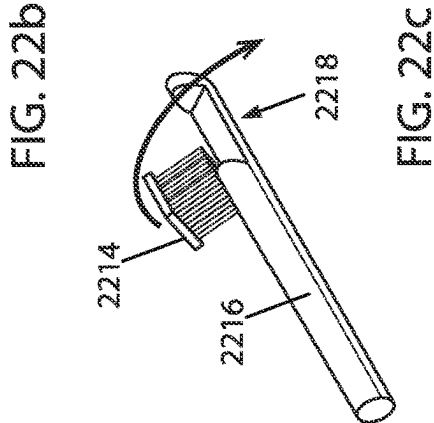
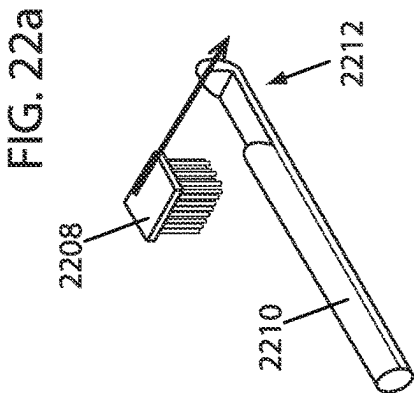
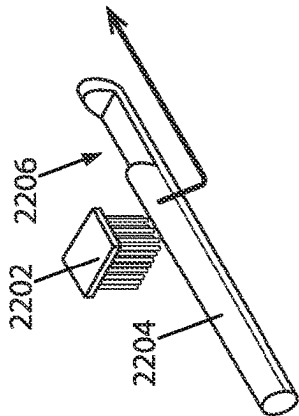
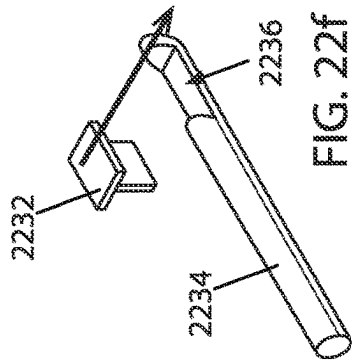
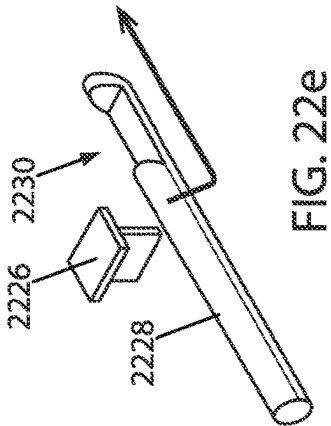
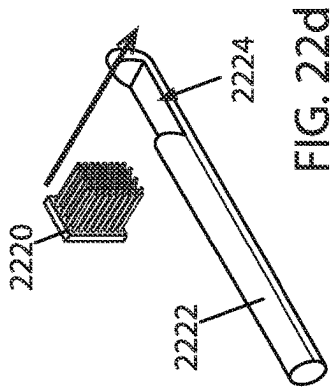
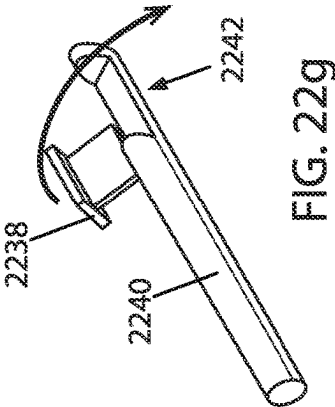
FIG. 20

Element	Path	Orientation to Path
Brush	Axial	Tangent
Wiper (soft)	Radial	Axial
Wiper (Hard)	Rotational	Axial
Pressure (air)		
Suction		

FIG. 21a

Element	Path	Orientation to Path
Brush	Axial	Tangent
Brush	Radial	Tangent
Brush	Rotational	Tangent
Brush	Axial	Axial
Brush	Radial	Axial
Brush	Rotational	Axial
Wiper (soft)	Axial	Tangent
Wiper (soft)	Radial	Tangent
Wiper (soft)	Rotational	Tangent
Wiper (soft)	Axial	Axial
Wiper (soft)	Radial	Axial
Wiper (soft)	Rotational	Axial
Wiper (Hard)	Axial	Tangent
Wiper (Hard)	Radial	Tangent
Wiper (Hard)	Rotational	Tangent
Wiper (Hard)	Axial	Axial
Wiper (Hard)	Radial	Axial
Wiper (Hard)	Rotational	Axial

FIG. 21b



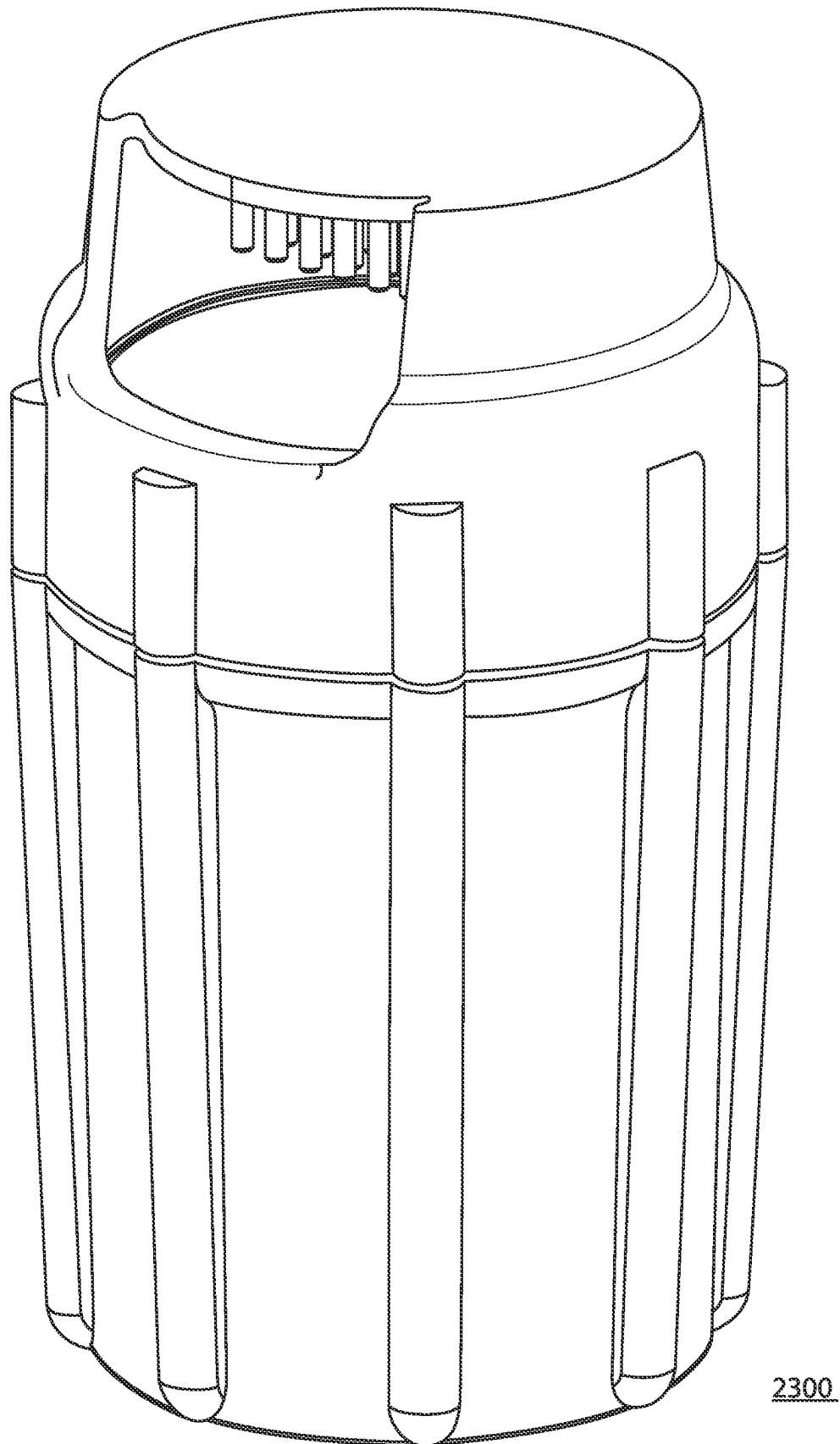


FIG. 23

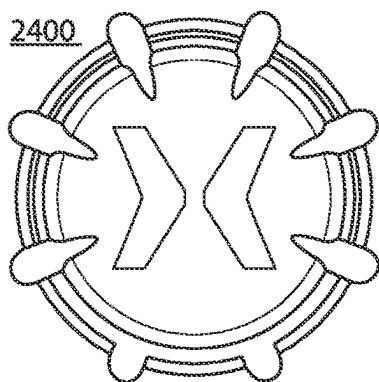


FIG. 24a

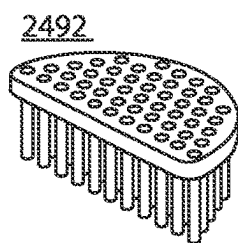


FIG. 24f

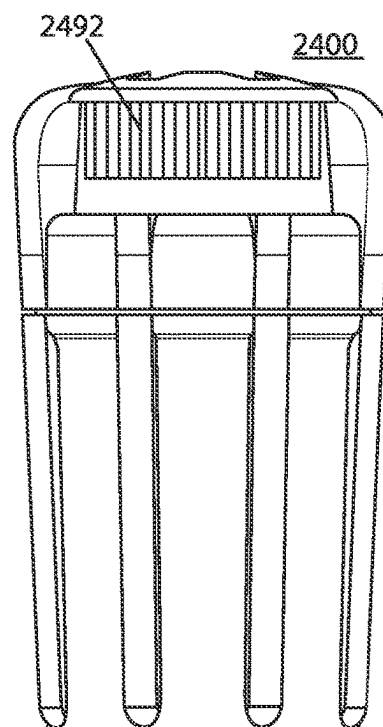


FIG. 24c

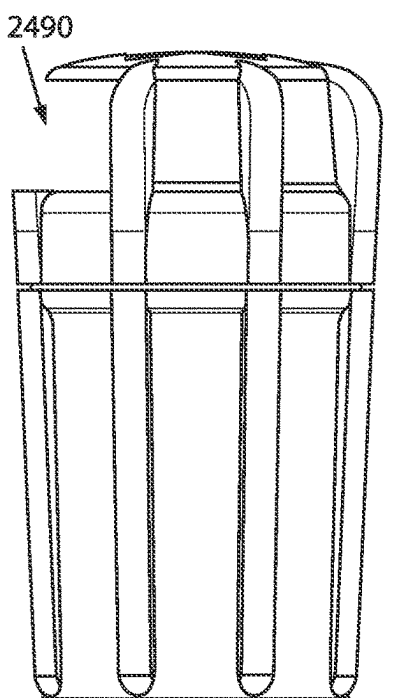


FIG. 24b

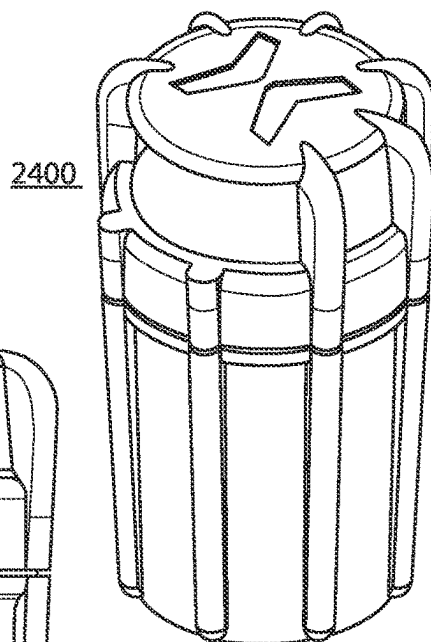


FIG. 24d

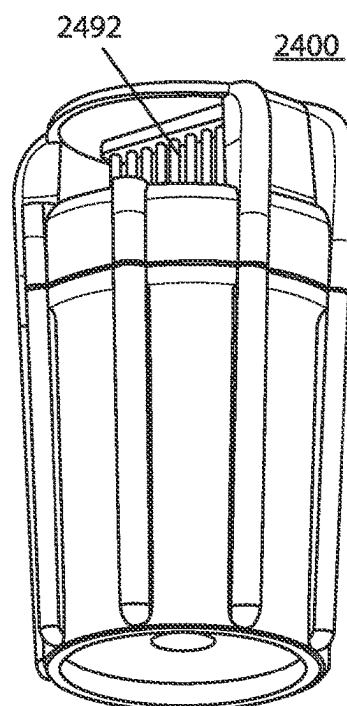


FIG. 24e

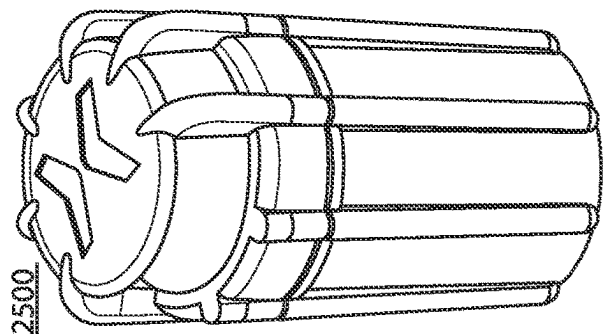


FIG. 25a

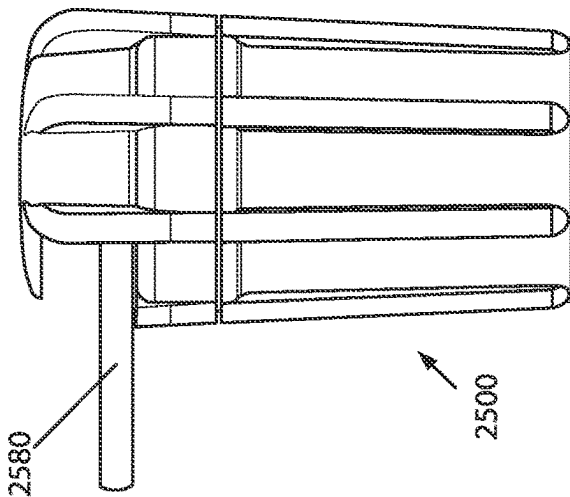


FIG. 25b

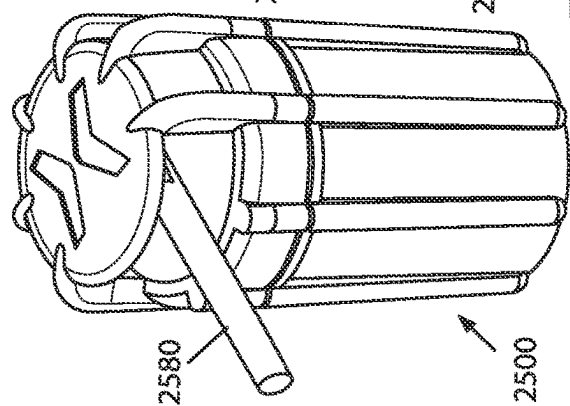


FIG. 25c

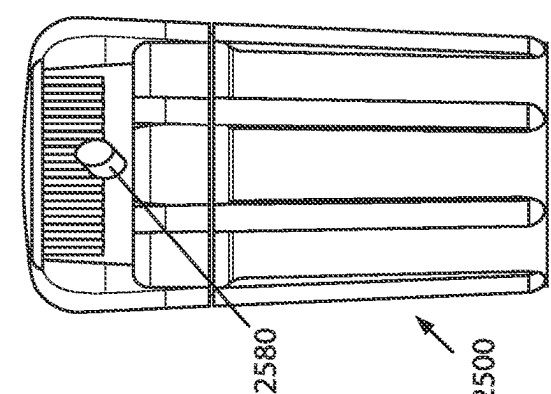


FIG. 25d

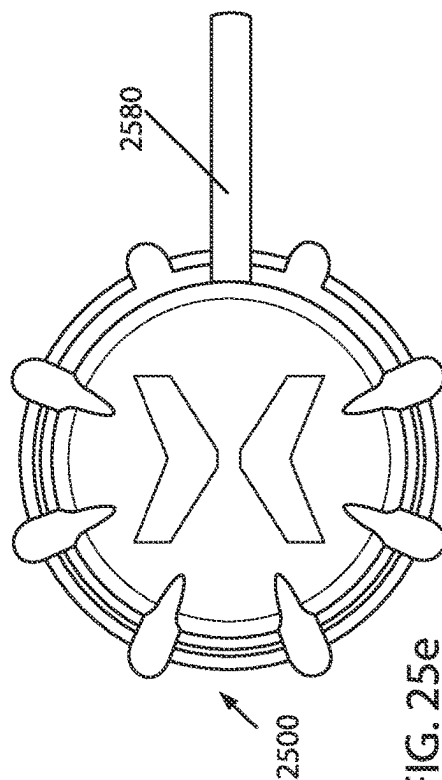


FIG. 25e

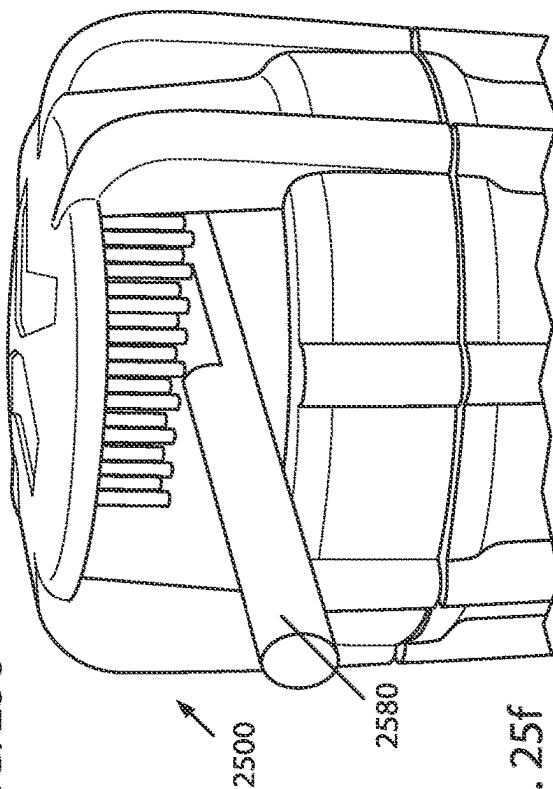


FIG. 25f

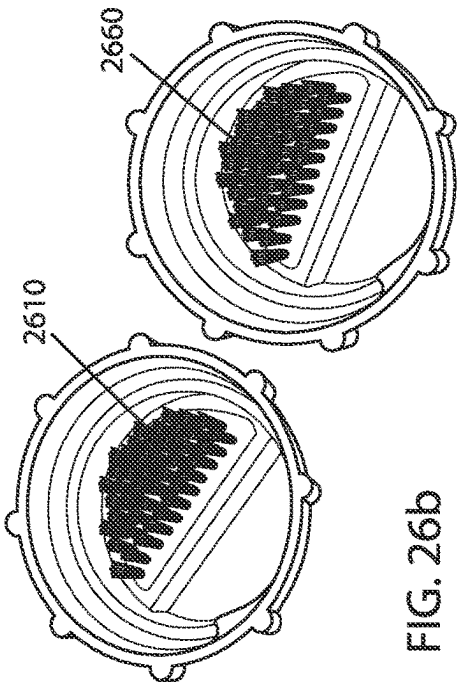


FIG. 26b

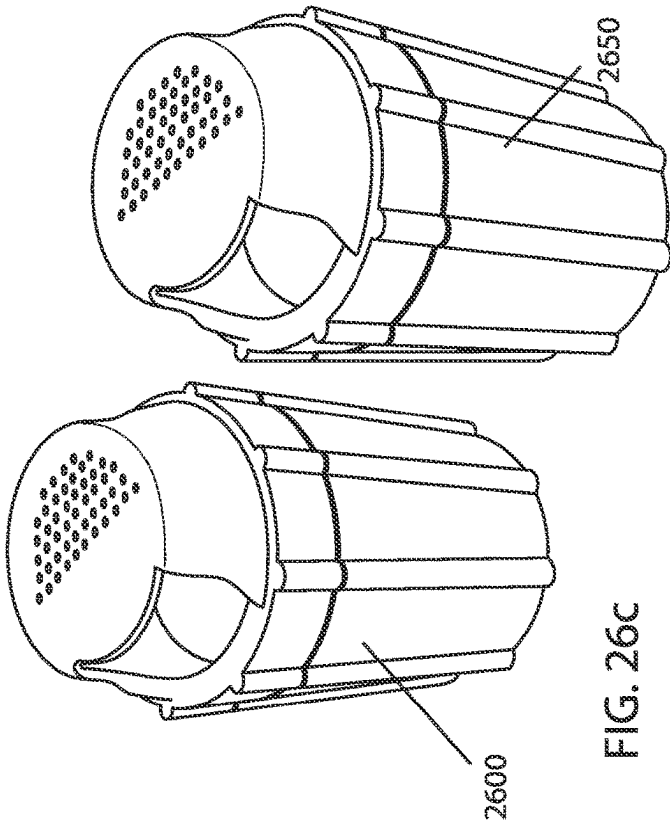


FIG. 26c

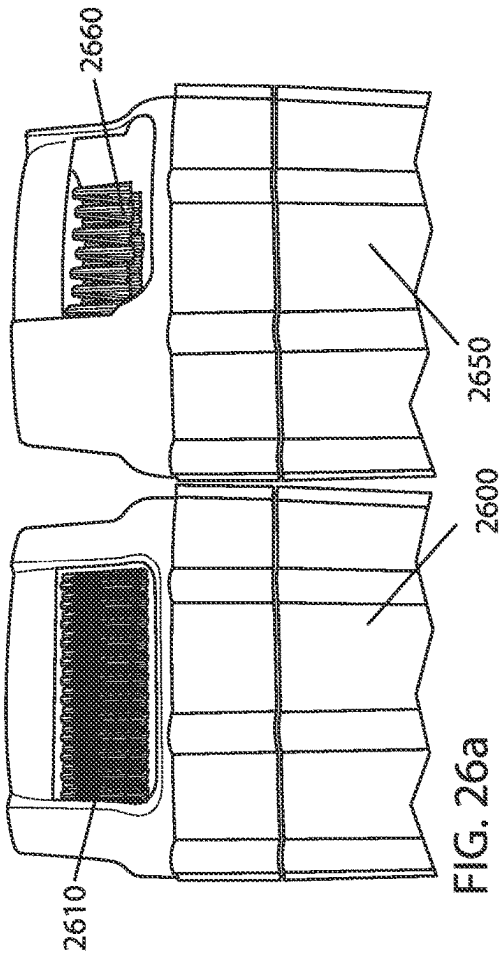


FIG. 26a

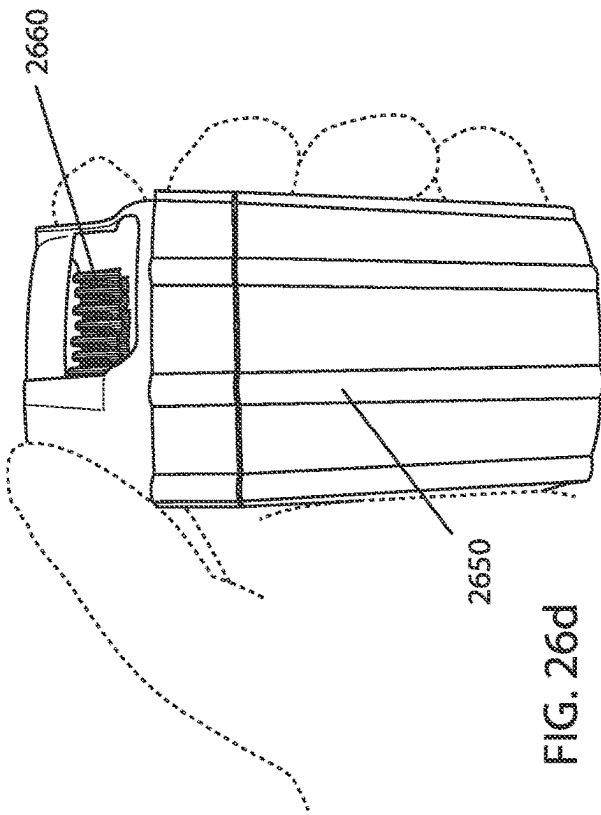
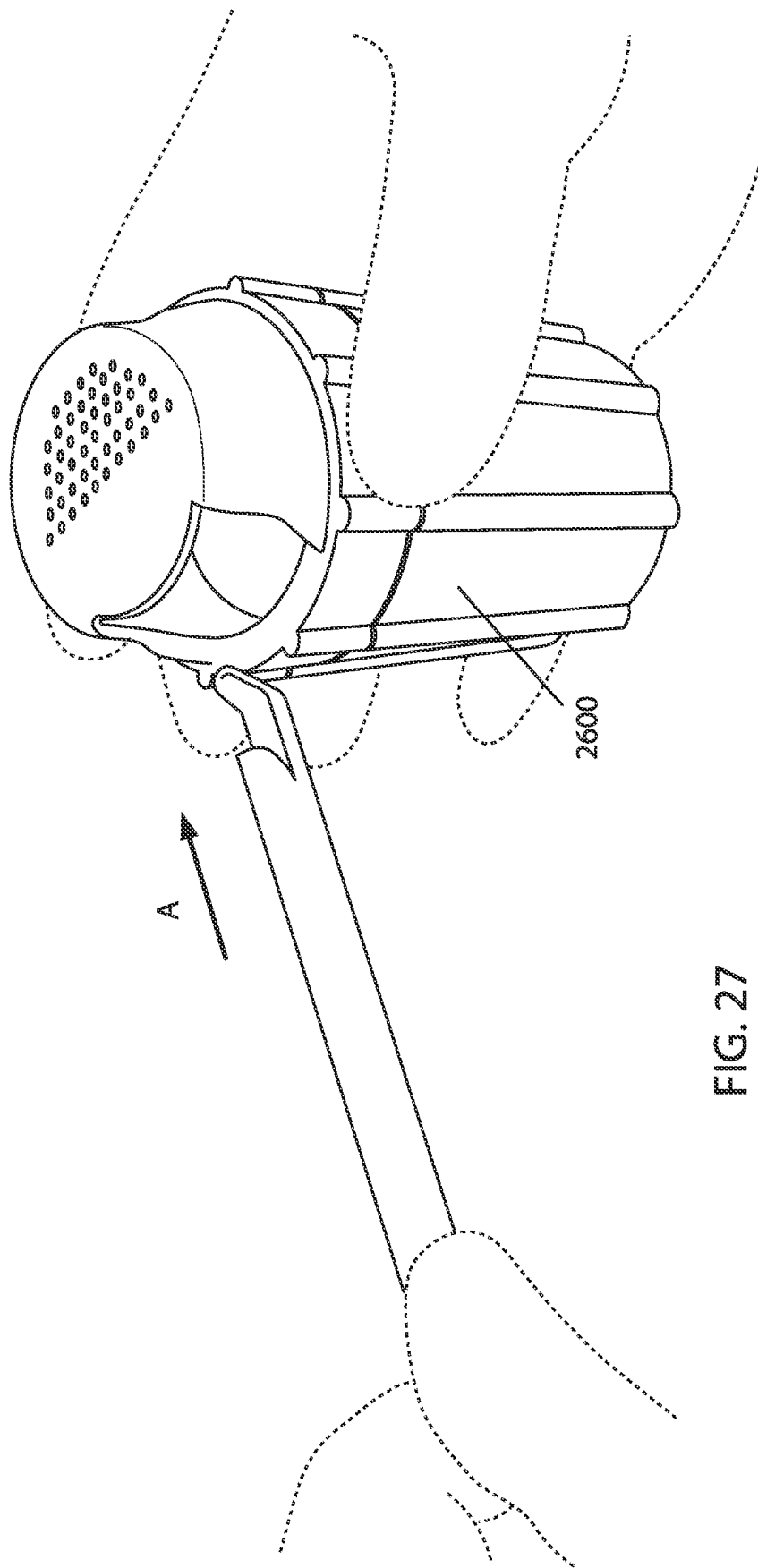


FIG. 26d



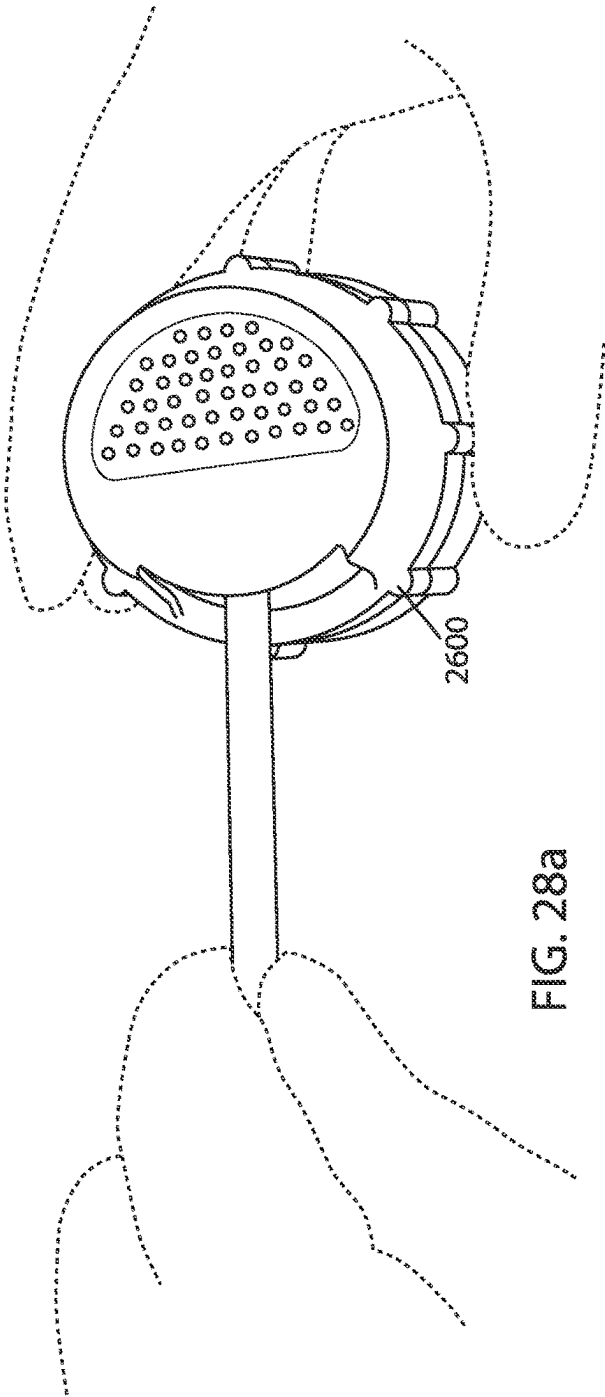


FIG. 28a

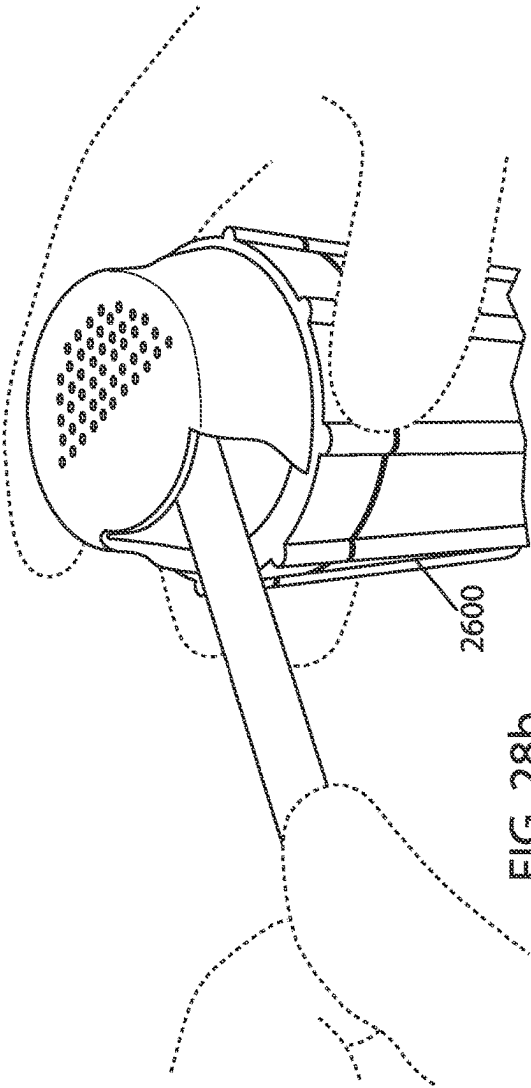
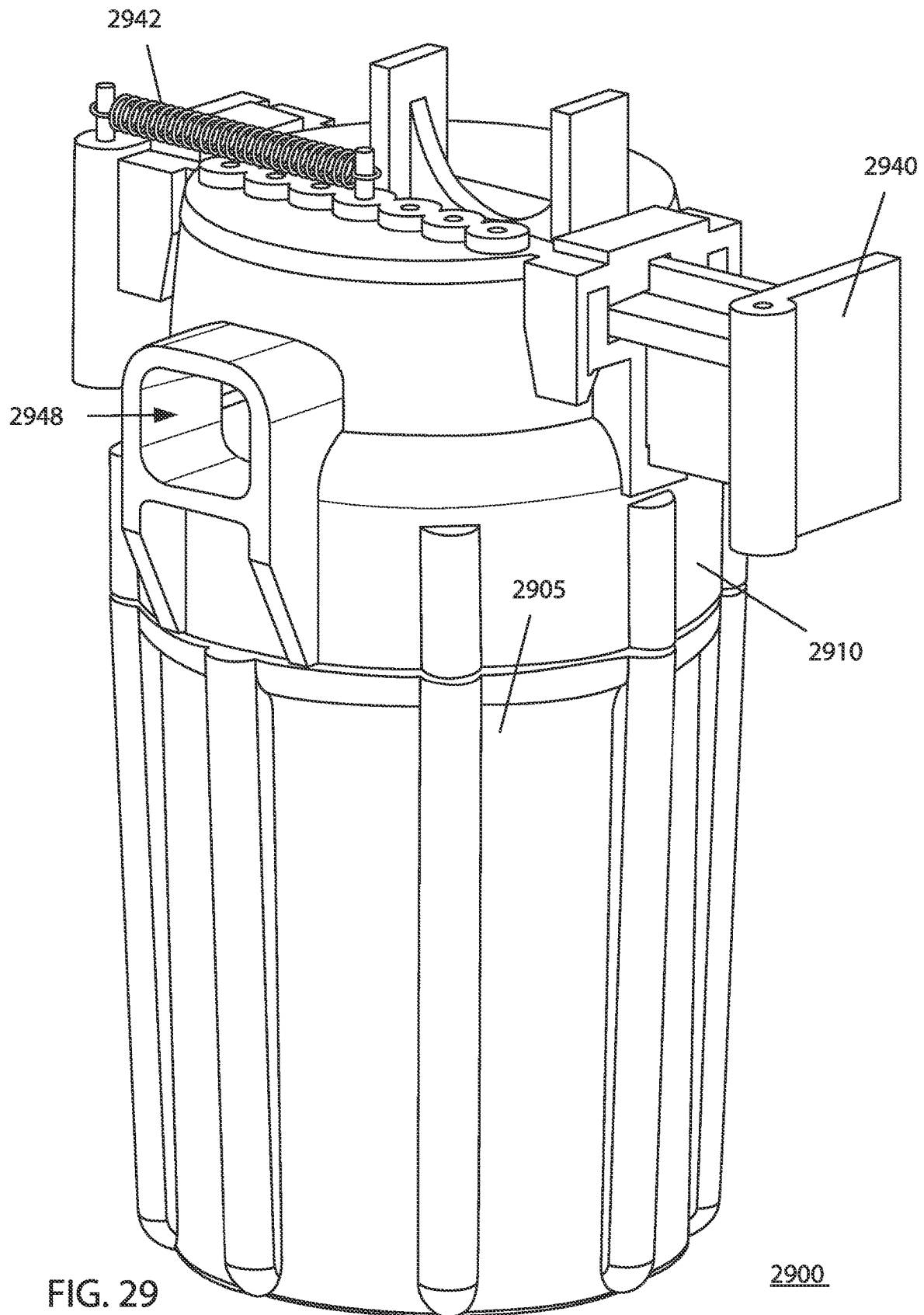
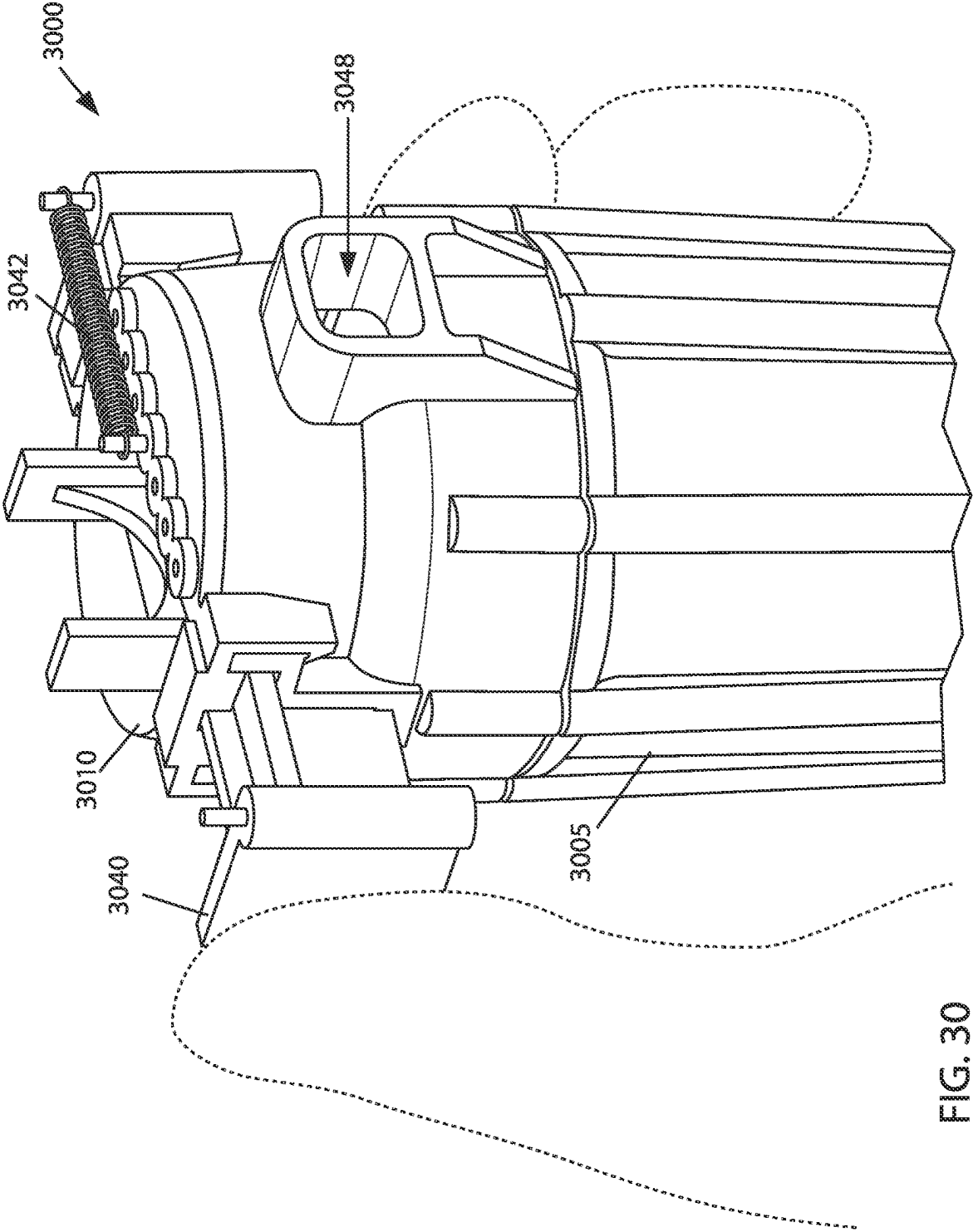
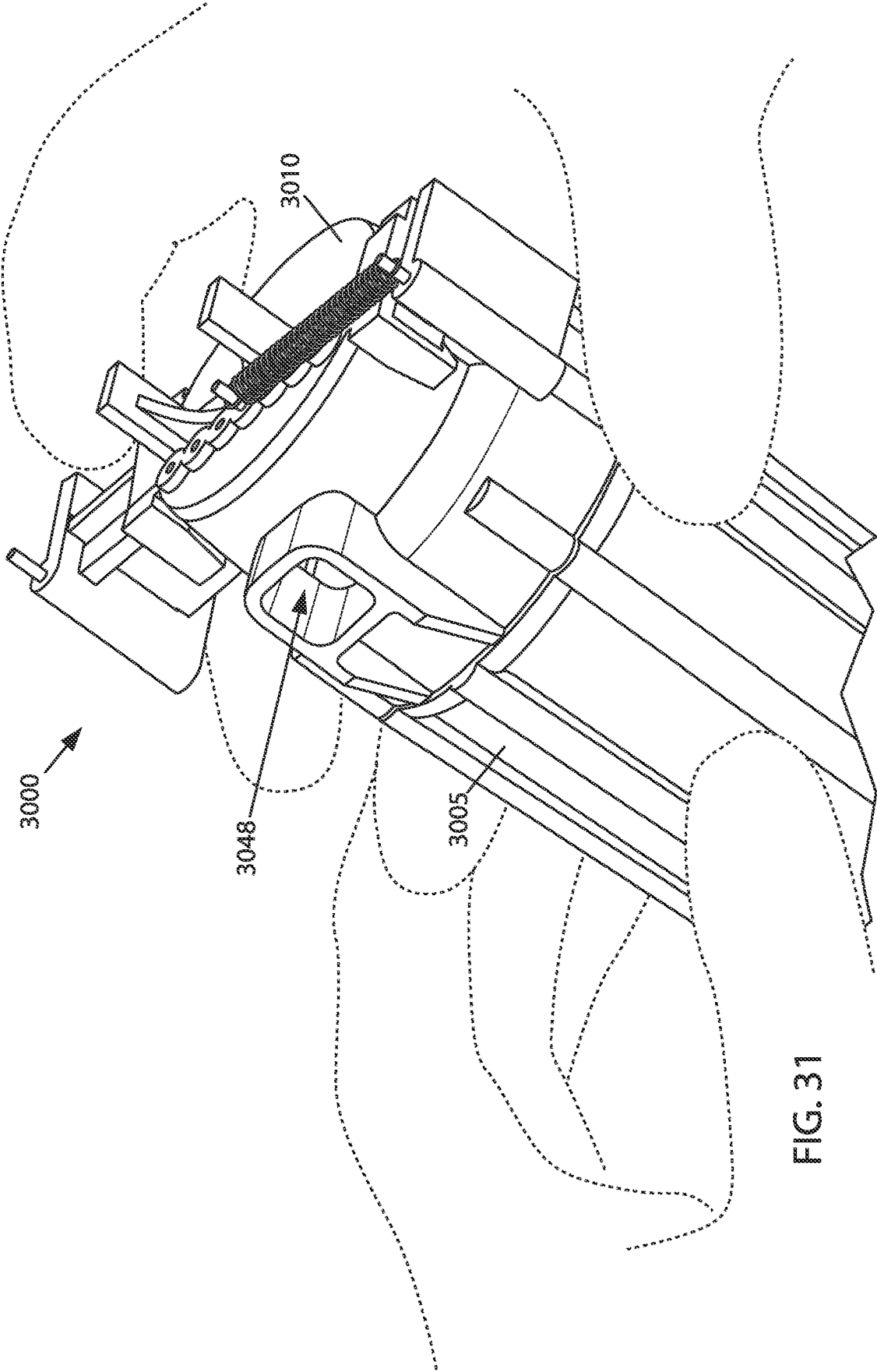


FIG. 28b







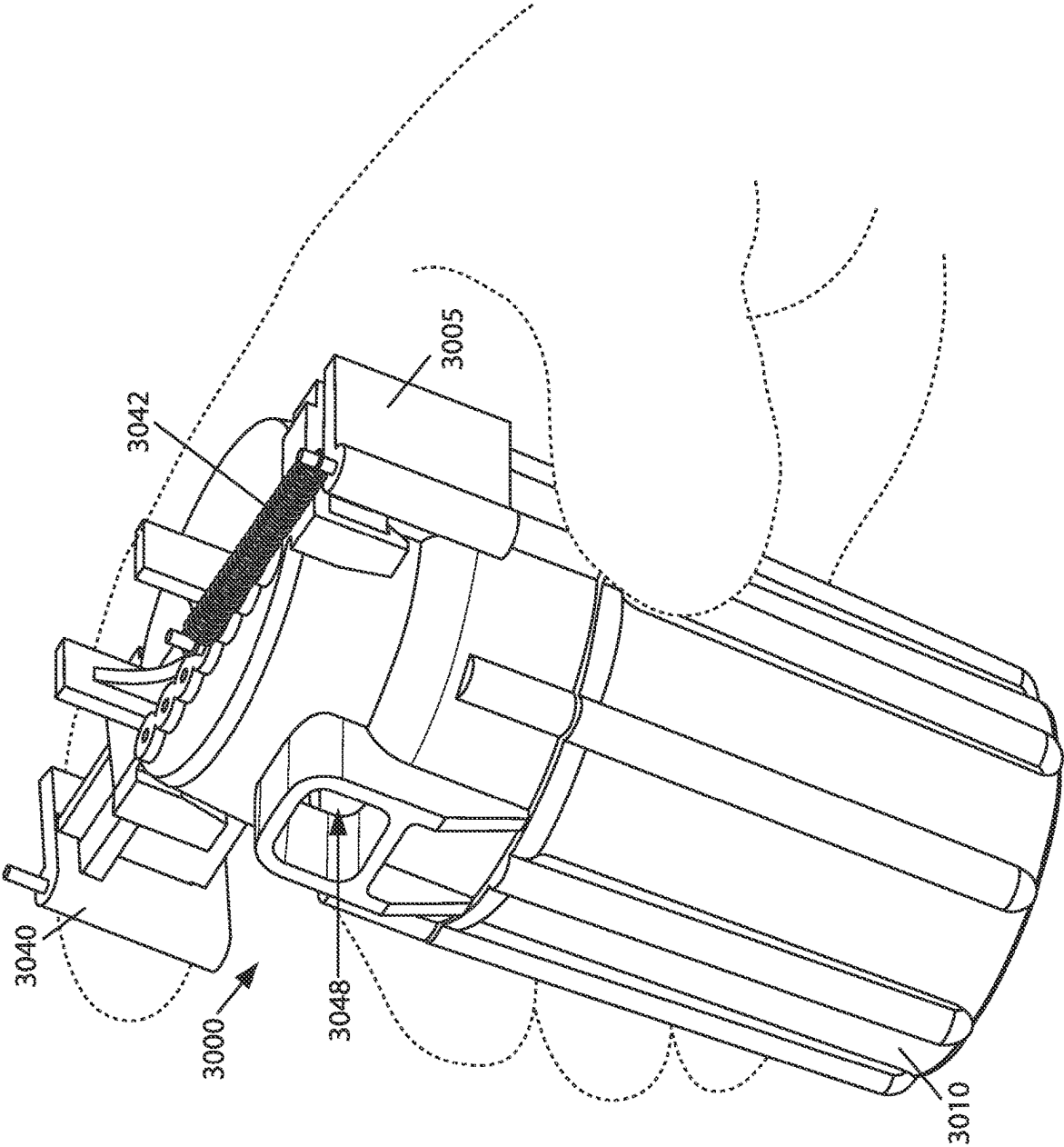
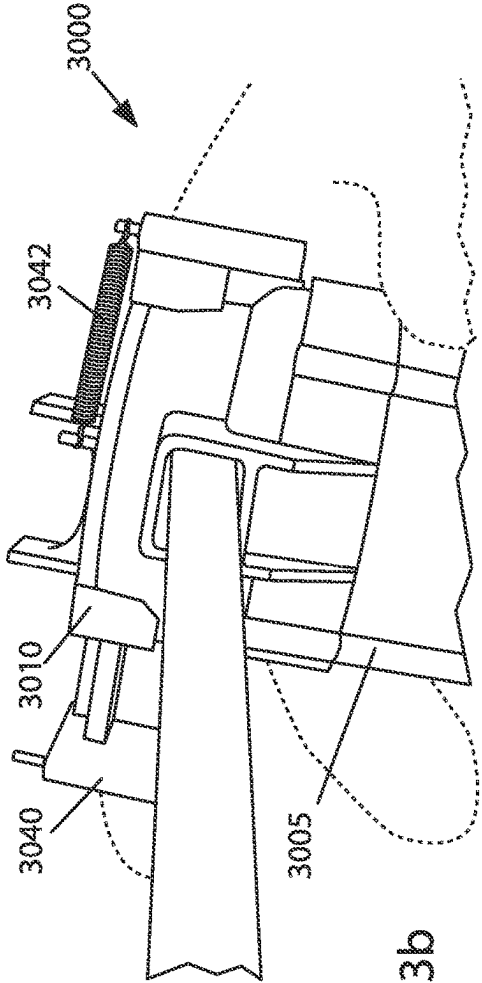
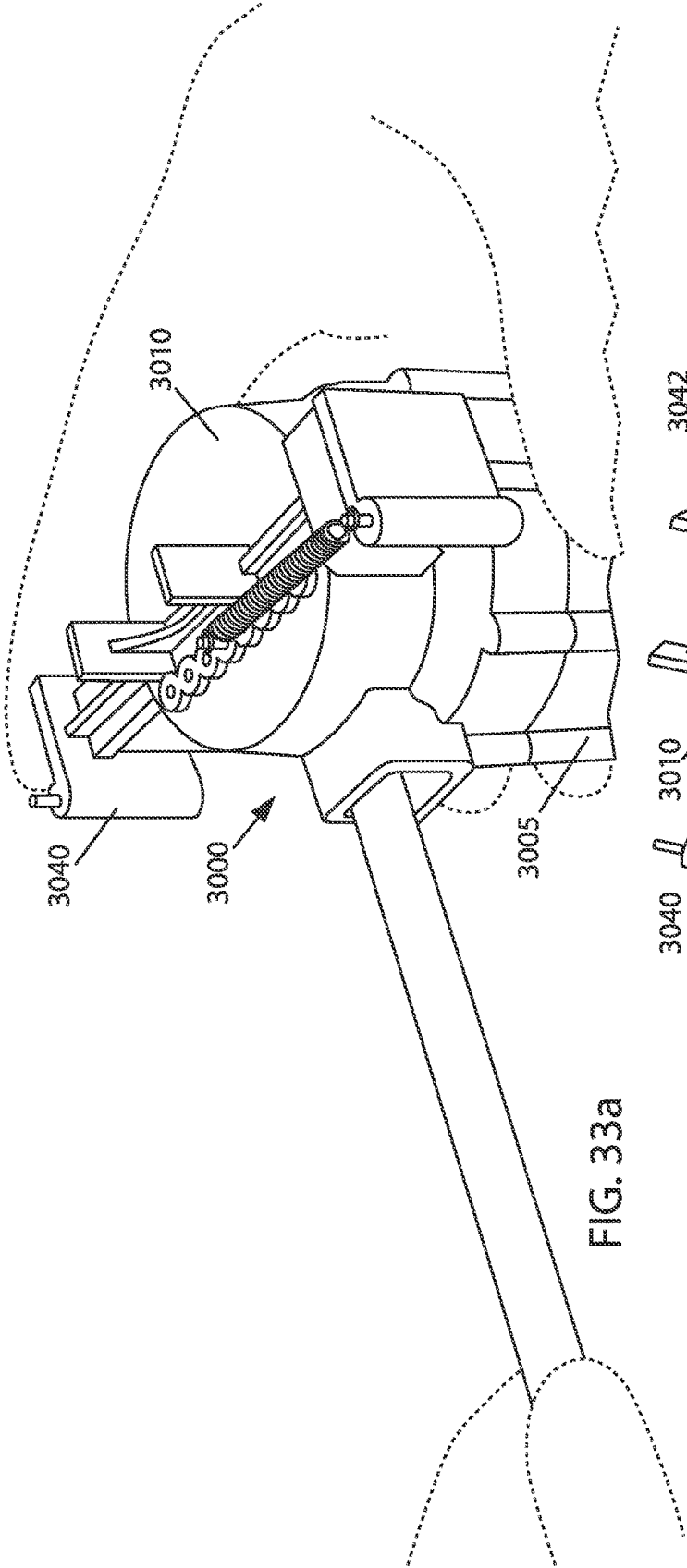
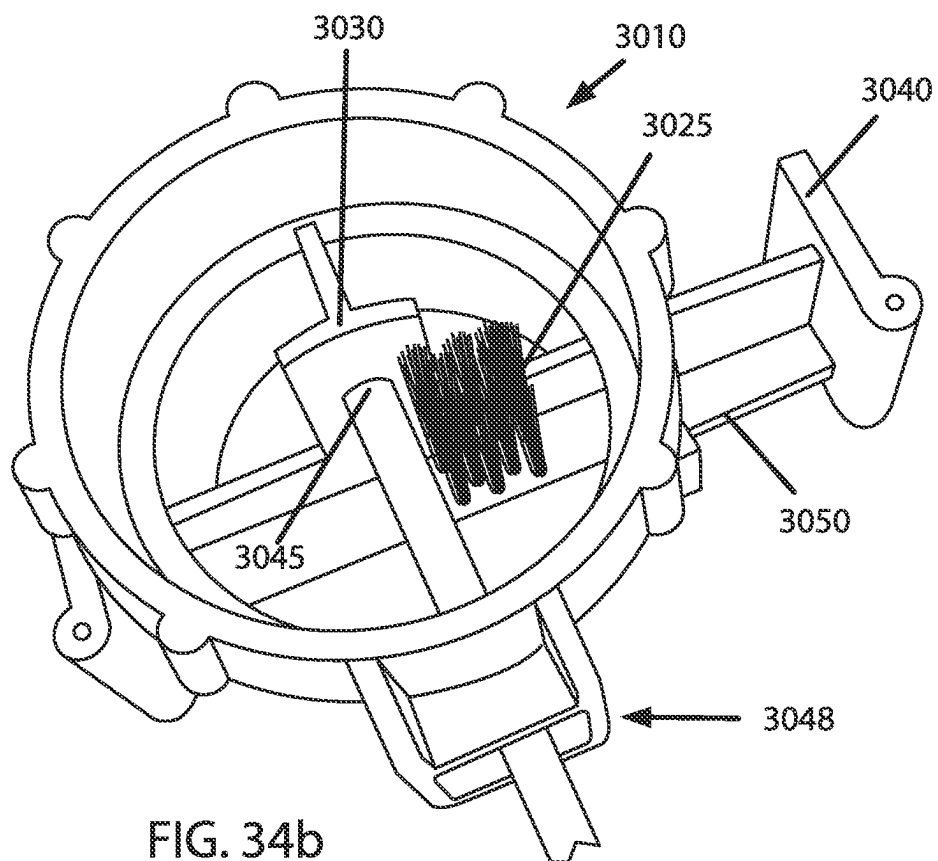
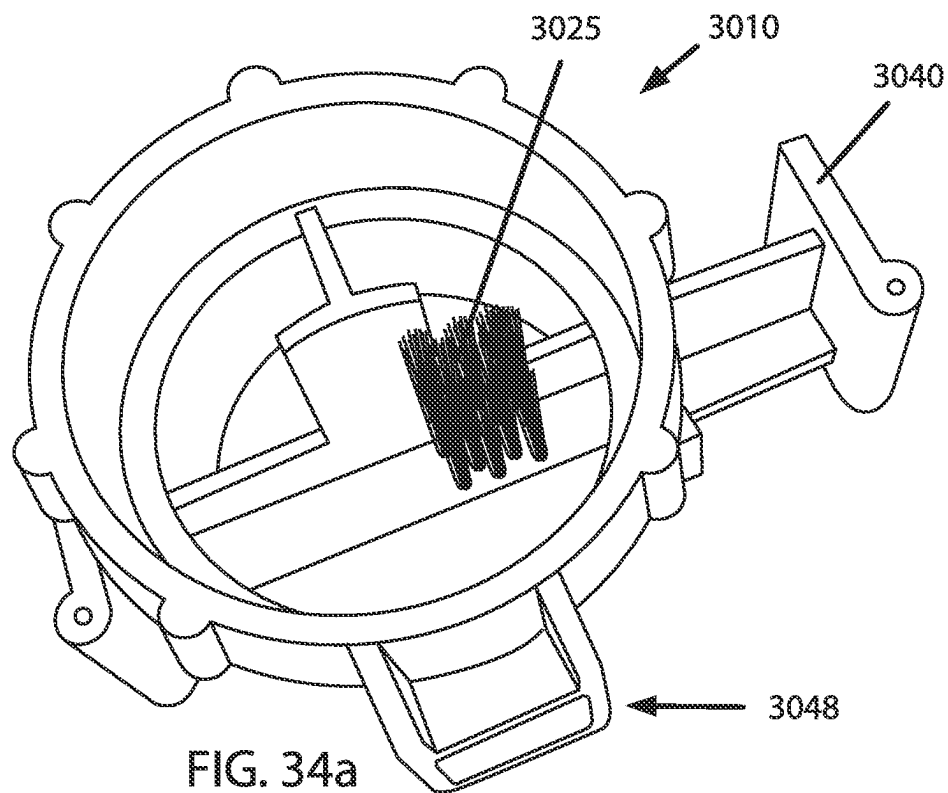


FIG. 32





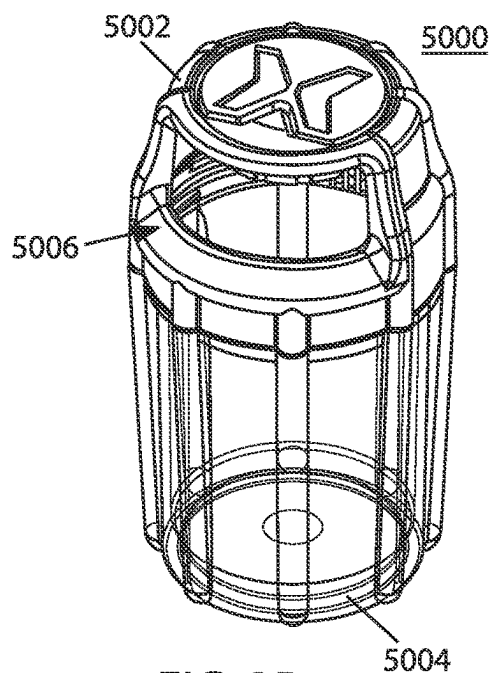


FIG. 35

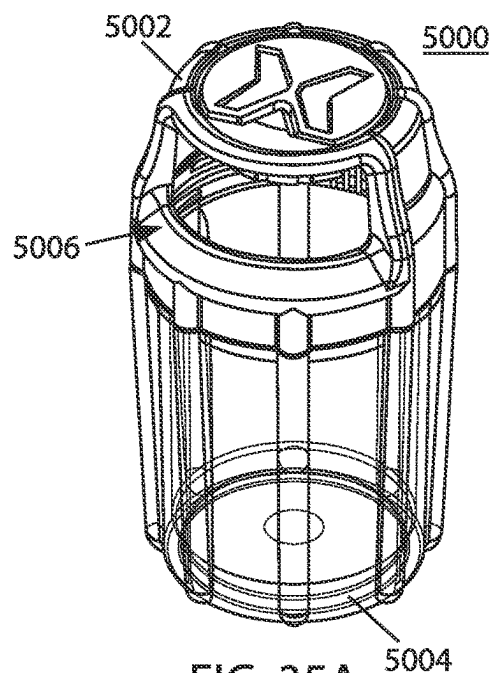


FIG. 35A

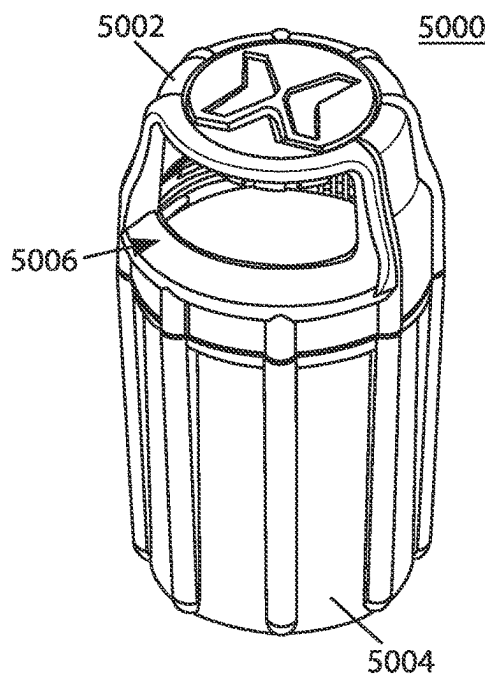


FIG. 35B

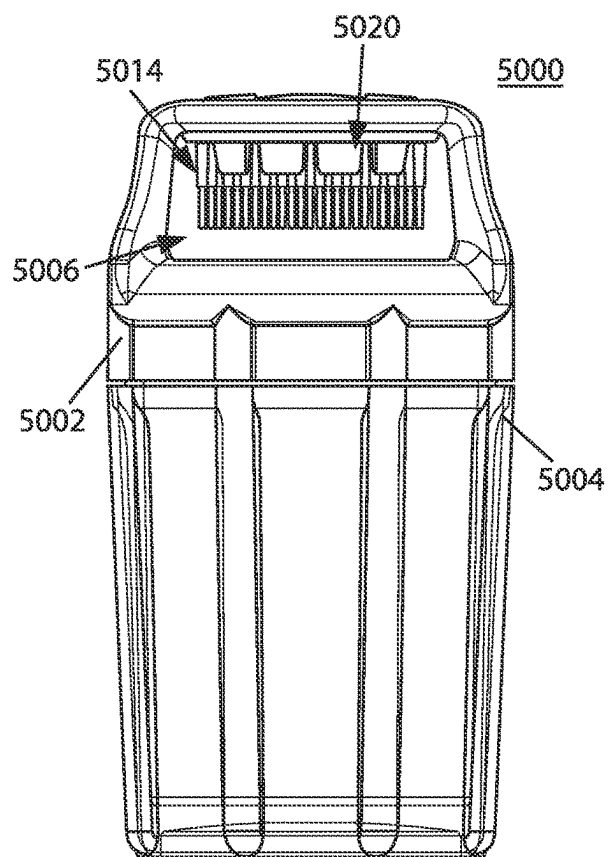


FIG. 36

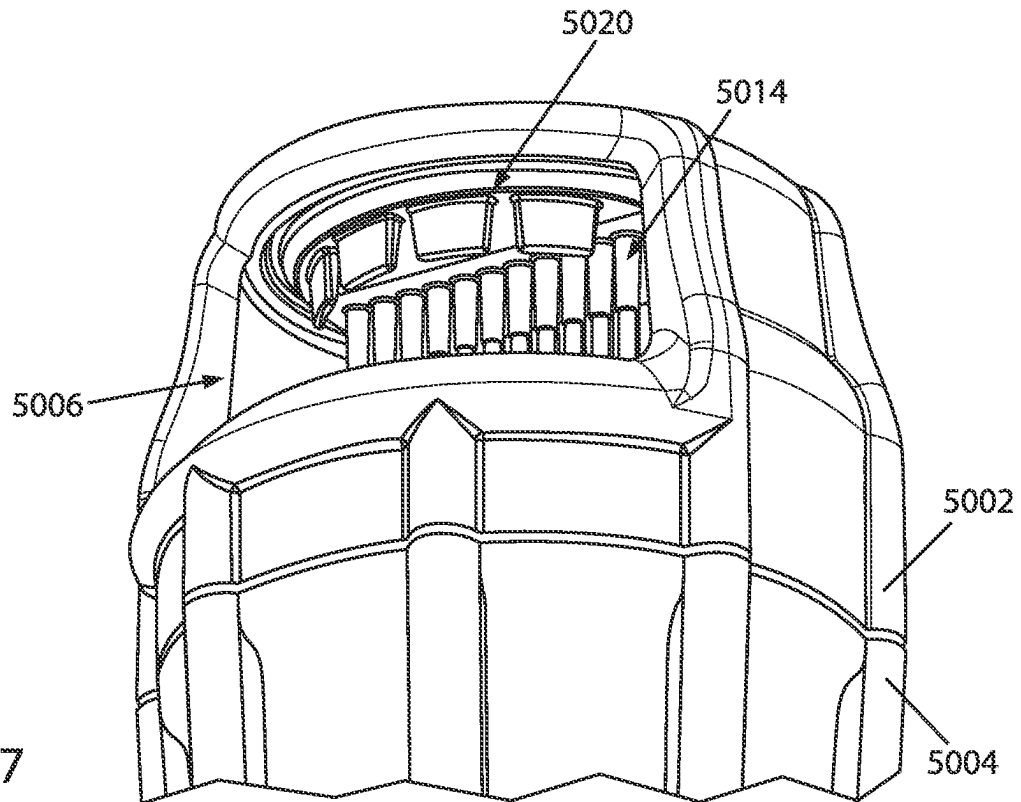


FIG. 37

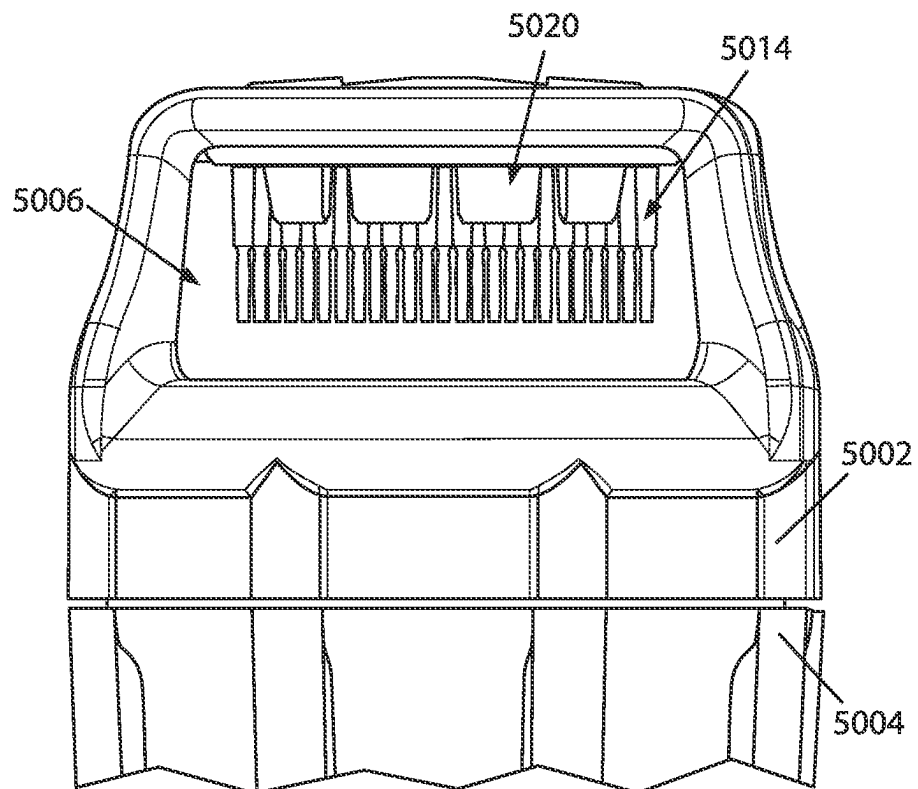


FIG. 38

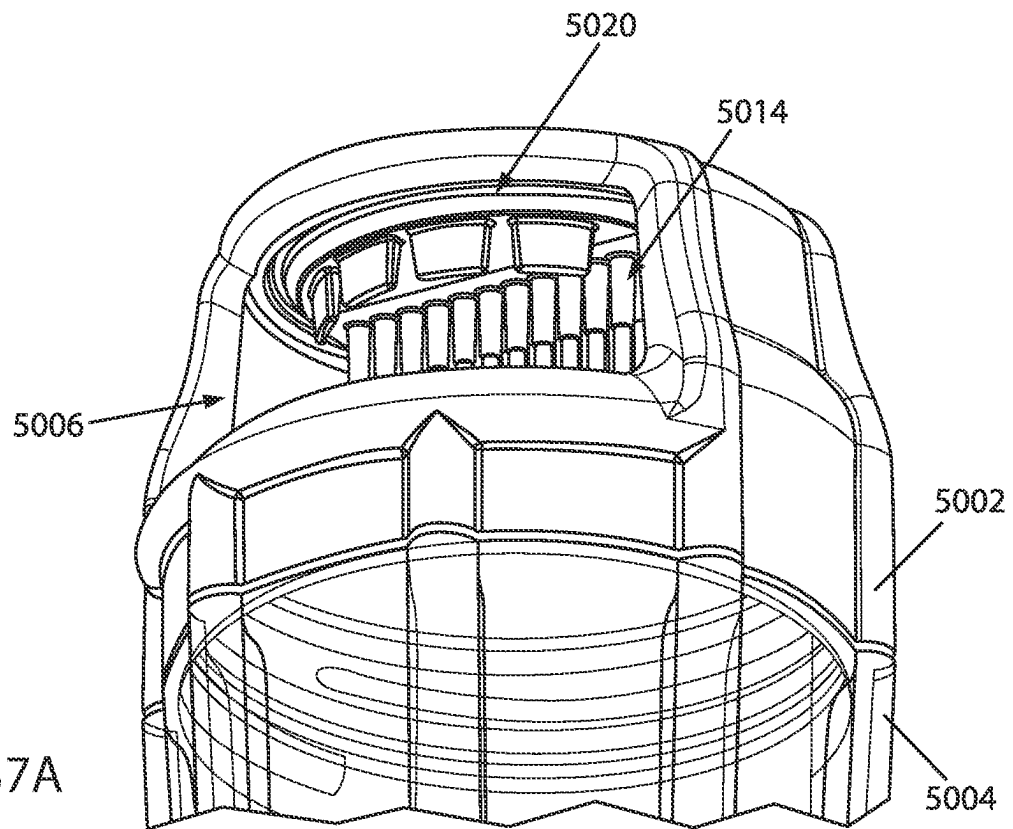


FIG. 37A

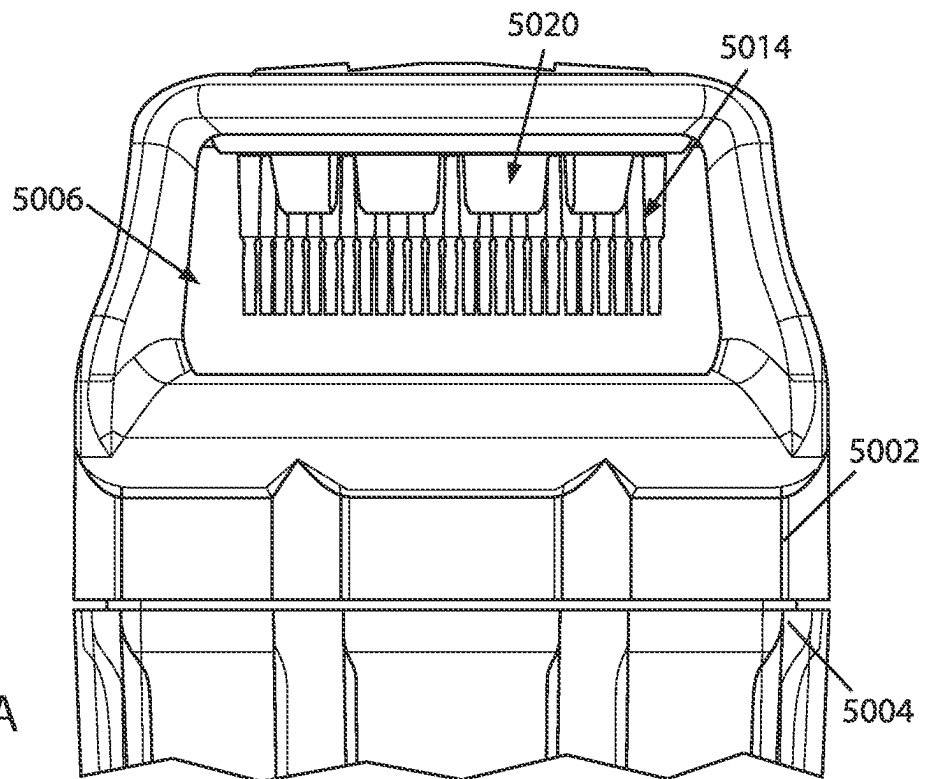


FIG. 38A

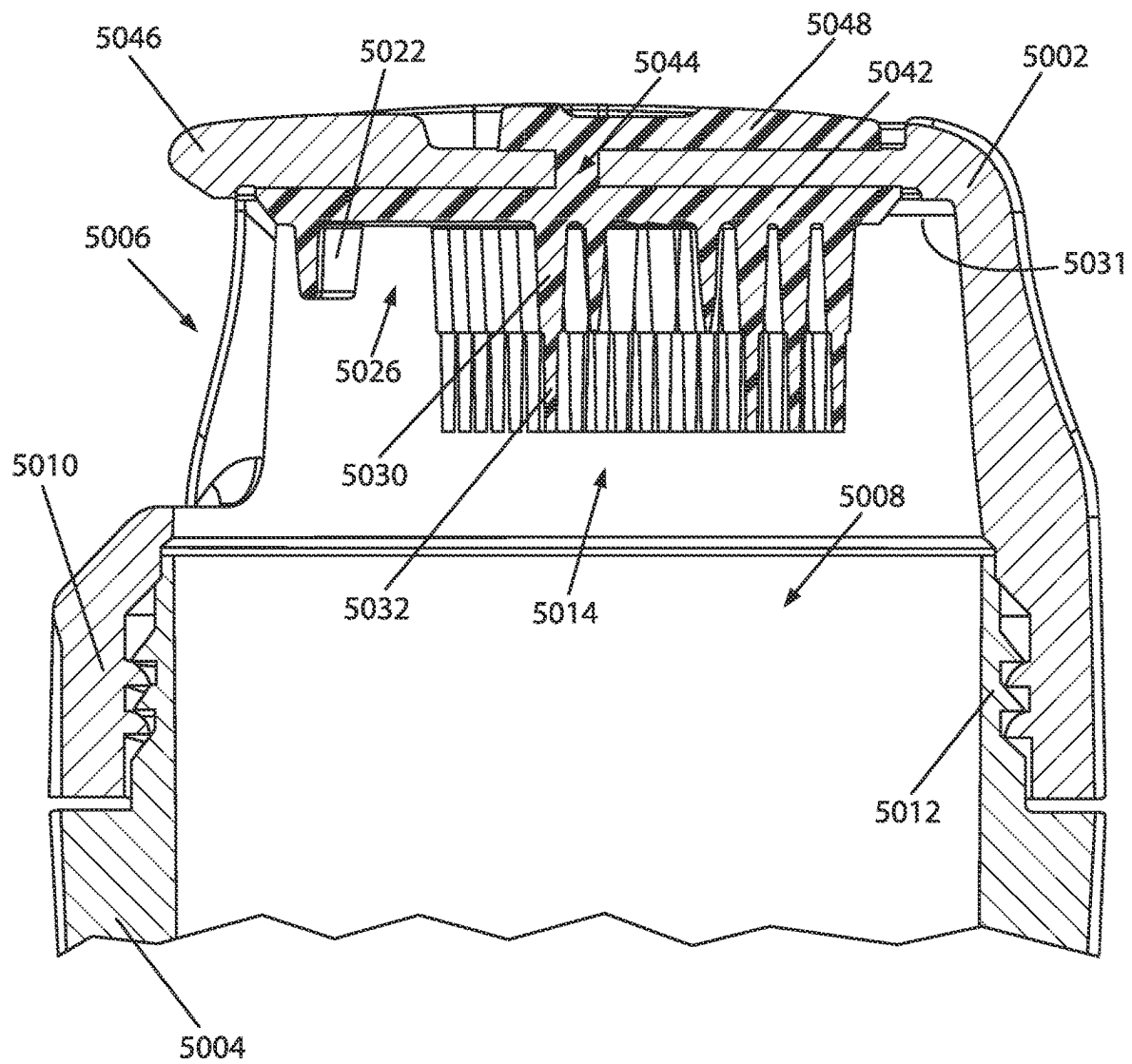
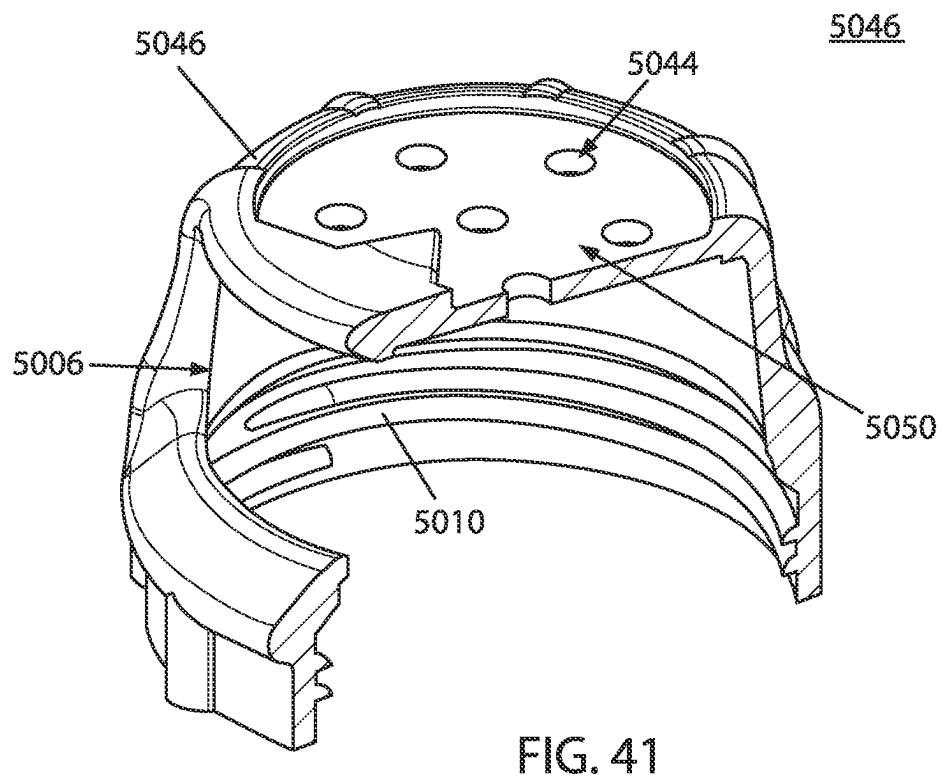
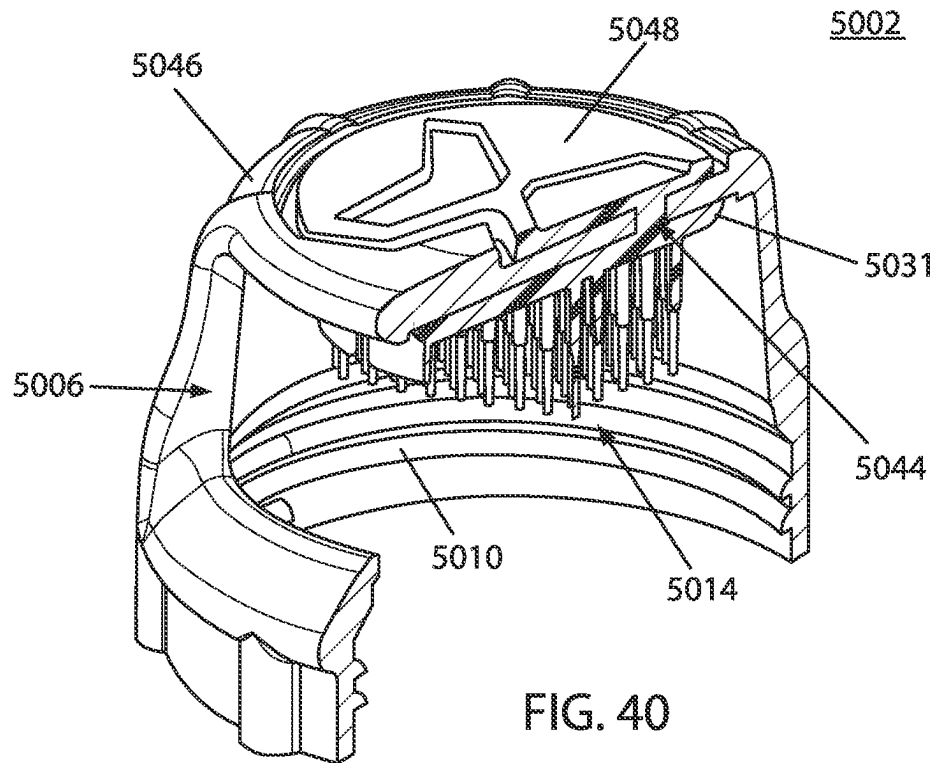
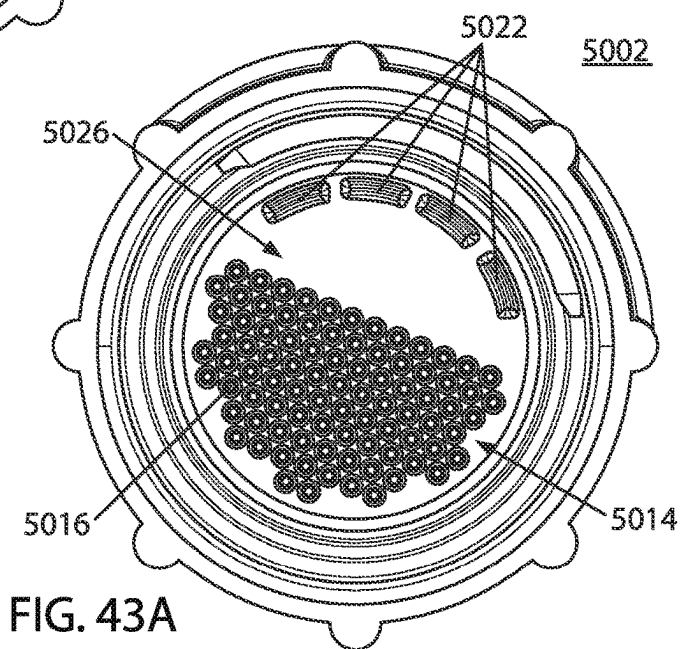
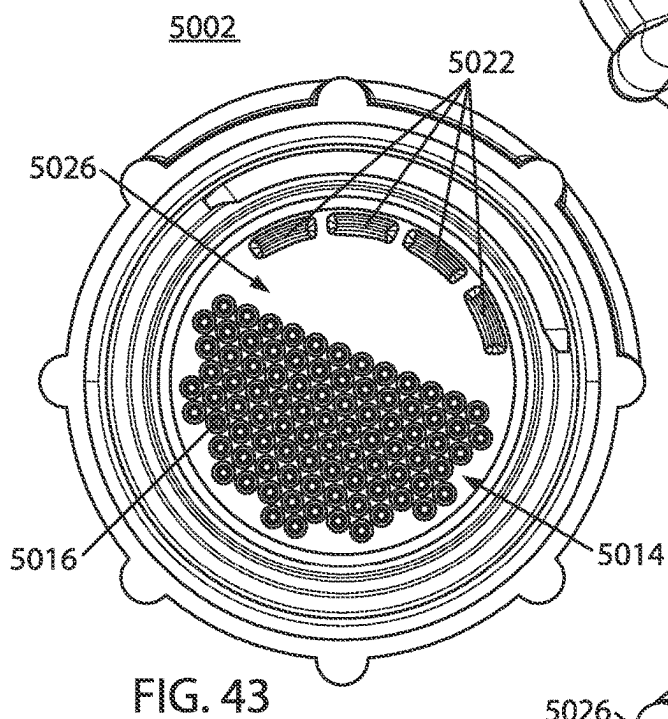
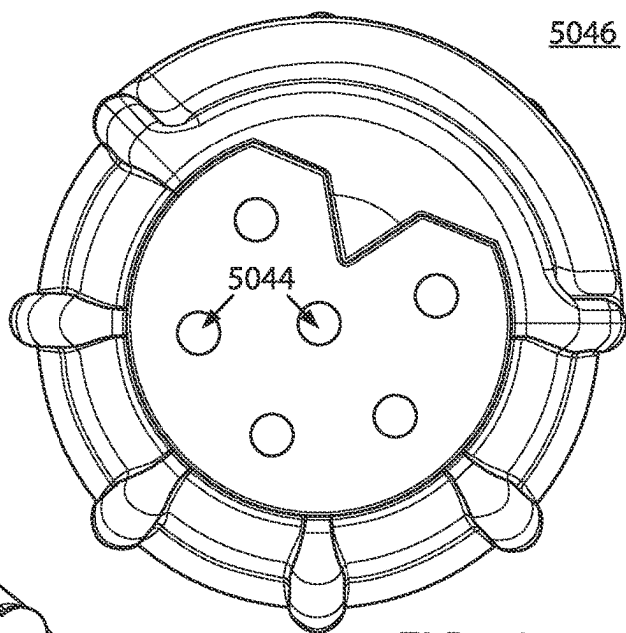


FIG. 39





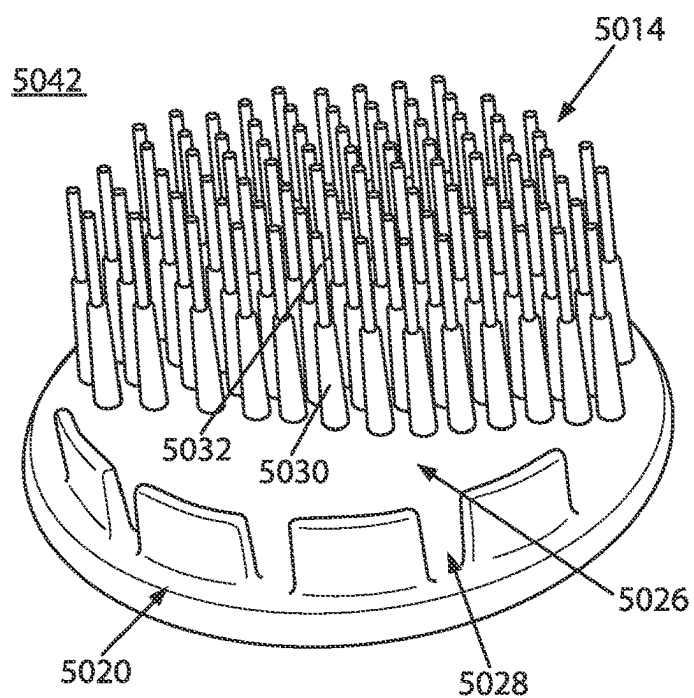
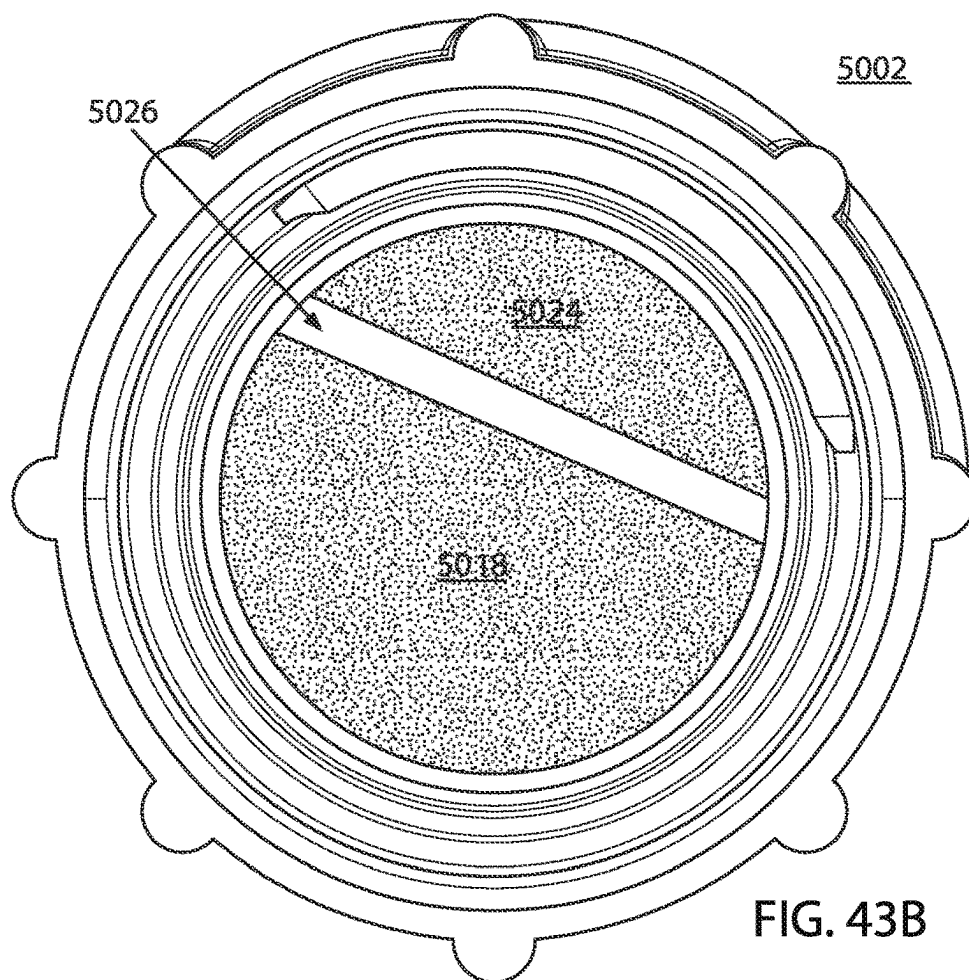


FIG. 44

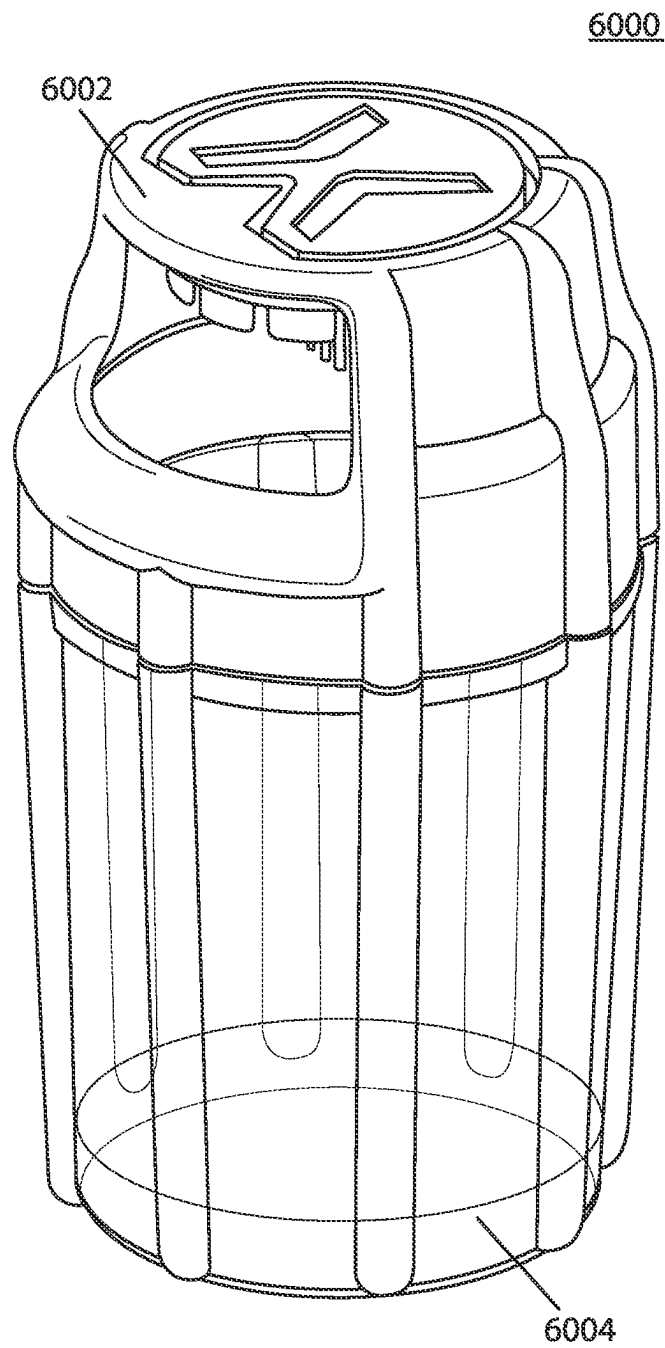


FIG. 45

6002

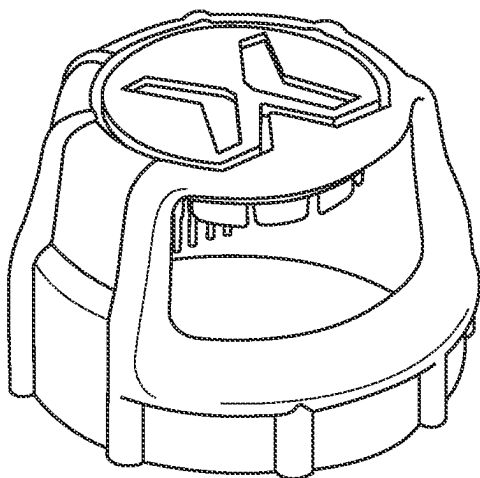


FIG. 46

6002

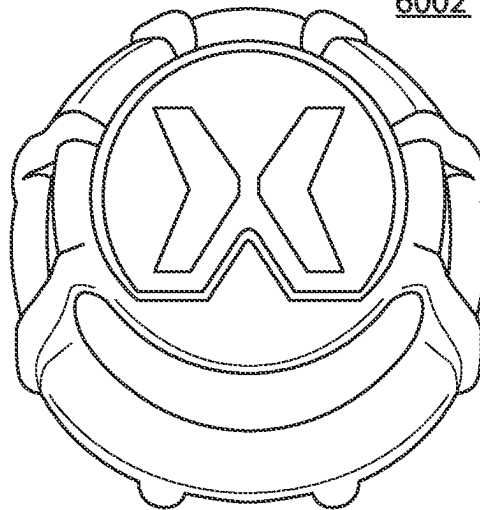


FIG. 47

6002

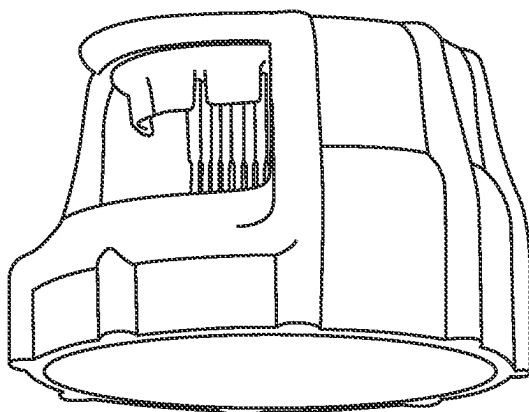


FIG. 48

6002

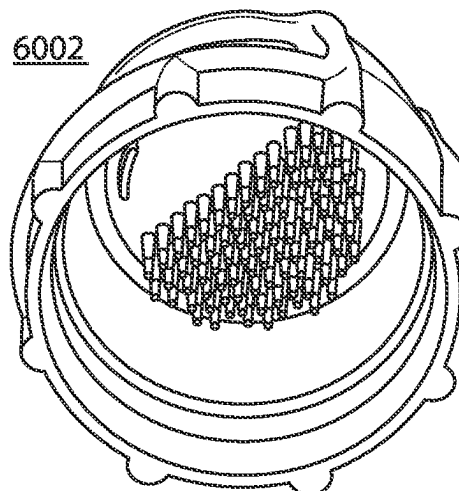


FIG. 49

6002

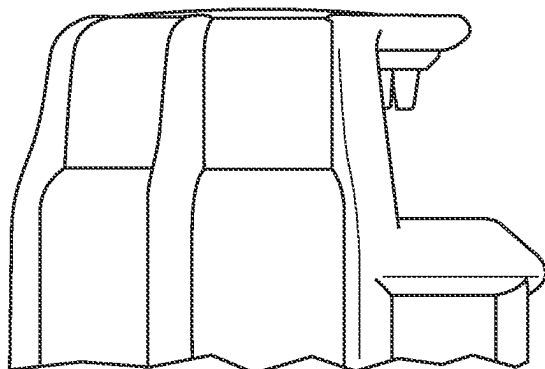


FIG. 50

6002

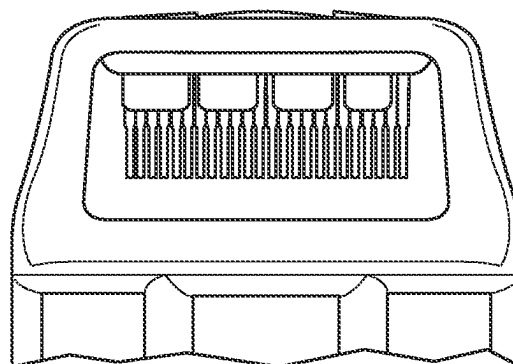
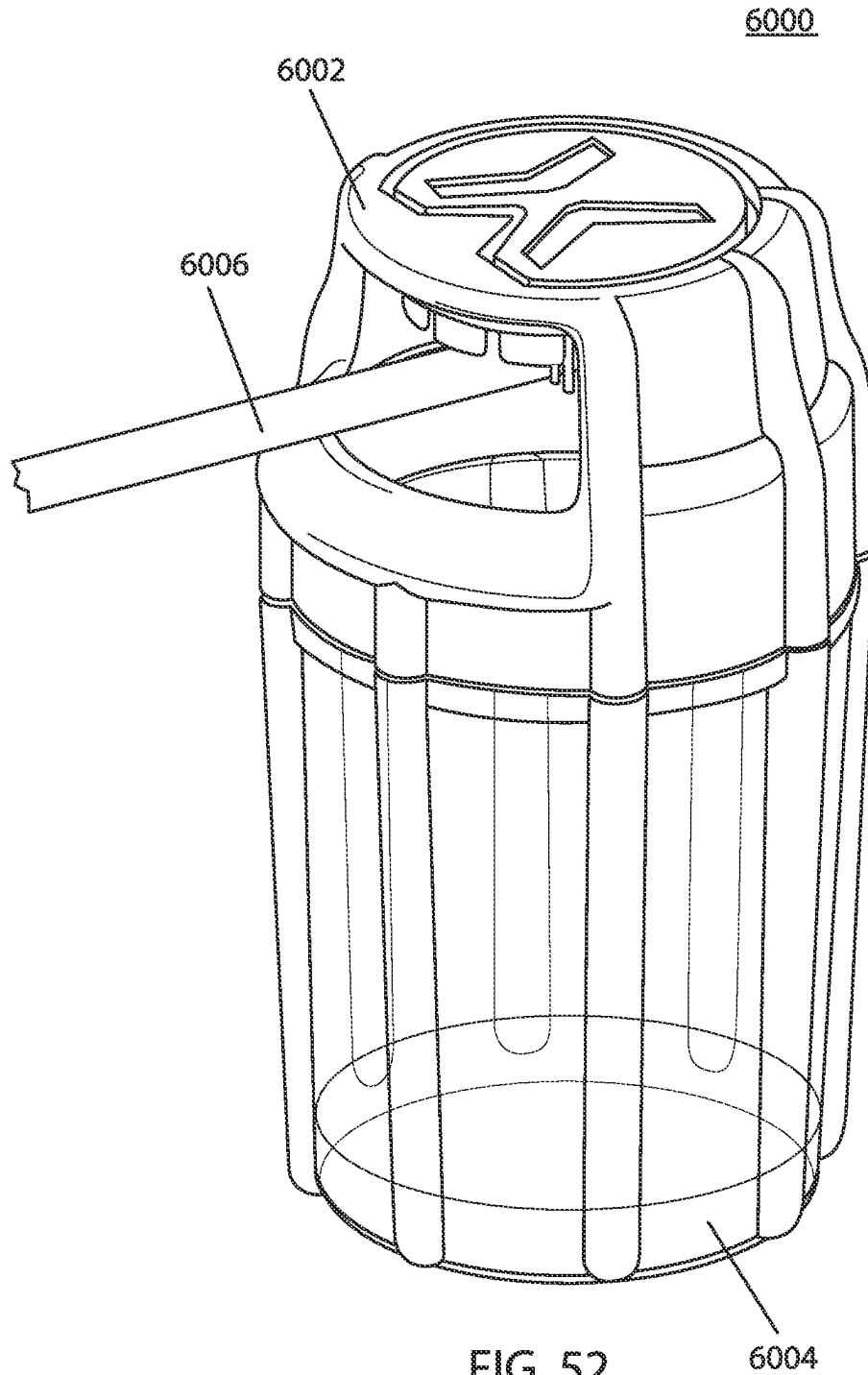


FIG. 51



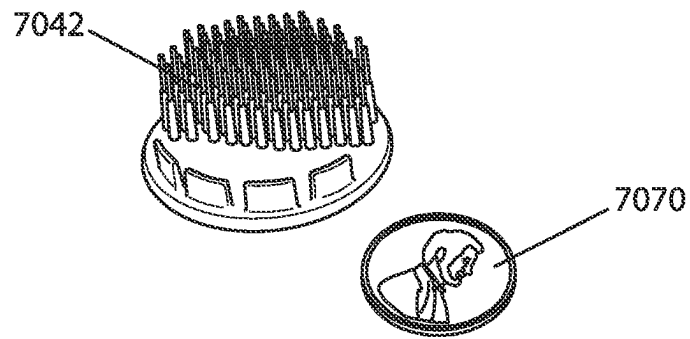


FIG. 53

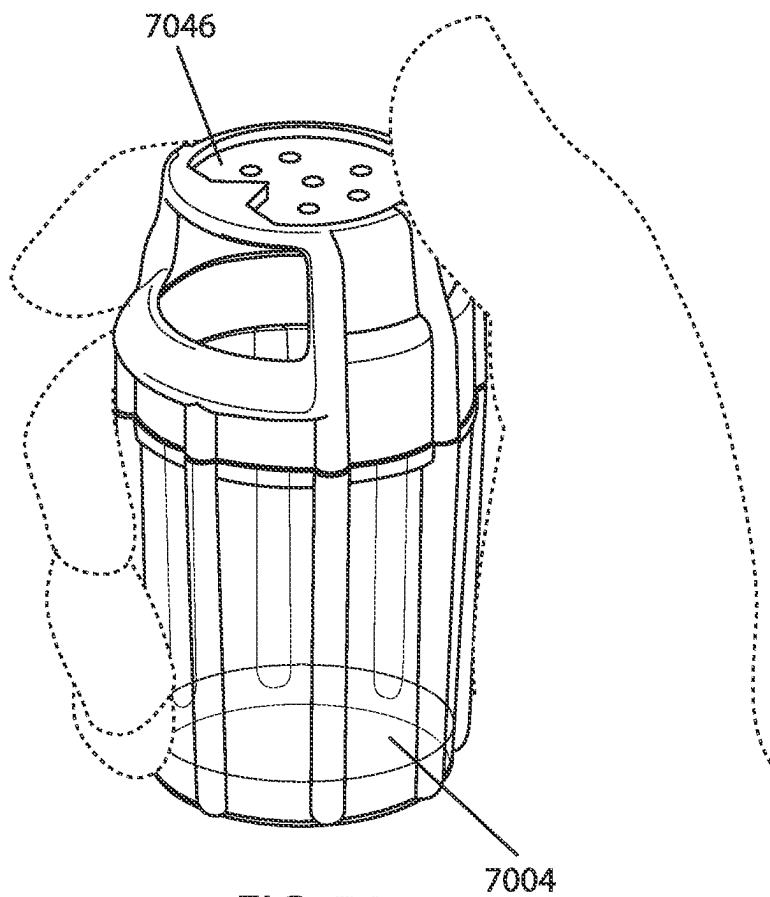


FIG. 54

COLLECTING AND HARVESTING CUT BONE FROM RONGEUR

CROSS-REFERENCE TO RELATED APPLICATION

Each of U.S. patent application publication no. 2015/0282816 A1 and WO 2015/154060 is incorporated by reference herein.

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Computer Program Listing

Submitted concurrently herewith via the USPTO's electronic filing system, and incorporated herein by reference, are one or more computer program files of the computer program listing of the present application. A table setting forth the name and size of each file included in the computer program listing is included below.

File Name	Creation Date	File Size (bytes)
ascify.txt	May 10, 2016 8:44	37473
drawing.txt	May 10, 2016 8:44	1343316
readme.txt	May 10, 2016 8:44	2593

One of these files, "readme.txt", contains instructions for extracting information from another of the files, "drawing.txt". This other file represents a compressed binary file that has been converted to ascii format. This file can be converted back to a compressed binary file utilizing assembly conversion program source code contained in the file "ascify.txt". The readme file includes instructions for compiling and running this conversion program source code, and instructions for converting the "drawing.txt" to the compressed binary file. The compressed binary file comprises a.zip archive including one or more eDrawings files representing one or more computer models that can be opened using the publicly available eDrawings software from SolidWorks.

BACKGROUND OF THE INVENTION

The present invention generally relates to the collection of cut material from a rongeur and, in particular, the collection of bone from a kerrison rongeur.

Rongeurs are surgical instruments for the cutting away of human tissue, and most commonly, cartilage and/or bone. "Kerrison" rongeurs are utilized in spinal surgery to remove bone and to thereby gain access to the spinal canal, and are well-known within conventional medical knowledge. Patent references disclosing and discussing kerrison rongeurs and their use in surgery include U.S. Pat. No. 3,902,498; 5,026,375; 4,722,338; 4,777,940; 4,777,948 and U.S. patent application publication 2003/0216740.

With reference to FIG. 1a, an exemplary prior art kerrison rongeur is illustrated. The kerrison rongeur includes a first jaw member 10 that slides parallel to line A relative to, and

on top of, a second jaw member 20. The first jaw member 10 included a distal cutting end having a cutting edge 15. The second jaw member 20 includes a stop 40 for placement beneath the tissue to be cut, which is generally bone or cartilage. A cutting area 30 is defined between the cutting edge 15 of the first jaw member 10 and the stop 40 of the second jaw member 20. The first jaw member 10 includes an open, interior cross-section defining an enclosed area within which the bone tissue is received when cut. Such an exemplary cross-section is illustrated in FIG. 1b. As illustrated, the cross-section of the first jaw member 10 has a generally inverted U shape.

In an exemplary use of a kerrison rongeur, a surgeon places the bone to be cut, such as the leading edge of the lamina of a vertebra, within the open portion of the distal end of the rongeur. The surgeon then squeezes the handle of the rongeur, which advances a moveable jaw member of the rongeur through and amputates a portion of bone. A surgeon may wait until the jaw member becomes full of bone, at which time the rongeur must be completely removed from the surgical site and passed to a scrub nurse or assisting technician for bone removal. The removal from the instrument of the cut portion of bone often requires that the scrub nurse or assisting technician use a small rigid hook, or toothed forceps, and often further requires that the physician temporarily relinquish the instrument entirely to make such bone removal possible. Once cleaned, the instrument is returned to the surgeon who, in returning it to the surgical site, must then reorient himself to the task at hand. This sequence must then be repeated over and over again. Moreover, in a typical spinal fusion, this sequence may require as many as 50 to 100 such repetitions.

Alternatively, a surgeon may present the distal end of the rongeur to a scrub nurse or assisting technician after each cut for removal of the bone, thereby avoiding bone build-up within the jaw member of the rongeur. This can be done without the surgeon relinquishing the rongeur and without the surgeon removing his or her attention and focus from the surgical site. Conventionally, the scrub nurse or assisting technician manually swipes the end of the kerrison rongeur with a sterile material for removing the cut bone. The scrub nurse or assisting technician may repeat the swiping of the cut bone multiple times before harvesting the swiped bone from the material used to perform the swiping. A gauze sponge may be used for swiping. Rather than swiping the bone from the rongeur, a gloved hand may be used by the scrub nurse or assisting technician to directly remove the bone from the rongeur. Thereafter, the harvested bone from the patient may be used in the surgical procedure on the patient, e.g., in autografts.

In view of the foregoing, it is believed that a need exists for a safe, expedient, and efficient way for the cut bone to be removed from the kerrison rongeur and harvested by a scrub nurse or assisting technician. This and other needs are believed to be addressed by one or more aspects and features of the invention.

SUMMARY OF THE INVENTION

The present invention includes many aspects and features. Moreover, while many aspects and features relate to, and are described in, the context of collection of bone from a kerrison rongeur, which is preferred, the present invention is not limited to only such use, as will become apparent from the following summaries and detailed descriptions of aspects, features, and one or more embodiments of the present invention.

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Accordingly, in one aspect of the present invention, a collector used to collect cut bone comprises: a container body defining an interior containment space for receiving and retaining collected bone, and having at least one open end for access and removal of collected bone from the interior containment space; and a cap in covering relation to the open end of the container body such that access to the interior containment space for removal of collected bone is inhibited. The collector comprises an intake port defining an opening for receiving therein a distal end of a kerrison-type rongeur for collecting cut bone from a cutting area thereof, and the cap comprises at least one scraper for engaging and dislodging cut bone from the cutting area of the distal end of the rongeur when received within the intake port.

In a feature of this aspect, the collector defines a passage through which cut bone dislodged by a scraper from the cutting area of the distal end of the rongeur is received within the interior containment space.

In additional features, the cap comprises a stop by which the extent to which the distal end of the rongeur received within the intake port is limited; and the stop is located at a spacing relative to the intake port such that, upon receipt of the distal end of the rongeur and abutment thereof with the stop, the one or more scrapers when actuated extend over and cover the distal end of the rongeur proximate the cutting area in which cut bone would be found.

In another feature, the container body comprises a stop by which the extent to which the distal end of the rongeur received within the intake port is limited.

In additional features, one or more of which may or may not be mutually exclusive: the collector further comprises a stop by which the extent to which the distal end of the rongeur received within the intake port is limited: the stop comprises a depressible button: the depressible button, when depressed, blocks the passage and inhibits receipt of dislodged bone within the interior containment space; and wherein the depressible button, when not depressed, does not inhibit receipt of dislodged bone within the interior containment space: the stop comprises a wall: the wall blocks the passage and inhibits receipt of dislodged bone within the interior containment space; and the wall includes an opening therein, the wall being movable between a first position in which the opening aligns with the passage and the wall does not inhibit receipt of dislodged bone within the interior containment space, dislodged bone passing through the opening in the wall, and a second position in which the opening is out of alignment with the passage and the wall inhibits receipt of dislodged bone within the interior containment space.

In additional features, one or more of which may or may not be mutually exclusive: the cap is detachable from the container body such that access to the interior space for removal of collected bone retained therein is uninhibited by the cap: the cap is attached to the container body in a frictional fit engagement: the cap is attached to the container body in a thread engagement; and the cap screws onto the container body.

In another feature, a portion of the cap is rotatable relative both to the container body and to another portion of the cap.

In additional features, one or more of which may or may not be mutually exclusive: the cap is movable relative to the container body detachable from the container body such that access to the interior space for removal of collected bone retained therein is uninhibited by the cap: a hinge connects the cap to the container body for movement relative thereto: the collector is disposable: one or more components of the collector are disposable; and one or more components of the

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collector are designed to be sterilized for reuse with different patients in different procedures.

In another feature, the collector is configured to collect cut bone from a kerrison rongeur.

In another feature, the collector is hand-held and lightweight.

In additional features, one or more of which may or may not be mutually exclusive: the container body comprises a generally elongate body or tube having opposite open-ends, wherein the cap comprises a first end cap covering a first of the opposite open-ends of the container body, and further comprising a second end cap covering a second of the opposite open-ends of the container body; and the first and second end caps are generally cylindrical in shape and are axially-aligned along a longitudinal axis of the collector.

In another feature, the one or more scrapers comprise a single scraper, the only one of which that is included in the collector.

In another feature, the one or more scrapers comprise more than one scraper.

In another feature, one of the one or more scrapers comprises bristles.

In another feature, one of the one or more scrapers comprises a brush.

In another feature, one of the one or more scrapers comprises a protuberance.

In another feature, one of the one or more scrapers comprises a barb.

In another feature, one of the one or more scrapers comprises a finger.

In another feature, one of the one or more scrapers is flexible and resilient, and sufficiently rigid so as to hold form when not engaged by the distal end of the rongeur and deflecting and bending upon abutment by and engagement with the distal end of the rongeur.

In another feature, one of the one or more scrapers is made from metal.

In another feature, one of the one or more scrapers is made from polypropylene.

In another feature, the one or more scrapers extend from an underside of the cap toward the interior containment space.

In another feature, the one or more scrapers not are located within the intake port.

In additional features, one or more of which may or may not be mutually exclusive: the one or more scrapers are located within the intake port; and the intake port is squeezable and springy such that the intake port can be squeezed by hand causing the one or more scrapers to enter to cutting area of a kerrison rongeur received within the intake port for dislodging bone carried therein.

In another feature, the container body comprises an opaque wall portion and a transparent wall portion.

In additional features, one or more of which may or may not be mutually exclusive: the container body comprises a movable panel: the panel is connected by a hinge for pivoting movement; and the panel is removable. In another feature, the intake port is located in the container body.

In another feature, the intake port is located in a side wall of the container body generally halfway in-between opposite ends of the container body.

In another feature, the intake port is located in the cap.

In another feature, the intake port comprises a large, wide opening in the side of the cap configured to receive the tip of a rongeur of various conventional sizes.

In additional features, one or more of which may or may not be mutually exclusive: the intake port is located on the

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top of the cap: the top of the cap defines a funnel-shaped surface with the intake opening at the center thereof: the top of the cap defines a flange for catching dislodged bone when the collector is turned sideways and the distal end of the rongeur is received within the intake port: the cap defines a cutout for visual alignment of the collector when receiving the distal end of the rongeur; and the cap comprises a protuberance for visual alignment of the collector when receiving the distal end of the rongeur.

In another feature, the intake port is located on a side of the cap.

In another feature, the intake port includes a hawk bill profile.

In additional features, the intake port defines a scraping tip; and the scraping tip is rigid.

In another feature, the intake port includes an angular profile.

In another feature, the intake port includes a rounded profile.

In additional features, one or more of which may or may not be mutually exclusive: the one or more scrapers are manually movable into a position for engaging and dislodging cut bone from the cutting area of the distal end of the rongeur when received within the intake port: the one or more scrapers comprise at least two opposed scrapers that are manually moved in directions toward one another so as to converge within the cutting area of the distal end of the rongeur when received within the intake port: opposite ends of the collector each includes a depressible portion by which, when depressed, the opposed scrapers are manually moved so as to converge within the cutting area of the distal end of the rongeur when received within the intake port: the depressible portion is spring-biased against depression: the opposed scrapers are rotatable about a longitudinal axis of the container body by manual rotation of the depressible portions of the end caps; and the one or more scrapers comprise a scraper that is manually moved within the cutting area of the distal end of the rongeur when received within the intake port by twisting of the cap on the container body.

In additional features, one or more of which may or may not be mutually exclusive: the cap further comprises a handle; and the handle extends generally downwardly proximate a side of the container body, a distal end of the handle being located closer to an end of the container body opposite the cap.

In additional features, one or more of which may or may not be mutually exclusive: the collector further comprises a suction port for suctioning dislodged bone into receipt within the interior containment space of the container body: the cap defines the suction port: the suction port is located on a side of the cap: the cap further comprises a handle: the suction port is located on the handle: the handle extends generally downwardly proximate a side of the container body, a distal end of the handle being located closer to an end of the container body opposite the cap; and the suction port is located at the distal end of the handle, with an interior passage extending through the handle from the suction port and opens into the interior containment space defined by container body.

In another feature, the collector is configured to be held in a sideways position when used to collect cut bone from a kerrison rongeur.

In yet another feature, the container body includes an interior containment space having a graduated containment volume of thirty cubic centimeters.

In another feature, the collector is configured to be held in a horizontal position when used to collect cut bone from a

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kerrison rongeur, the cap and the container body being horizontally oriented relative to one another.

In another feature, the collector is configured to be held in an upright position when used to collect cut bone from a kerrison rongeur.

In another feature, the collector is configured to be held in a vertical position when used to collect cut bone from a kerrison rongeur, the cap and the container body being vertically oriented relative to one another with the cap being at a vertical elevation greater than the container body.

In another feature, the container body is box-shaped with generally rectangular sides and wherein the cap comprises rectangular sides.

In another feature, the container body is transparent. Preferably the container body is made from an inert material conventionally used with medical containers for holding human tissue for use in a body. In at least some embodiments, the container body comprises a molded plastic body, and the cap comprises a molded plastic body.

In another aspect, a device for collecting autologous bone fragments comprises a container and a cap. The container comprises a visibly transparent material and may include graduations to indicate volume of collected tissue comprising bone in the container. The cap is attachable to the container by way of threads on both the container and the cap, whereby the cap screws onto the container. Preferably the cap is symmetrical with the exception of an opening that is wide with respect to the diameter of the cap. The opening is configured to allow the passage of the tip of a kerrison rongeur of multiple sizes along with bone and other tissue matter carried on the tip. The cap further comprises a brush disposed on the underside of the cap facing the interior of the container. The brush comprises a plurality of bristles which may comprise individual monofilament bodies. Moreover, the bristles preferably comprise a material that is absorbable by the human body: a bio-absorbable bristle is preferred in the event that a bristle becomes dislodged and mixed with the harvested matter from the tip of the kerrison rongeur, and thereafter is inadvertently inserted into the patient.

In use, the tip of the kerrison rongeur carrying the matter cut from the patient is inserted through the opening of the cap, and the brush is used to dislodge the matter from the kerrison tip whereby the matter falls into the interior space of the container and is thereby collected. Dislodging the matter may be effected by moving the kerrison tip upward away from the container and into the brush in the underside of the cap and/or translating the kerrison tip perpendicularly to the length of the kerrison body. Additional or alternative movements can be used such as, for example, rotating the brush relative to the tip of the kerrison rongeur for dislodging the cut matter from the tip. The kerrison rongeur thereafter is removed for further use. Later when the harvested matter is needed, the cap is unscrewed from the container and the matter, i.e., bone in preferred implementations, is retrieved from the container.

In another aspect of the invention, a collector used to collect cut bone comprises: a container body defining an interior containment space for receiving and retaining collected bone and having an open end for access and removal of collected bone from the interior containment space; and a cap in covering relation to the open end. The cap comprises an opening for receiving therein a distal end of a kerrison rongeur: comprises a first plurality of scrapers in the form of fingers for engaging and dislodging cut bone from a cutting area of the distal end of the rongeur when the distal end of the rongeur is received within the collector; and comprises a second plurality of scrapers in the form of wipers for

engaging and dislodging cut bone from the cutting area of the distal end of the rongeur when the distal end of the rongeur is withdrawn through the opening from the collector, the second plurality of scrapers being arranged so as to permit insertion of the distal end of the rongeur through the opening into the collector without engaging the distal end of the rongeur.

In a feature, the scrapers of the first and second pluralities extend downwardly from an underside of the cap.

In a feature, the opening is located on a first side of the cap, and wherein the first plurality of scrapers extends downwardly on an opposite side of the cap relative to the opening.

In a feature, each of the first plurality of scrapers includes a proximate portion relative to the underside of the cap and a distal portion relative to the underside of the cap, the proximate portion being less flexible than the distal portion in engaging and dislodging cut bone from the cutting area of the distal end of the rongeur.

In a feature, the first plurality of scrapers forms a bristle field.

In a feature, each of the first plurality of scrapers extend from an underside of the cap toward the interior containment space.

In a feature, the opening is located in the cap.

In a feature, the opening is located on a side of the cap.

In another aspect, a handheld collector used to collect cut bone from a kerrison rongeur comprises: a cap; and a container. The cap comprises an opening dimensioned to receive there through a distal end of a kerrison rongeur. The container of the collector comprises a generally elongate body that is cylindrical in shape, and walls of the body define an interior containment space of the container into which bone falls when dislodged from the distal end of a received kerrison rongeur. The cap includes at least two distinct areas of different pluralities of scrapers for dislodging bone from the distal end of a kerrison rongeur.

In a feature, the cap is generally circular at a lower perimeter thereof and wherein the opening extends along the perimeter an arc having an obtuse angle. The arc preferably has an angle of between 130 degrees and 140 degrees.

In a feature, the cap is attached to the container in an upper portion of the collector and is removable from the container.

In a feature, the cap comprises a threaded portion that engages and mates with a threaded portion of the container when the cap and container are screwed together.

In a feature, a gap extends between the different areas.

In a feature, a first plurality of scrapers similar to each other is located in a first area of the underside of the cap, which first area is in close proximity to the opening, and a second plurality of scrapers similar to each other is located in a second area of the underside of the cap, which second area is further from the opening than the first area. The first and second areas preferably are arranged in spaced relation to each other such that a gap comprising an absence of scrapers extends between the first plurality and the second plurality of scrapers. Furthermore, preferably each scraper of the first plurality comprises a finger insofar as each scraper comprises a protuberance that is elongate with generally oval cross-section: each such finger has a stepped diameter between a proximal portion thereof and a distal portion thereof relative to the underside of the cap; and each scraper of the first plurality comprises a larger width at a proximal portion thereof and a smaller width at a distal portion thereof. The width of each scraper of the first plurality may taper along the proximal portion, decreasing

as a height-wise extent increases in a direction away from the underside of the cap: the width of the distal portion may taper along the proximal portion, decreasing as a height-wise extent increases in a direction away from the underside of the cap: or both.

In a feature, the width of each scraper of the first plurality tapers along its overall height from the underside of the cap to its distal end. Alternatively, the width of each scraper of the first plurality does not taper along its overall height from the underside of the cap to its distal end.

In a feature, the first area of the first plurality of scrapers comprises a grouping of the scrapers that collectively form a bristle field.

In a feature, each scraper of the second plurality comprises a scraper is seen to comprise a wiper having a length and a height that substantially exceeds a width thereof.

In a feature, the second area of the second plurality of scrapers comprises a grouping thereof collectively forming an arrangement of teeth.

In a feature, the second area of the second plurality of scrapers comprises a grouping thereof collectively forming a row of teeth, with each tooth being a wiper and with a gap extending between adjacent teeth. The row of teeth preferably is arranged along an arc generally extending in close relation to a circumferential boundary of the underside of the cap, and the second area of the second plurality of scrapers preferably comprises a grouping thereof collectively forming a single row of teeth. In other variations, the second area of the second plurality of scrapers comprises a grouping thereof collectively forming multiple rows of teeth.

In a feature, each scraper of the first plurality is more than twice the height of each scraper of the second plurality in extending away from the underside of the cap. Preferably the height of the lower portion of each scraper of the first plurality is greater than the overall height of each scraper of the second plurality.

In a feature, each scraper of the first and second pluralities is sufficiently rigid so as to generally hold form when not engaged by a distal end of a kerrison rongeur and to generally dislodge bone found in a cutting area of a kerrison rongeur when the scraper is moved into or through the cutting area.

In a feature, each scraper of the first and second pluralities is sufficiently flexible and resilient so as to generally deflect and bend—to various extents—upon abutment by and engagement with the distal end of a kerrison rongeur.

In a feature, each scraper of the second plurality is less flexible and resilient when compared to a proximal portion of each scraper of the first plurality.

In a feature, each scraper of the second plurality is less flexible and resilient when compared to a distal portion of each scraper of the first plurality.

In a feature, a proximal portion of each scraper of the first plurality is less flexible and resilient when compared to a distal portion of each scraper of the first plurality.

In a feature, each scraper of the second plurality, and the proximal portion and the distal portion of each scraper of the first plurality, together are all formed from the same material by molding.

In a feature, each scraper of the second plurality, and a proximal portion and a distal portion of each scraper of the first plurality, together are all formed from the same material by injection molding.

In a feature, the scrapers of the first plurality and the second plurality are integrally formed as a single component of the cap. Preferably the first component extends through a plurality of channels formed in a second component of the

cap which channels extend between an underside of the cap and a topside of the cap: the first component comprises a topside portion located within a recess formed in a top surface on the topside of the second component; and the topside portion comprises an ornamentation including branding or a decorative element.

In another aspect, a method of manufacturing a cap for a collector—the cap comprising pluralities of scrapers that differ based on flexibility and resiliency as well as arrangement and spacing between adjacent scrapers, comprises the steps of: providing a base component; and injection molding an additional component onto the base component to form the cap, the additional component being molded in a shape defining a first area having a first plurality of scrapers defining a bristle field and a second area having a second plurality of scrapers defining a row of teeth, with a gap comprising an absence of scrapers extending between the first and second areas.

In a feature, the method of providing the base component comprises first molding the base component.

In a feature, the first component is overmolded on the base component.

In a feature, one or more of the base and additional components are molded from one or more inert plastic materials.

In a feature, the material from which the base component is molded comprises a bio-absorbable material.

In a feature, each of the first plurality of scrapers protrude at least twice the extent to which each of the second plurality of scrapers protrude.

In a feature, portions of the additional component extend through channels in the base component of the cap, the channels extending between a topside and an underside of the cap. Preferably first and second portions of the additional component respectively extend on opposite sides of the base component, the first portion of the additional component forming the first and second pluralities of scrapers. The scrapers preferably are permanently affixed to the base component on the underside of the cap and are not removable from the cap without tearing of the additional component.

In another aspect, a collector comprises a container and a cap, wherein the cap comprises pluralities of scrapers that differ based on flexibility and resiliency as well as arrangement and spacing between adjacent scrapers.

In a feature, a first plurality of scrapers each is shaped and configured in an arrangement to resemble a bristle field like that of a brush; and wherein a second plurality of scrapers each is shaped and configured in an arrangement to resemble a row of teeth, with the row of teeth extending along an opening in the cap, with the bristle field located distal to the opening, and with a gap extending between the row of teeth and the bristle field.

In another aspect, a method of collecting cut bone from a kerrison rongeur using a collector—the collector comprising a container having an interior containment space for catching bone and a cap attached thereto in covering relation over the interior containment space, comprises the steps of: inserting a distal end of a kerrison rongeur through an opening in the cap of the collector: causing the distal end of the kerrison rongeur to engage and be moved and rotated in engagement with scrapers of a first plurality of scrapers, each of the first plurality extending from an underside of the cap for dislodging bone from the kerrison rongeur; and withdrawing the distal end of the kerrison rongeur from the cap while engaging the distal end of the kerrison rongeur with scrapers of a second plurality of scrapers, each of the

second plurality extending from an underside of the cap proximate a perimeter of the opening for dislodging any remaining bone from the kerrison rongeur, a gap comprising an absence of scrapers extending between the first and second pluralities of scrapers.

In a feature, the distal end of the rongeur is inserted through the opening in the cap without engaging any of the second plurality of scrapers.

In a feature, the distal end of the rongeur is inserted through the opening in the cap at an angle of between about 30 degrees and about 40 degrees.

In a feature, the first plurality comprises fingers.

In a feature, the first plurality of scrapers forms a bristle field.

In a feature, each of the second plurality of scrapers is in the form of a tooth.

In a feature, the second plurality of scrapers are arranged so as to form a row of teeth extending from the underside of the cap proximate the opening in the cap.

In a feature, each of the first plurality of scrapers comprises a bio-absorbable material.

In a feature, each of the second plurality of scrapers comprises a bio-absorbable material.

In another aspect, a kit comprises an aforementioned collector and a rongeur, wherein the collector is configured to collect cut bone from the rongeur of the kit.

Another aspect comprises a method of using an aforementioned collector to collect cut bone.

Another aspect comprises a method of using an aforementioned collector to collect cut bone from a rongeur.

Another aspect comprises a method of using an aforementioned collector to collect cut bone from a kerrison rongeur.

In addition to the aforementioned aspects and features of the present invention, it should be noted that the present invention further encompasses the various possible combinations and subcombinations of such aspects and features. Thus, for example, any aspect may be combined with an aforementioned feature in accordance with the present invention without requiring any other aspect or feature.

BRIEF DESCRIPTION OF THE DRAWINGS

One or more preferred embodiments of the present invention now will be described in detail with reference to the accompanying drawings.

FIG. 1a is a side elevational view of an exemplary prior art kerrison rongeur.

FIG. 1b illustrates a cross-sectional shape of the first jaw member 10 of the kerrison rongeur of FIG. 1a.

FIG. 2a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with a first embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 2b is a perspective view illustrating a sequence of steps for using the collector of FIG. 2a in collecting bone from a kerrison rongeur, wherein the collector is illustrated in a generally horizontal orientation.

FIG. 2c is a perspective view of a portion of the collector of FIG. 2a after bone has been collected from a kerrison rongeur.

FIG. 3a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with a second embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 3b is a perspective view illustrating a sequence of steps for using the collector of FIG. 3a in collecting bone

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from a kerrison rongeur, wherein the collector is illustrated in a generally horizontal orientation.

FIG. 3c is a top plan view of a cap of the collector of FIG. 3a.

FIG. 3d is a side elevational view of the cap of the collector of FIG. 3a.

FIG. 4a is a perspective view representative of a sequence of steps for using another collector in accordance with a third embodiment of the present invention, wherein the collector is illustrated in a generally horizontal orientation receiving the distal end of a kerrison rongeur.

FIG. 4b is a top plan view of the collector and distal tip of the kerrison rongeur of FIG. 4a.

FIG. 5a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 5b is a perspective view illustrating a sequence of steps for using the collector of FIG. 5a in collecting bone from a kerrison rongeur, wherein the collector is illustrated in a generally horizontal orientation with the distal end of the kerrison rongeur being received within the collector.

FIG. 5c is a side elevational view of an alternative embodiment to that of FIG. 5a, in which the height of the collector is less than that shown in FIG. 5a, the collector of FIG. 5c thus having a shorter profile.

FIG. 6a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention, wherein the collector is illustrated in a generally horizontal orientation.

FIG. 6b is a side elevational view of the collector of FIG. 6a.

FIG. 7a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention, wherein the collector is illustrated in a generally horizontal orientation.

FIG. 7b is a side elevational view of the collector of FIG. 7a.

FIG. 7c is another side elevational view of the collector of FIG. 7a.

FIG. 8a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention, wherein the collector is illustrated in a generally horizontal orientation.

FIG. 9a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention, wherein the collector is illustrated in a generally horizontal orientation.

FIG. 9b is a perspective view of part of a release mechanism of the collector of FIG. 9a.

FIG. 9c is a schematic illustration representing the arrangement and operation of the release mechanism of FIG. 9b.

FIG. 10a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation and includes a suction tube attached thereto.

FIG. 10b is a side plan view of the collector of FIG. 10a and includes a close-up, cross-sectional view of an intake port of the collector of FIG. 10a within which a distal tip of a kerrison rongeur is received.

FIG. 11a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation and includes a suction tube attached thereto.

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FIG. 11b is a side plan view of the collector of FIG. 11a.

FIG. 12a is a side plan view in cross-section of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation and includes a suction tube attached thereto, and wherein the collector includes multiple interior scrapers.

FIG. 12b is a side plan view in cross-section of another collector similar to that of FIG. 12a, and differs in that the collector of FIG. 12b includes a single interior scraper.

FIG. 12c is a perspective view in cross-section of a portion of the collector of FIG. 12a illustrating a screen for filtering of outflow from the collector through the suction tube.

FIG. 13a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 13b is another perspective view of an opposite side of the collector as seen in FIG. 13a.

FIG. 14a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 14b is a perspective view illustrating a use of the collector of FIG. 14a in collecting bone from a kerrison rongeur, wherein the collector is illustrated in a generally horizontal orientation with the distal end of the kerrison rongeur about to be received through an intake port of the collector.

FIG. 15a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 15b is another perspective view of an opposite side of the collector as seen in FIG. 15a, wherein the intake port of the collector is better seen.

FIG. 16a is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 16b is another perspective view of an opposite side of the collector as seen in FIG. 16a, wherein the intake port of the collector is better seen.

Each of FIGS. 17a-17f illustrates in elevational plan view, taken in cross-section, a shape of an intake port for a collector in accordance with various embodiments of the present invention.

Each of FIGS. 18a-18c illustrates in plan view an opening of an intake port for a collector in accordance with various embodiments of the present invention.

Each of FIGS. 19a-19c illustrates in plan view a generally oval (preferably circular) intake port opening of an intake port for a collector in accordance with various embodiments of the present invention.

FIG. 20 is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector is illustrated in a generally vertical orientation.

FIG. 21a illustrates a table listing values of properties giving rise to some possible variations between some embodiments of the invention.

FIG. 21b illustrates a table listing some possible combinations of some of the elements of the table of FIG. 21a.

FIG. 22a is a perspective view of a tip and scraper showing path and orientation properties representative of some embodiments of the invention.

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FIG. 22*b* is a perspective view of a tip and scraper showing path and orientation properties representative of some embodiments of the invention.

FIG. 22*c* is a perspective view of a tip and scraper showing path and orientation properties representative of some embodiments of the invention.

FIG. 22*d* is a perspective view of a tip and scraper showing path and orientation properties representative of some embodiments of the invention.

FIG. 22*e* is a perspective view of a tip and scraper in the form of a wiper showing path and orientation properties representative of some embodiments of the invention.

FIG. 22*f* is a perspective view of a tip and scraper in the form of a wiper showing path and orientation properties representative of some embodiments of the invention.

FIG. 22*g* is a perspective view of a tip and scraper in the form of a wiper showing path and orientation properties representative of some embodiments of the invention.

FIG. 23 illustrates a perspective view of a collector in accordance with another embodiment of the invention.

FIG. 24*a* illustrates a top plan view of a cap of a collector 2400 in accordance with another embodiment of the invention.

FIG. 24*b* is a side elevational view of the collector 2400. FIG. 24*c* is another side elevational view of the collector 2400.

FIG. 24*d* is a top perspective side view of the collector 2400.

FIG. 24*e* is a bottom perspective side view of the collector 2400.

FIG. 24*f* is a perspective view of the scraper of the collector 2400, which scraper is located under and attached to the cap, as seen through the opening in the cap in FIGS. 24*c* and 24*e*.

FIG. 25*a* is a perspective view of a collector in accordance with another embodiment of the invention.

FIG. 25*b* is a side elevational view of the collector of FIG. 25*a*, further illustrating a tip of a rongeur inserted into the cap thereof.

FIG. 25*c* is a perspective view of the collector of FIG. 25*a*, further illustrating a tip of a rongeur inserted into the cap thereof.

FIG. 25*d* is another elevational side view of the collector of FIG. 25*a*, further illustrating a tip of a rongeur inserted into the cap thereof.

FIG. 25*e* is a top perspective view of the collector of FIG. 25*a*, further illustrating a tip of a rongeur inserted into the cap thereof.

FIG. 25*f* is a perspective close-up view of the top of the collector of FIG. 25*a* with a tip of a rongeur inserted therein, and further illustrating bristles of the brush extending from the underside of the top of the cap downwardly into engagement with the tip of the rongeur.

FIG. 26*a* is a side-by-side elevational view of the tops of two prototype collectors, each in accordance with a respective embodiment of the invention.

FIG. 26*b* is a perspective view of the caps of the collectors of FIG. 26*a* after the caps have been unscrewed from the containers and placed upside down on a surface to expose views of the brushes attached to the underside of the caps.

FIG. 26*c* is a side-by-side top perspective view of the collectors of FIG. 26*a*.

FIG. 26*d* is a side elevational view of one of the collectors of FIG. 26*a*.

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FIG. 27 is a perspective view showing movement of the tip of a rongeur in the direction of arrow A so as to extend through a port of a collector in accordance with an embodiment of the invention.

FIG. 28*a* is another perspective view showing the tip of the rongeur of FIG. 27 received within the cap of the collector of FIG. 27.

FIG. 28*b* is another perspective view also showing the tip of the rongeur of FIG. 27 received within the cap of the collector of FIG. 27.

FIG. 29 is a perspective side view of a collector in accordance with yet another embodiment of the invention.

FIG. 30 is a perspective side view of a prototype collector representative of the embodiment of FIG. 29.

FIG. 31 is another perspective side view of the prototype collector of FIG. 30.

FIG. 32 is another perspective side view of the prototype collector of FIG. 30.

FIG. 33*a* is a perspective side view of the prototype collector of FIG. 30 with the tip of a rongeur inserted through the port thereof into the collector.

FIG. 33*b* is another perspective side view of the prototype collector of FIG. 30 with the tip of the rongeur inserted through the port thereof into the collector.

FIG. 34*a* is a perspective view of the underside of the cap of the prototype collector of FIG. 30, wherein the scraper attached to the slidable plunger for clearing material from the cutting area perhaps is best seen.

FIG. 34*b* is a perspective view of the underside of the cap of the prototype collector of FIG. 30, wherein the stop of the cap, against which the tip of the rongeur rests when fully inserted, is perhaps best seen.

FIG. 35 is a perspective view of a collector used to collect cut bone from a Kerrison rongeur in accordance with an embodiment of the present invention.

FIG. 35A is a perspective shaded view of the collector of FIG. 35.

FIG. 35B is another perspective shaded view of the collector of FIG. 35.

FIG. 36 is a side elevational view of the collector of FIG. 35.

FIG. 37 is a perspective view of an upper portion of the collector of FIG. 35—the upper portion including a cap.

FIG. 37A is a perspective shaded view of the upper portion of the collector of FIG. 35.

FIG. 38 is a side elevational view of the upper portion of the collector of FIG. 35.

FIG. 38A is a side elevational shaded view of the upper portion of the collector of FIG. 35.

FIG. 39 is a side cross-sectional view of the upper portion of the collector of FIG. 35.

FIG. 40 is a perspective cross-sectional view of the cap of the upper portion of the collector of FIG. 35.

FIG. 41 is a perspective cross-sectional view of a first component of the cap of the upper portion of the collector of FIG. 35.

FIG. 42 is a top view of the first component of FIG. 41.

FIG. 43 is a bottom view of the cap of the upper portion of the collector of FIG. 35.

FIG. 43A is a bottom shaded view of the cap of FIG. 43.

FIG. 43B is a bottom view of the cap of FIG. 41 with areas of different pluralities of scrapers schematically illustrated, including a void or gap that exists between the areas of different pluralities of scrapers.

FIG. 44 is a perspective view of a second component of the cap of the upper portion of the collector of FIG. 35.

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FIG. 45 is a perspective view of a collector used to collect cut bone from a kerrison rongeur in accordance with an embodiment of the present invention.

FIG. 46 is a perspective view of a cap of the collector of FIG. 45.

FIG. 47 is a top view of the cap of FIG. 46.

FIG. 48 is a side perspective view of the cap of FIG. 46.

FIG. 49 is a bottom perspective view of the cap of FIG. 46.

FIG. 50 is a side elevational view of the cap of FIG. 46.

FIG. 51 is another side elevational view of the cap of FIG. 46.

FIG. 52 is a perspective view of the collector of FIG. 45 and portion of a kerrison rongeur including distal tip thereof which has been inserted into and received within the cap of the collector.

FIG. 53 is a photograph of a representative first component of a cap placed adjacent a penny to demonstrate preferred size.

FIG. 54 is a photograph of a representative container being held by hand with a representative second, base component having been screwed onto the container.

DETAILED DESCRIPTION

As a preliminary matter, it will readily be understood by one having ordinary skill in the relevant art ("Ordinary Artisan") that the present invention has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the invention and may further incorporate only one or a plurality of the above-disclosed features. Furthermore, any embodiment discussed and identified as being "preferred" is considered to be part of a best mode contemplated for carrying out the present invention. Other embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure of the present invention. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the invention and may further incorporate only one or a plurality of the above-disclosed features. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present invention.

Accordingly, while the present invention is described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present invention, and is made merely for the purposes of providing a full and enabling disclosure of the present invention. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded the present invention in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection afforded the present invention be defined by reading into any claim a limitation found herein that does not explicitly appear in the claim itself.

Thus, for example, any sequence(s) and/or temporal order of steps of various processes or methods that are described herein are illustrative and not restrictive. Accordingly, it should be understood that, although steps of various processes or methods may be shown and described as being in a sequence or temporal order, the steps of any such processes or methods are not limited to being carried out in any particular sequence or order, absent an indication otherwise.

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Indeed, the steps in such processes or methods generally may be carried out in various different sequences and orders while still falling within the scope of the present invention. Accordingly, it is intended that the scope of patent protection afforded the present invention is to be defined by the issued claim(s) rather than the description set forth herein.

Additionally, it is important to note that each term used herein refers to that which the Ordinary Artisan would understand such term to mean based on the contextual use of such term herein. To the extent that the meaning of a term used herein—as understood by the Ordinary Artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by the Ordinary Artisan should prevail.

Regarding applicability of 35 U.S.C. § 112 (f), no claim element is intended to be read in accordance with this statutory provision unless the explicit phrase "means for" or "step for" is actually used in such claim element, whereupon this statutory provision is intended to apply in the interpretation of such claim element.

Furthermore, it is important to note that, as used herein, "a" and "an" each generally denotes "at least one," but does not exclude a plurality unless the contextual use dictates otherwise. Thus, reference to "a picnic basket having an apple" describes "a picnic basket having at least one apple" as well as "a picnic basket having apples." In contrast, reference to "a picnic basket having a single apple" describes "a picnic basket having only one apple."

When used herein to join a list of items, "or" denotes "at least one of the items," but does not exclude a plurality of items of the list. Thus, reference to "a picnic basket having cheese or crackers" describes "a picnic basket having cheese without crackers", "a picnic basket having crackers without cheese", and "a picnic basket having both cheese and crackers." Finally, when used herein to join a list of items, "and" denotes "all of the items of the list." Thus, reference to "a picnic basket having cheese and crackers" describes "a picnic basket having cheese, wherein the picnic basket further has crackers," as well as describes "a picnic basket having crackers, wherein the picnic basket further has cheese."

Additionally, as used herein, "cap" denotes "a lid configured to be attached to an object in covering relation to an interior containment space of the object".

As used herein, a "scraper" is a brush, a group of bristles, a protuberance, a barb, or a finger; and is sufficiently rigid so as to generally hold form when not engaged by the distal end of a kerrison rongeur and to generally dislodge bone found in a cutting area of a kerrison rongeur when moved into or through the cutting area. Preferably, a scraper also is sufficiently flexible and resilient so as to generally deflect and bend to some extent upon abutment by and engagement with the distal end of a kerrison rongeur. A scraper may be relatively hard or soft within this range. Furthermore, a scraper preferably comprises a bio-absorbable material in at least some embodiments of the invention. In this respect, a scraper may comprise a brush with bio-absorbable bristles which, if inserted into the body, are absorbed by the body.

Referring now to the drawings, one or more preferred embodiments of the present invention are next described. The following description of one or more preferred embodiments is merely exemplary in nature and is in no way intended to limit the invention, its implementations, or uses.

A First Embodiment

Accordingly, a perspective view of a collector 100 used to collect cut bone from a kerrison rongeur is shown in FIG. 2a.

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The collector **100** is in accordance with a first embodiment of the present invention and is illustrated in a generally vertical orientation in FIG. **2a**. A perspective view of a sequence of steps for using the collector **100** in collecting bone from a kerrison rongeur is represented in FIG. **2b**, in which figure the collector **100** is illustrated in a generally horizontal orientation.

As will be appreciated, the collector **100** is hand-held and comprises a container body **101** (shown in FIG. **2a** and FIG. **2b** to be a generally elongate body or tube that is cylindrical in shape), and having opposite, axially-aligned end caps **102,104**. The container body **101** defines an interior containment space and comprises an opaque wall portion **103** and a transparent wall portion **105**.

In use, the collector **100** is manually held at opposite ends in a generally horizontal position, as shown in FIG. **2b**, in which it is positioned to receive the distal end **108** of a kerrison rongeur containing cut bone **110**. The distal end **108** is received through an intake port opening **106** that is located in side wall portion **103** halfway in-between the opposite ends of the collector **100**. In this respect, the collector **100** is held sideways and is said to be side-loaded.

Preferably in use, a scrub nurse, assisting technician, or similar person (hereinafter generally "assistant") moves or loads the collector **100** onto the distal end **108** of the kerrison rongeur when extended from the surgical site by the surgeon, who preferably maintains focus and attention on the surgical site and does not look away for purposes of aligning the distal end **108** with the intake port opening **106** (which alignment is the assistant's responsibility): it will be appreciated that the intake port opening **106** is maneuvered relative to and aligned with the distal end **108** by the assistant.

Upon receipt of the distal end **108** with the cut bone **110** through the opening **106** into the interior of the collector **100**, the assistant depresses (pushes inwardly in direction of arrows **A**) on spring-loaded, axially-aligned end buttons **112,114** located on the opposite end caps **102,104** of the collector **100**, respectively. In this respect, the buttons preferably are urged or biased by springs (not visible in FIG. **2a** or **2b**) in directions opposite to the arrows **A** in FIG. **2a** so as to extend outwardly from the body **101** for grasping and holding of the collector **100** by the assistant during maneuvering of the collector onto the distal end **108** of the kerrison rongeur. This causes opposed interior scrapers **116,118** located in the interior of the collector **100** and connected to the end caps **102,104** to converge onto and cover the distal end **108** of the kerrison rongeur when received within the interior of the collector **100** through the port opening **106**. This action-especially when performed multiple times-should serve to dislodge the bone **110** from the kerrison rongeur, resulting in the bone **110** either becoming lodged within the scrapers or falling onto an interior wall of the collector **100** defining at least in part the interior containment space. Moreover, the scrapers **116,118** optionally may be rotatable about a longitudinal axis **L** of the container body **101** by respective manual rotation of the end buttons **112,114**. The transparency of the wall portion **105** will enable the assistant to view whether the bone **110** has been collected yet from the kerrison rongeur, and to continue actuating the end buttons **112,114** either until the bone **110** has been collected or a reasonable number of attempts has been tried.

Once collected, the assistant removes (unloads) the collector **100** from the distal end **108** of the kerrison rongeur with the bone **110** remaining within the interior of the collector **100**, either loosely retained therein or lodged within one of the opposed scrapers **116,118**. Preferably the

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assistant then acknowledges to the surgeon the completion of the collection of the bone from kerrison rongeur, whereupon the surgeon returns the kerrison rongeur to the surgical site for continued cutting. This sequence continues until cutting by the surgeon is completed.

During the procedure, the bone **110** may be kept within the collector **100** until needed or may be removed as desired. The bone **110** preferably is harvested by removing each of the end caps **102,104** from the elongate body **101** of the collector **100**. FIG. **2c** is a perspective view of a scraper **118** of an end cap **114** of the collector **100** after bone has been collected from a kerrison rongeur and the end cap **114** has been detached from the body **101**. In this respect, each of the end caps **112,114** preferably is received within an end of the elongate body **101** and retained thereby by a friction fit, for example, between the wall of the body **101** and a rubber or silicone seal **115**; and/or a threaded engagement by screwing (not shown). Bone that is lodged within a scraper **116,118** of an end cap **102,104** can be removed by manually agitating the scraper **116,118** and pulling on the bone, with the scraper **116,118** extending over a sterile cloth, towel, pad, or table surface to catch the falling bone. Similarly, the bone within the body **101** of the collector **100** that is loose may be dumped onto a sterile cloth, towel, or pad, or directly onto a sterile surface of a table. The transparent portion **105** of the body **101** further may comprise a hinged or removable panel, by which the interior of the body **101** may be accessed for manual removal of any bone that may become stuck within the interior containment space of the body **101**. The harvested bone then can be used during the procedure, such as for example, in an autograft procedure.

The materials from which the components of the collector are made may be any desired, suitable material for use in the foregoing procedure. For instance, a scraper may comprise metal or other rigid or semi-rigid material that is effective in dislodging bone from the distal end of the kerrison rongeur. It is further contemplated that one or more of the materials may be molded from, for example, an inert plastic material. Furthermore, the materials preferably are lightweight such that the collector is readily manipulated by hand. Furthermore, the collector may be disposable, in that the collector may be used during a single medical procedure for a patient and then discarded in accordance with applicable HAZMAT protocols. Alternatively, one or more components of the collector (including all of the components) are designed to and are able to be sterilized for reuse with another patient during another procedure.

A Second Embodiment

FIG. **3a** is a perspective view of a collector **200** used to collect cut bone from a kerrison rongeur in accordance with a second embodiment of the present invention. The collector **200** is illustrated in a generally vertical orientation in FIG. **3a**. A perspective view of the collector **200** representing a sequence of steps for using the collector **200** to collect bone from a kerrison rongeur is shown in FIG. **3b**, wherein the collector **200** is illustrated in a generally horizontal orientation.

The collector **200** is hand-held and comprises a container body **201**, shown to be generally cylindrical in shape. The container body **201** defines an interior containment space and includes a transparent sidewall portion **203** through which the interior of the container body **201** is visible.

The collector **200** further comprises a cap **205** that is attached to the container body **201**. FIG. **3c** is a top plan

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view of the cap 205 of the collector 200, and FIG. 3d is a side elevational view of the cap 205 of the collector 200.

The top of the cap 205 includes an intake port opening 213 centrally located in the top of the cap 205, perhaps as best shown in FIG. 3c. The top of the cap 205 includes a funnel-shaped surface 202 that is radially inclined toward the opening 213, as perhaps best shown in FIG. 3b. The cap 205 further includes a platform 207 that extends into the interior of the container body 201, as perhaps best shown in FIG. 3b. The platform 207 includes a stop 209 and scrapers 211. The scrapers 211—two of which are shown in the figures, but any number of which may be included—extend from the platform and are flexible and resilient, preferably sufficiently rigid or semi-rigid so as to hold form when not engaged by the distal end of a kerrison rongeur and deflecting and bending upon abutment by and engagement with the distal end of a kerrison rongeur. The scrapers 211 may be made from metal or other material, and it is contemplated that a scraper may be made of polypropylene.

It will be appreciated that the funnel-shaped top surface 202 of the cap 205 is designed to direct a distal end of a kerrison rongeur toward the opening 213 upon abutment therewith, and the stop 209 is intended to abut and physically preclude further advancement of the distal tip of the kerrison rongeur when received through the opening 213.

The cap 205 is freely rotatable relative to the container body 201, as indicated by the arrow R in FIG. 3b. Rotation—or twisting—of the cap 205 on the container body 201 is facilitated by external ribs 215 on the side of the cap 205 for manual grasping. Rotation of the cap 205 does not detach the cap 205 from the container body 201, which preferably is received and held within a mouth of the container body 201 in a frictional fit between the container body and a seal 217 of the cap 205. The cap 205 preferably is removable by pulling of the cap 205 off of the container body 201 with a reasonable amount of effort.

The stop 209 is located at a spacing relative to the opening 213 such that, upon receipt of the distal end of the kerrison rongeur and abutment thereof with the stop 209, the scrapers 211 extend over and cover the distal end of the kerrison rongeur proximate the cutting area in which cut bone would be found. Furthermore, rotation of the cap results in movement of the scrapers 211 around, about and through the cutting area of the distal end of the kerrison rongeur so as to dislodge and remove for collection any cut bone carried therein.

In this respect, the collector 200 is held horizontally generally as shown in FIG. 3b.

Preferably in use, an assistant moves or loads the collector 200 onto the distal end of a kerrison rongeur when extended from the surgical site by the surgeon, who preferably maintains focus and attention on the surgical site and does not look away for purposes of aligning the distal end with the intake port opening 213, which is generally located in the center of the cap 205. Alignment is the assistant's responsibility, and it will be appreciated that the intake port opening 213 is maneuvered relative to and aligned with the distal end of the kerrison rongeur by the assistant. The funnel-shaped surface 202 facilitates the alignment with and receipt of the distal tip within the opening 213.

Upon receipt of the distal end of the kerrison rongeur into the interior of the collector 200 and its abutment with the stop 209, the assistant twists (rotates in direction of arrow R) the cap 205 relative to the container body 201. This causes the scrapers 211 to engage and remove any cut bone carried by the kerrison rongeur, resulting in the bone either falling onto an interior wall of the container body 201 or becoming

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lodged within a scraper. Moreover, the transparency of the container body 201 will enable the assistant to view whether the bone has been collected yet from the kerrison rongeur, and to continue twisting the cap 205 on and relative to the container body 201 either until the bone has been collected or a reasonable number of rotations has occurred.

Once collected from the kerrison rongeur, the assistant removes (unloads) the collector 200 from the distal end of the kerrison rongeur with the bone remaining within the interior of the collector 200, either loosely retained therein or lodged within a scraper. Preferably the assistant then acknowledges to the surgeon the completion of the collection of the bone from kerrison rongeur, whereupon the surgeon returns the kerrison rongeur to the surgical site for continued cutting. This sequence continues until cutting by the surgeon is completed.

During the procedure, the bone may be kept within the collector 200 until needed or may be removed as desired. The bone preferably is harvested by removing the cap 205 from the container body 201. Bone that is lodged within a scraper can be removed by manually agitating the scraper and pulling on the bone so that the bone falls onto a sterile cloth, towel, pad, or table surface. The bone within the container body 201 that is loose may be dumped onto the sterile cloth, towel, or pad, or directly onto the sterile surface of the table. Bone within the container body 201 that is not loose may be manually removed. The harvested bone then can be used during the procedure, such as for example, in an autograft procedure.

The materials from which the components of the collector are made may be any desired, suitable material for use in the foregoing procedure. For instance, a scraper may comprise metal or other rigid or semi-rigid material that is effective in dislodging bone from the distal end of the kerrison rongeur. It is further contemplated that one or more of the materials may be molded from, for example, an inert plastic material. Furthermore, the materials preferably are lightweight such that the collector is readily manipulated by hand. Furthermore, the collector may be disposable, in that the collector may be used during a single medical procedure for a patient and then discarded. Alternatively, one or more components of the collector (including all of the components) are designed to and are able to be sterilized for reuse with another patient during another procedure.

A Third Embodiment

FIG. 4a is a perspective view illustrating a sequence of steps for using another collector 300 in accordance with a third embodiment of the present invention, wherein the collector 300 is illustrated in a generally horizontal orientation receiving the distal end of a kerrison rongeur. FIG. 4b is a top plan view of the collector 300 and distal tip of the kerrison rongeur of FIG. 4a. The collector 300 and use thereof is similar to collector 200 and its use, with the exception that a cap 305 of the collector includes a cut out 308 in the side thereof by which an assistant may better visualize alignment with and receipt of the distal end of a kerrison rongeur within the intake port opening in the cap 305, and with the further exception that the cap further does not include a funnel-shaped top surface.

A Fourth Embodiment

FIG. 5a is a perspective view of a collector 400 used to collect cut bone from a kerrison rongeur for harvesting in accordance with another embodiment of the present inven-

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tion, wherein the collector **400** is illustrated in a generally vertical orientation. FIG. **5b** is a perspective view representative of a sequence of steps for using the collector **400** in collecting bone from a kerrison rongeur, wherein the collector **400** is illustrated in a generally horizontal orientation with the distal end of the kerrison rongeur being received within the collector **400**.

The collector **400** is hand-held and comprises a container body **401**, shown to be generally cylindrical in shape. The container body **401** defines an interior containment space and includes a transparent sidewall portion **403** through which the interior of the container body **401** is visible. The collector **400** further comprises a cap **405** that is attached to the container body **401**. The cap **405** includes an interior ledge **411** that defines an intake port opening **413** through which the distal end of a kerrison rongeur is received. The cap **405** further includes a guide member **417** that projects from the ledge **411** and that is shaped and contoured for receiving and guiding therein a portion of the distal end of the kerrison rongeur. The guide member **417** preferably includes a U-shape configuration, as shown. An saddle member **419** extends from a bottom **409** of the container body **401** and, in conjunction with the guide member **417**, assists in aligning the distal end of the kerrison rongeur for collection of bone, as shown in FIG. **5b**. The saddle member **419** preferably includes a C-shape configuration, as shown.

A scraper **421** is provided proximate the bottom **409** of the container body **401** and extends toward the saddle member **419** into the cutting area of the distal end of a kerrison rongeur when aligned by the saddle member **419**. The scraper **421** is secured to a base **423** that extends from the side wall **403** of the container body **401** in abutment with the bottom **409** of the container body **401**. Further in this regard, the bottom **409** serves as a stop for the distal end of the kerrison rongeur.

The cap **405** is freely rotatable relative to the container body **401**, as indicated by the arrows R in FIG. **5b**. Rotation—or twisting—of the cap **405** on the container body **401** is facilitated by external ribs **416** on the side of the cap **405** for manual grasping. Rotation of the cap **405** does not detach the cap **405** from the container body **401**, which preferably is received and held within a mouth of the container body **401** in a frictional fit between the container body and a seal (now shown, but similar to seal **217** found in collector **200**). The cap **405** preferably is removable by pulling of the cap **405** off of the container body **401** with a reasonable amount of effort.

The scraper **421** is located at a spacing relative to the saddle member **419** such that, upon receipt of the distal end of the kerrison rongeur and abutment thereof with the bottom **409** serving as a stop, the scraper **411** extends over and covers the distal end of the kerrison rongeur proximate the cutting area thereof in which cut bone would be found. Furthermore, rotation of the container body **401** relative to the cap **405** results in movement of the scraper **421** around, about and through the cutting area of the distal end of the kerrison rongeur so as to dislodge and remove for collection any cut bone carried therein. It will further be appreciated that the saddle member **419** also rotates with the container body **401**, whereas the guide member **417** is an integral molded part of—and remains fixed relative to—the cap **405**.

It will further be noted that the cap **405** defines an inner surface **430** that, when the collector is positioned for receiving the distal end of the kerrison rongeur, as shown in FIG. **5b**, the inner surface **430** will form a flange that catches bone that may fall from the kerrison rongeur while attempting to

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receiving the distal end of the kerrison rongeur within the guide member **417** and through the intake opening **413**.

Preferably in use, an assistant moves or loads the collector **400** onto the distal end of a kerrison rongeur when extended from the surgical site by the surgeon, who preferably maintains focus and attention on the surgical site and does not look away for purposes of aligning the distal end with the intake port opening **413**, which is generally located in the center of the cap **405**. Alignment is the assistant's responsibility, and it will be appreciated that the intake port opening **413** is maneuvered relative to and aligned with the distal end of the kerrison rongeur by the assistant. The guide member **417** further aids the assistant in positioning of the distal end of the kerrison rongeur within the saddle member **419** after being inserted through the opening **413**.

Upon receipt of the distal end of the kerrison rongeur into the interior of the collector **400** and its abutment with the bottom **409** serving as a stop, the assistant twists (rotates back and forth in the directions of arrows R) the container body **401** relative to the cap **405**. This causes the scraper **421** to engage and remove any cut bone carried by the kerrison rongeur, resulting in the bone either falling onto an interior wall of the container body **401** or becoming lodged within the bristles of the scraper **421**. Moreover, the transparency of the side wall **403** container body **401** enables the assistant to view whether the bone has been collected yet from the kerrison rongeur, and to continue twisting the container body **401** relative to the cap **405** back and forth either until the bone has been collected or a reasonable number of attempts have been made by the assistant.

Once collected from the kerrison rongeur, the assistant removes (unloads) the collector **400** from the distal end of the kerrison rongeur with the bone remaining within the interior of the collector **400**, either loosely retained therein or lodged within the bristles. Preferably the assistant then acknowledges to the surgeon the completion of the collection of the bone from kerrison rongeur, whereupon the surgeon returns the kerrison rongeur to the surgical site for continued cutting without taking focus away from the surgical site. This sequence continues until cutting by the surgeon is completed.

During the procedure, the bone may be kept within the collector **400** until needed or may be removed as desired. The bone preferably is harvested by removing the cap **405** from the container body **401**. Bone that is lodged within a scraper can be removed by manually agitating the bristles and pulling on the bone so that the bone falls onto a sterile cloth, towel, pad, or table surface. The bone within the container body **401** that is loose may be dumped onto the sterile cloth, towel, or pad, or directly onto the sterile surface of the table. Bone within the container body **401** that is not loose may be manually removed by hand or with an instrument. The harvested bone then can be used during the procedure, such as for example, in an autograft procedure.

The materials from which the components of the collector are made may be any desired, suitable material for use in the foregoing procedure. For instance, a scraper may comprise metal or other rigid or semi-rigid material that is effective in dislodging bone from the distal end of the kerrison rongeur. It is further contemplated that one or more of the materials may be molded from, for example, an inert plastic material. Furthermore, the materials preferably are lightweight such that the collector is readily manipulated by hand. Furthermore, the collector may be disposable, in that the collector may be used during a single medical procedure for a patient and then discarded. Alternatively, one or more components of the collector (including all of the components) are

designed to and are able to be sterilized for reuse with another patient during another procedure.

FIG. 5c is a side elevational view of an alternative collector 400' generally similar to the collector 400, but in which the height of the collector 400' is less than the height of the collector 400 and, thus, the collector 400' has a smaller profile.

A Fifth Embodiment

FIG. 6a is a perspective view of a collector 500 used to collect cut bone from a kerrison rongeur for harvesting in accordance with another embodiment of the present invention, wherein the collector 500 is illustrated in a generally horizontal orientation. FIG. 6b is a side elevational view of the collector 500.

The collector 500 includes a container body 501 that is box-shaped with generally rectangular sides and a cap 505 with generally rectangular sides. The container body 501 defines an interior containment space and preferably includes transparent side walls 503. The container body 501 is utilized to store collected bone. The cap 505 includes a stop comprising a wall 519 that extends between opposed interior side surfaces of the cap 505 and divides passage through the cap to the interior of the container body 501 into two side passages 508 around opposite sides of the wall 519. Scrapers 521 are located in front of the wall and extend from one of the interior side surfaces of the cap 505 from to which the wall 519 connects.

Preferably in use, an assistant moves or loads the collector 500 onto the distal end of a kerrison rongeur when extended from the surgical site by the surgeon, who preferably maintains focus and attention on the surgical site and does not look away for purposes of aligning the distal end of the kerrison rongeur with the opening in the cap 505. Alignment is the assistant's responsibility, and it will be appreciated that the collector 500 is maneuvered relative to and aligned with the distal end of the kerrison rongeur by the assistant such that the distal end of the kerrison rongeur, when received within the cap 505, will come into abutment with the wall 519 and cease further advancement within the cap 505 toward the container body 501. The spacing of the wall 519 to the scrapers 521 is such that the scrapers will pass through the cutting area of the distal end of the kerrison rongeur when the collector 500 is moved back and forth in lateral directions relative to the longitudinal extent of the kerrison rongeur. Furthermore, the wall 519 may be maintained in abutment with the tip of the kerrison rongeur during such back and forth movement for proper alignment of the scrapers 521 relative to the cutting area of the kerrison rongeur, thereby facilitating dislodgment of cut bone found in the cutting area by the scrapers 521. Such back and forth movement is facilitated by longitudinally protruding ribs 515 on the cap 505, which enable better grasping by hand of the cap 505. Additionally, one or more ribs 515—such as the middle pair of ribs shown in FIGS. 6a and 6b, for example—may be used for sighting by the assistant during loading of the distal end of the kerrison rongeur, as represented by the direction and alignment of arrow A relative to the middle pair of ribs seen in FIG. 6b.

Once collected from the kerrison rongeur, the assistant removes (unloads) the collector 500 from the distal end of the kerrison rongeur with the bone remaining within the interior of the cap 505, either loosely retained therein or lodged within the bristles. Tilting of the collector 500 to a vertical orientation and light shaking of the collector 500 should cause any loose bone collected from the kerrison

rongeur to fall through one of the side passages into the container body 501 where the bone yield can be viewed through the transparent walls 503.

Preferably the assistant acknowledges to the surgeon the completion of the collection of the bone from kerrison rongeur, whereupon the surgeon returns the kerrison rongeur to the surgical site for continued cutting without taking focus away from the surgical site. This sequence continues until cutting by the surgeon is completed.

During the procedure, the bone may be kept within the container body 501 of the collector 500 until needed or may be removed as desired. The bone preferably is harvested by removing the cap 505 from the container body 501. Bone that is lodged within any bristles of the scrapers 521 can be removed by manually agitating the bristles and pulling on the bone so that the bone falls onto a sterile cloth, towel, pad, or table surface. The bone within the container body 501 that is loose may be dumped onto the sterile cloth, towel, or pad, or directly onto the sterile surface of the table. Bone within the container body 501 that is not loose may be manually removed by hand or with an instrument. The harvested bone then can be used during the procedure, such as for example, in an autograft procedure.

The materials from which the components of the collector are made may be any desired, suitable material for use in the foregoing procedure. For instance, a scraper may comprise metal or other rigid or semi-rigid material that is effective in dislodging bone from the distal end of the kerrison rongeur. It is further contemplated that one or more of the materials may be molded from, for example, an inert plastic material. Furthermore, the materials preferably are lightweight such that the collector is readily manipulated by hand. Furthermore, the collector may be disposable, in that the collector may be used during a single medical procedure for a patient and then discarded. Alternatively, one or more components of the collector (including all of the components) are designed to and are able to be sterilized for reuse with another patient during another procedure.

A Sixth Embodiment

FIG. 7a is a perspective view of a collector 600 used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention, wherein the collector 600 is illustrated in a generally horizontal orientation. Each of FIGS. 7b, 7c is a respective side elevational view of the collector 600. The collector 600 is similar in construction and use as the collector 500 discussed above with reference to FIGS. 6a and 6b. A primary difference over the collector 500, however, is that the wall 521 is replaced with a depressible button 621. The button 621 preferably is spring-loaded and biased into an extended position, shown in FIG. 7a in solid line and additionally shown in FIG. 7b, in which extended position passage through the cap 605 to an interior containment space of the container body 601 is not blocked. Upon being fully depressed, the button extends into and blocks passage through the cap 605, as shown in phantom in FIG. 7a. When in the depressed position, the button serves as a stop in same manner as wall 519 in collector 500, described above.

A Seventh Embodiment

FIG. 8a is a perspective view of a collector 700 used to collect cut bone from a kerrison rongeur for harvesting in accordance with another embodiment of the present invention, wherein the collector 700 is illustrated in a generally

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horizontal orientation. The collector **700** is similar in construction and use as the collector **600** discussed above with reference to FIG. **7a**, FIG. **7b**, and FIG. **7c**. A primary difference over the collector **600**, however, is that the cap includes a ramp **704** that assists in guiding the tip of the end portion of a kerrison rongeur into proper position within the cap. The height of the passage is also less than that of the height of the passage through cap **605**, and the length of travel when depressing the button is consequently less as well.

An Eighth Embodiment

FIG. **9a** is a perspective view of a collector **800** used to collect cut bone from a kerrison rongeur for harvesting in accordance with another embodiment of the present invention, wherein the collector **800** is illustrated in a generally horizontal orientation. FIG. **9b** is a perspective view of part of a release mechanism **840** of the collector **800**, and FIG. **9c** is a schematic illustration representing the arrangement and operation of the release mechanism **840** relative to the scrapers **821** and cap **805** of the collector **800**.

The collector **800** includes a release mechanism **840** having a button **823** connected by an arm **842** to a wall **843**. The wall **843** defines a release opening **844** therein. The release mechanism **840** is contained within the cap **805** and includes a spring **845** that biases the release mechanism **840** into a closed configuration, wherein bone collected within intake port **813** using scrapers **821** is blocked from access to an interior containment space of the container body **801**. When the button **823** is depressed against the biasing of the spring **845** in the direction of arrow A, the wall **843** moves (slides) so as to align the release opening **844** with the intake port opening **813** thereby permitting the collected bone to pass to the interior of the container body **801**. When in the closed configuration, the wall **843** serves as a stop for the distal end of a kerrison rongeur, in which position the scrapers **821** pass through the cutting area when the cap **805** is rotated about its longitudinal axis relative to the kerrison rongeur.

Additional Embodiments Utilizing Suction

Additional collectors used to collect cut bone from a kerrison rongeur for harvesting in accordance with still yet more embodiments of the present invention embodiments are disclosed with reference to FIGS. **10a** through **19c**. Each of these additional embodiments utilizes suction and includes a suction port for attachment thereto of a suction hose or tube. The hose or tube in turn is connected to a suction source, such as those commonly found in operating rooms.

A collector **900** in accordance with a first such embodiment is shown in FIGS. **10a** and **10b**, wherein a perspective view of the collector **900** used to collect cut bone from a kerrison rongeur for harvesting is illustrated in a generally vertical orientation and includes a suction tube **950** attached to a suction port **952** located on the cap **905** of the collector **900**. FIG. **10b** is a side plan view of the collector **900** that perhaps better shows in a close-up, cross-sectional view, the configuration of the intake port **913** of the collector **900** within which a distal tip of a kerrison rongeur is received. In accordance with this embodiment, the intake port includes one or more scrapers located within the interior thereof. Additionally, the intake port **913** itself forms a "hawk bill" scraping tip. The scrapers and scraping tip are used to dislodge and scrape bone from the kerrison rongeur, with the

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suction applied through the suction tube drawing into the collector the dislodged and scraped bone. The bone is caught and retained within an interior containment space of the container body **901** for later harvesting and use.

FIG. **11a** is a perspective view of a collector **1000** used to collect cut bone from a kerrison rongeur for harvesting in accordance with yet another embodiment of the present invention, wherein the collector **1000** is illustrated in a generally vertical orientation and includes a suction tube **1050** attached to a suction port **1052** located on the cap **1005** of the collector **1000**. FIG. **11b** is a side plan view of the collector **1000**.

Unlike collector **900**, the cap **1005** of collector **1000** forms a long, ergonomic handle **1075** for gripping where the suction port **1052** is located. Like collector **900**, the intake opening **1013** of collector **1000** is formed in the cap **1005** and includes scrapers **1021** located within the interior thereof, as seen in FIG. **11a**.

FIG. **12a** is a side plan view, in cross-section, of a collector **1100** used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector **1100** is illustrated in a generally vertical orientation and includes a suction tube **1150** attached to a suction port **1152** located on the cap **1105**. As shown in FIG. **12a**, the collector **1100** includes multiple interior scrapers **1121** extending from the cap **1105** downwardly within the interior thereof for dislodging bone from the distal end of a kerrison rongeur.

FIG. **12b** is a side plan view in cross-section of another collector **1200** similar to the collector **1100** of FIG. **12a**, and differing only in that a single interior scraper **1221** is provided rather than a plurality of scrapers.

FIG. **12c** is a perspective view in cross-section of a portion of the collector **1100** of FIG. **12a** illustrating a porous screen mesh or screen **1190** for filtering of outflow from the collector **1100** through the suction tube **1150** so that precious bone is not inadvertently suctioned away from the collector **1100**. Indeed, a screen is preferred as the collector is most likely will be accidentally tipped over onto its side at some point during its use such that bone would be suctioned away if not for the screen.

FIG. **13a** is a perspective view of a collector **1300** used to collect cut bone from a kerrison rongeur for harvesting in accordance with yet another embodiment of the present invention, wherein the collector **1300** is illustrated in a generally vertical orientation. FIG. **13b** is another perspective view of an opposite side of the collector **1300** as seen in FIG. **13a**. The collector **1300** includes a suction tube **1350** attached to a suction port **1352** located at a distal end of a handle **1375** that, itself, is located proximate the bottom of the collector **1300**. An airflow channel **1377** is defined within and extends through the interior of the handle **1375** between the suction port **1313** and an interior containment space of the container body **1301** (which is represented in phantom in FIGS. **13a** and **13b**). As shown in FIGS. **13a** and **13b**, the handle **1375** and suction port are part of the cap **1305**, which is removably attached to the container body **1301**. Intake port **1313** is also part of the cap **1305**.

FIG. **14a** is a perspective view of a collector **1400** used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector **1400** is illustrated in a generally vertical orientation. FIG. **14b** is a perspective view illustrating a use of the collector **1400** in collecting bone from a kerrison rongeur for harvesting, wherein the collector **1400** is illustrated in a generally horizontal orientation with the distal end of the kerrison rongeur about to be received through an

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intake port **1413** of the collector **1400** that is located on the top of the cap **1405** of the collector **1400**.

FIG. **15a** is a perspective view of a collector **1500** used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector **1500** is illustrated in a generally vertical orientation. FIG. **15b** is another perspective view of an opposite side of the collector **1500** as seen in FIG. **15a**, wherein an intake port **1513** of the collector **1500** is better seen. The intake port **1513** is located on a side of the cap **1505** of the collector **1500**.

FIG. **16a** is a perspective view of a collector **1600** used to collect cut bone from a kerrison rongeur in accordance with yet another embodiment of the present invention, wherein the collector **1600** is illustrated in a generally vertical orientation. FIG. **16b** is another perspective view of an opposite side of the collector **1600** as seen in FIG. **16a**, wherein an intake port **1613** of the collector **1600** is better seen.

FIG. **17a** illustrates, in a cross-sectional view, a profile of an intake port **1701** for a collector in accordance with various embodiments of the present invention, which intake port has a rounded, angled tip.

FIG. **17b** illustrates, in a cross-sectional view, a profile of an intake port **1702** for a collector in accordance with various embodiments of the present invention, which intake port has a rounded, angled "hawk bill" scraping tip.

FIG. **17c** illustrates, in a cross-sectional view, a profile of an intake port **1703** for a collector in accordance with various embodiments of the present invention, which intake port has a rounded, angled round hawk bill scraping tip.

FIG. **17d** illustrates, in a cross-sectional view, a profile of an intake port **1704** for a collector in accordance with various embodiments of the present invention, which intake port has a rounded, rectilinear tip.

FIG. **17e** illustrates, in a cross-sectional view, a profile of an intake port **1705** for a collector in accordance with various embodiments of the present invention, which intake port has a rectilinear tip.

FIG. **17f** illustrates, in a cross-sectional view, a profile of an intake port **1706** for a collector in accordance with various embodiments of the present invention, which intake port has an angled tip.

FIG. **18a** illustrates in plan view an opening of an intake port **1801** for a collector in accordance with various embodiments of the present invention, which intake port includes multiple sets of bristle scrapers mounted on a top and sides of the port.

FIG. **18b** illustrates in plan view an opening of an intake port **1802** for a collector in accordance with various embodiments of the present invention, which intake port includes side scrapers combined with one large top scraper.

FIG. **18c** illustrates in plan view an opening of an intake port **1803** for a collector in accordance with various embodiments of the present invention, which intake port includes a single large top mounted scraper that scrapes both sides and top simultaneously, and which port has a narrowed entrance to improve scraping efficiency upon insertion of a kerrison rongeur.

FIG. **19a** illustrates in plan view a generally oval (preferably circular) intake port **1901** of for a collector in accordance with various embodiments of the present invention, which intake port includes top and side mounted scrapers.

FIG. **19b** illustrates in plan view a generally oval (preferably circular) intake port **1902** for a collector in accor-

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dance various embodiments of the present invention, which intake port includes multiple top scrapers combined with side scrapers.

FIG. **19c** illustrates in plan view a generally oval (preferably circular) intake port **1903** for a collector in accordance with various embodiments of the present invention, which intake port includes a single, top mounted scraper.

More Embodiments and Prototypes

Still another collector in accordance with an embodiment of the present invention is shown in a perspective view in FIG. **20**. As shown therein, the intake port **2013** comprises a squeezable and springy material capable of being squeezed by hand. When the distal end of a kerrison rongeur is received within the intake port **2013**, the intake port is manually squeezed such that scrapers located within the intake port **2013** engage and dislodge any cut bone found within the cutting area of the kerrison rongeur. The locations, types, and configurations of the scrapers may be in accordance with the disclosures of scrapers found hereinabove.

FIG. **21a** illustrates a table listing values of properties giving rise to some possible variations between some embodiments of the invention, and FIG. **21b** illustrates a table listing some possible combinations of some of the elements of the table of FIG. **21a**. In this respect, the first column of the table of FIG. **21a** lists elements that maybe used in one or more embodiments. The listed elements include a brush: a soft wiper (i.e., soft scraper); a hard wiper (i.e., hard scraper); pressure (air); and suction. Each of these elements represents a way of removing or dislodging cut matter from a tip of a rongeur. The second column lists paths of movement of the tip of the rongeur relative to the element for the brush and different wipers. The third column lists possible orientation to the path.

Exemplary combinations of some of these elements, including the brush and wipers, are set forth in FIG. **21b**. FIGS. **22a** through **22g** are perspective views of tips and scrapers showing various paths and orientations.

In particular, FIG. **22a** shows scraper **2202**, rongeur **2204** with cutting area **2206**, and a path of movement of the scraper **2202** relative to the rongeur **2204** and cutting area **2206**; FIG. **22b** shows scraper **2208**, rongeur **2210** with cutting area **22012**, and a path of movement of the scraper **2208** relative to the rongeur **2210** and cutting area **2212**; FIG. **22c** shows scraper **2214**, rongeur **2216** with cutting area **2218**, and a path of movement of the scraper **2214** relative to the rongeur **2216** and cutting area **2218**; FIG. **22d** shows scraper **2220**, rongeur **2222** with cutting area **2224**, and a path of movement of the scraper **2220** relative to the rongeur **2222** and cutting area **2224**; FIG. **22e** shows a scraper in the form of a wiper **2226**, rongeur **2228** with cutting area **2230**, and a path of movement of the wiper **2226** relative to the rongeur **2228** and cutting area **2230**; FIG. **22f** shows a scraper in the form of a wiper **2232**, rongeur **2234** with cutting area **2236**, and a path of movement of the wiper **2232** relative to the rongeur **2234** and cutting area **2236**; and FIG. **22g** shows a scraper in the form of a wiper **2238**, rongeur **2240** with cutting area **2242**, and a path of movement of the wiper **2238** relative to the rongeur **2240** and cutting area **2242**.

A perspective view of a collector **2300** in accordance with yet another embodiment of the invention is illustrated in FIG. **23**.

A perspective view of a collector **2400** in accordance with yet another embodiment of the invention—and structurally

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similar in design to the embodiment of FIG. 23—is illustrated in FIGS. 24a-24f, wherein: FIG. 24a is a top view of the cap of the collector 2400; FIG. 24b is a side elevational view of the collector 2400; FIG. 24c is another side elevational view of the collector 2400; FIG. 24d is a top perspective side view of the collector 2400; FIG. 24e is a bottom perspective side view of the collector 2400; and FIG. 24f is a perspective view of the scraper of the collector 2400 located under and attached to the cap, as seen through the opening in the cap in FIGS. 24c and 24e.

A perspective view of a collector 2500 in accordance with yet another embodiment of the invention is illustrated in FIGS. 25a-25f. A portion 2580 of the end of a kerrison rongeur including the tip also is illustrated relative to the container 2500 in FIGS. 25b-25f.

A perspective view of two collectors 2600, 2650 each in accordance with another respective embodiment of the invention are illustrated in FIGS. 26a-24f. The two collectors 2600, 2650 are structurally the same with the exception that brush 2610 of collector 2600 comprises bristles of the same length, whereas brush 2660 of collector 2650 has varying length bristles. In particular, FIG. 26a is a side-by-side elevational view of the tops of the collectors 2600, 2650 looking through the ports (wide side openings) in the caps thereof at the brushes 2610, 2660, where the varying length bristles of brush 2660 can be seen. FIG. 26b is a perspective view of the caps of the collectors 2600, 2650 after the caps have been unscrewed from the containers and placed upside down on a surface to expose views of the brushes 2610, 2660. As perhaps best shown in FIG. 26b, the brushes are secured to a generally semicircular area of the underside of the respective caps. Continuing with reference to collectors 2600, 2650. FIG. 26c is a side-by-side top perspective view of these collectors, and FIG. 26d is a side elevational view of collector 2650 being held with a left hand. The varying length of the bristles of brush 2660 also are seen in FIG. 26d as well.

FIGS. 27-28b demonstrate insertion of the tip of a kerrison rongeur through the port and into the interior of the cap of the collector 2600 in engagement with the brush 2610. Specifically, FIG. 27 shows movement of the tip in the direction of arrow A so as to extend through the port; and FIGS. 28a and 28b each shows the tip being received within the cap of the collector 2600 in engagement with the brush 2610.

A perspective view of a collector 2900 in accordance with yet another embodiment of the invention is illustrated in FIG. 29 and comprises a container 2905 and cap 2910. The cap includes a slidable plunger that is spring-biased into an open position by spring 2942. A scraper is located on the plunger inside the cap 2910 and, when the plunger is moved (slides) into the closed position by pressing on tab 2940 with a finger or thumb, the scraper (not shown) moves across the cutting area of a rongeur tip that has been inserted through port 2948 so as to dislodge and clear matter carried on the rongeur in the cutting area.

FIGS. 30-34b show a prototype collector 3000 as represented by collector 2900 of FIG. 29. In particular, FIG. 30 shows the collector 3000 held in a person's left hand with thumb on the tab 3040 of the plunger. An alternative method of holding the collector is with the right hand, as demonstrated in FIGS. 31 and 32.

The plunger is biased by spring 3042 into an open position. FIG. 34a shows the plunger 3050 in an open position. The spring is attached and secured at one end by a pin to the plunger, and at the other end by a pin to the main body of the cap 3010. In order to adjust the spring force, the

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pin can be placed in one of a plurality of holes formed in the cap 3010 as perhaps best seen in FIG. 30 and illustrated in representative collector 2900 of FIG. 29. The cap 3010 includes a port 3048 physically configured to receive the tip of a kerrison rongeur there through, as seen in FIGS. 33a and 33b.

When inserted through the port 3048, the distal end 3045 of the tip of the rongeur contacts a stop 3030, which is best seen in FIG. 34b. Indeed, in FIG. 34b the tip is shown having been inserted through port 3048 of cap 3010 and into abutment with stop 3030, whereat movement of the plunger by pressing on tab 3040 will force the scraper (in the form of a brush 3025) directly through the cutting area of the tip, thereby dislodging matter in the cutting area.

"Bristle Field and Teeth" Preferred Embodiments

FIG. 35 is a perspective view of a handheld collector 5000 used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the invention. Similarly, FIGS. 35A and 35B are perspective shaded views of the collector 5000, and FIG. 36 is a side elevational view of the collector 5000. As seen in these drawings, the collector comprises a cap 5002 and a container 5004. The cap includes an opening 5006 dimensioned to receive there through a distal end of a kerrison rongeur. The view of FIG. 36 directly faces such opening 5006.

Preferably, the cap is generally circular at its lower perimeter and the opening 5006 extends along the perimeter an arc having an obtuse angle and, more preferably, the opening 5006 extends along an arc having an angle of between 130 degrees and 140 degrees. As such, the opening is wide. The container of the collector comprises a generally elongate body that is cylindrical in shape, and walls of the body define an interior containment space 5008 of the container into which bone falls when dislodged from the distal end of a received kerrison rongeur.

An upper portion of the collector is seen in greater detail in FIGS. 37 and 38 and corresponding shaded views of FIGS. 37A and 38A. In this respect, FIGS. 37 and 37A show a perspective view of the upper portion of the collector including the cap, and FIGS. 38 and 38A show a side elevational view of the upper portion of the collector including the cap.

Additionally, FIG. 39 illustrates a side cross-sectional view of the upper portion of the collector; FIG. 40 is a perspective partial view in cross-section of the cap of the upper portion of the collector; and FIG. 43 illustrates a bottom view of the cap of the upper portion of the collector, of which FIG. 43A is a bottom shaded view thereof and 43B is a bottom view thereof schematically illustrating distinct areas of different pluralities of scrapers.

As perhaps best seen in FIG. 39, the cap includes a threaded portion 5010 that engages and mates with a threaded portion 5012 of the container when the cap and container are screwed together. The cap is thereby attached to the container in an upper portion of the collector and is removable from the container.

In the preferred collector 5000, the cap includes at least two areas of different pluralities of scrapers for dislodging bone from the distal end of a kerrison rongeur, with a gap extending between the different areas. With particular reference to FIGS. 39, 43, 43A, and 43B, a first plurality 5014 of similar scrapers 5016 is located in a first area 5018 of the underside of the cap that is in close proximity to the opening, and a second plurality 5020 of similar scrapers 5022 is located in a second area 5024 of the underside of the cap that

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is further from the opening than the first area **5018**. Moreover, the first and second areas **5018, 5024** are arranged in spaced relation to each other such that a gap **5026** comprising an absence of scrapers extends between the first plurality **5014** and the second plurality **5020**.

Referring now to FIGS. **39, 40, 43, 43A**, and **43B**, each scraper **5016** is seen to comprise a finger insofar as each scraper comprises a protuberance that is elongate with generally oval cross-section. Moreover, each such finger has a stepped diameter between a proximal portion **5030** thereof and a distal portion **5032** thereof relative to the underside **5031** of the cap (FIGS. **39** and **40**). In this respect, such finger includes a larger width at the proximal portion and a smaller width at the distal portion. The width of such finger further tapers along the proximal portion, decreasing as a height-wise extent increases away from the underside of the cap. The width of the distal portion similarly may taper, and in variations between different embodiments of the invention the stepped-diameter may be omitted with the finger tapering along its overall height from the underside of the cap to its distal end. In still other embodiments, it is contemplated that the finger may not taper along its height. In any event, the first area **5018** of the first plurality **5014** of scrapers **5016** itself preferably comprises a grouping of these fingers that collectively form a bristle field, much like that of a brush as seen in FIGS. **39, 40, 43**, and **43A**.

With continuing reference to FIGS. **39, 40, 43, 43A**, and **43B**, each scraper **5022** is seen to comprise a wiper having a length and a height that substantially exceeds a width thereof. Moreover, as perhaps best seen in FIG. **43B** the second area **5024** of the second plurality **5020** of such scrapers comprises a grouping thereof collectively forming a row of teeth with each tooth being a wiper and with a gap **5028** extending between adjacent teeth. The row of teeth is arranged along an arc generally extending in close relation to a circumferential boundary of the underside of the cap, perhaps as best seen in FIG. **43**. Additionally, preferably one row of teeth is formed; however, it is contemplated that multiple rows of teeth or different arrangements of the teeth may be utilized within the broader scope of the present invention.

In comparison of each scraper **5016** to a scraper **5022**, each scraper **5016** in extending from the underside of the cap is more than twice the height of each scraper **5022**, as perhaps best seen in FIG. **39**. Indeed, as shown the height of the lower portion of each scraper **5016** is greater than the height of each scraper **5022**.

Each scraper is sufficiently rigid so as to generally hold form when not engaged by a distal end of a kerrison rongeur and to generally dislodge bone found in a cutting area of a kerrison rongeur when the scraper is moved into or through the cutting area. Each scraper further is sufficiently flexible and resilient so as to generally deflect and bend—to various extents—upon abutment by and engagement with the distal end of a kerrison rongeur. A scraper may be relatively hard or soft within this range.

With respect to the preferred embodiment **5000**, each scraper **5022** is less flexible and resilient when compared to the proximal portion **5030** of each scraper **5016**; and each scraper **5022** is less flexible and resilient when compared to the distal portion **5032** of each scraper **5016**. Furthermore, the proximal portion **5030** of each scraper **5016** is less flexible and resilient when compared to the distal portion **5032** of each scraper **5016**. These differences in the flexibility and resiliency characteristics preferably arise from the differences in the relative thicknesses of the scraper **5022**, the proximal portion **5030**, and the distal portion **5032**. The

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differences also may arise from differences in shape of the scraper **5022**, the proximal portion **5030**, and the distal portion **5032**: from differences in the materials from which the scraper **5022**, the proximal portion **5030**, and the distal portion **5032** are formed; and any combination of such possibilities.

In the preferred embodiment, each scraper **5022** and the proximal portion **5030** and the distal portion **5032** of each scraper **5016** are all formed from the same material by molding and, preferably, by injection molding. Preferably, these scrapers are integrally formed as a single component **5042** of the cap **5002**, as perhaps best seen in FIG. **39**. Additionally, FIG. **44** is a perspective view of the single component.

The first component **5042** extends through a plurality of channels **5044**, which are formed in a second, base component **5046** of the cap, to a top surface of the second component **5046**, whereat the first component **5042** includes a topside portion **5048**. The topside portion preferably is located within a recess **5050** formed in the top surface of the second component **5046** and comprises an ornamentation including branding or a decorative element. FIG. **41** is a perspective cross-sectional view of the second component **5046** of the cap, and FIG. **42** is a top view of the second component **5046**.

In manufacturing the collector **5000**, the cap **5002** preferably is made by injection molding the first component **5042** onto the second component **5046**. The second component also preferably is first molded. Thus, one or more of these components and even the container itself may be molded from one or more inert plastic materials, for example. Furthermore, the materials preferably are lightweight such that the collector is readily manipulated by hand for dislodging bone from a distal end of a kerrison rongeur that is received therein. Additionally, the collector may be disposable, insofar as the collector may be used during a single medical procedure for a patient and then discarded in accordance with applicable HAZMAT protocols. Alternatively, one or more components of the collector (including all of the components) may be designed to be—and are able to be—sterilized for reuse with another patient during another procedure.

Still yet in at least some manufacturing methods, the material from which the first component—and specifically the scrapers thereof—is molded also comprises a bio-absorbable material which, if inserted into the body, is absorbed by the body.

By molding—and specifically overmolding—the component including the scrapers on both sides of the base component of the cap as well as through the channels in the base component of the cap, the scrapers are permanently affixed to the base component on the underside of the cap and are not removable from the cap without tearing of the overmolded component. As such, there is no need for assembly of the cap, and it is believed that the risk is reduced that one or more scrapers may become detached during aggressive engagement of the distal end of the rongeur with the scrapers. Such manufacturing method further is believed to protect against other product failures arising from other means of attaching the scrapers within the cap.

FIG. **45** is a perspective view of a collector **6000** used to collect cut bone from a kerrison rongeur in accordance with another embodiment of the present invention. The collector **6000** is similar in structure to the collector **5000** and includes a cap **6002** and container **6004**. FIG. **46** is a perspective view of the cap **6002**; FIG. **47** is a top view of the cap; FIG. **48** is a side perspective view of the cap; FIG. **49** is a bottom

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perspective view of the cap: FIGS. 50 and 51 each is a side elevational view of the cap; and FIG. 52 is a perspective view of the collector 6000 and portion of a Kerrison rongeur including distal tip thereof which has been inserted into and received within the cap 6002.

To demonstrate relative size, FIG. 53 is a photograph of a representative first component 7042 placed adjacent a penny 7070; and FIG. 54 is a photograph of a representative container 7004 being held by hand with a representative second base component 7046 screwed on the container.

Yet another collector in accordance with another embodiment of the present invention is similar in structure to collectors 5000, 6000 and is disclosed in the computer program listing of the appendix, which is incorporated by reference herein.

In view of the foregoing, it will be appreciated that in each of these preferred embodiments represented in FIGS. 35-52, the cap comprises pluralities of scrapers that differ based on flexibility and resiliency as well as arrangement and spacing between adjacent scrapers. Indeed, as seen herein, preferred embodiments each include a first plurality of scrapers shaped and configured in an arrangement to resemble a bristle field like that of a brush, and a second plurality of scrapers shaped and configured in an arrangement to resemble a row of teeth, with the row of teeth extending along the opening and with the bristle field located distal to the opening and with a gap extending between the row of teeth and the bristle field.

From experimentation it has been found that this combination of different plurality of scrapers, i.e., the bristle field and teeth, advantageously removes more bone during use of a collector than the bristle field alone. In particular, it has been found that the method of: inserting the distal end of the rongeur through the opening in the cap, preferably at an angle of 30-40 degrees, and preferably without engaging the teeth; then causing the distal end to engage and be moved and rotated within the bristle field; and, upon withdrawal of the distal end, then engaging the distal end with the teeth, results in a greater amount of bone being dislodged within the container than if the teeth were omitted from the collector.

In accordance with preferred embodiments of the present invention, a patient advantageously is afforded his or her own bone for the fusion when a collector is used to harvest bone cut from the patient using a Kerrison rongeur. For example, lamina chips are clinically proven to have both osteoinductive and osteoconductive properties conducive for bone fusion. By using a patient's own bone, there is less chance of rejection, infection, and significant cost by not having to rely on bone substitute.

Based on the foregoing description, it will be readily understood by those persons skilled in the art that the present invention is susceptible of broad utility and application. Many embodiments and adaptations of the present invention other than those specifically described herein, as well as many variations, modifications, and equivalent arrangements, will be apparent from or reasonably suggested by the present invention and the foregoing descriptions thereof, without departing from the substance or scope of the present invention.

Accordingly, while the present invention has been described herein in detail in relation to one or more preferred embodiments, it is to be understood that this disclosure is only illustrative and exemplary of the present invention and is made merely for the purpose of providing a full and enabling disclosure of the invention. The foregoing disclosure is not intended to be construed to limit the present

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invention or otherwise exclude any such other embodiments, adaptations, variations, modifications or equivalent arrangements, the present invention being limited only by the claims appended hereto and the equivalents thereof.

What is claimed is:

1. A method of collecting cut bone from a rongeur using a collector comprising a container having an open end and an opposite closed end, the container defining an interior containment space therebetween, and a cap having a single opening and scrapers attached within the cap, the cap extending over the interior containment space in covering relation to the open end of the container, the method comprising the steps of:

- (a) inserting a distal end of a rongeur into the interior space of the container through the single opening of the cap such that the distal end of the rongeur engages the scrapers, whereby cut bone carried on the rongeur is dislodged and falls into the container;
- (b) withdrawing the distal end of the rongeur from the cap; and
- (c) detaching the cap with the scrapers from the container for removing bone dislodged from the rongeur by the scrapers.

2. The method of claim 1, wherein the step of detaching the cap from the container comprises unscrewing the cap.

3. The method of claim 1, further comprising rotating the distal end of the rongeur relative to the container while the distal end of the rongeur is in engagement with the scrapers.

4. The method of claim 1, wherein the scrapers comprise a bristle field.

5. A method of collecting cut bone from a rongeur using a collector comprising a container having an open end and an opposite closed end, the container defining an interior containment space therebetween, and a cap having a single opening and means attached within the cap for engaging and dislodging cut bone from a cutting area of a distal end of the rongeur when received through the single opening, the cap extending over the interior containment space in covering relation to the open end of the container, the method comprising the steps of:

- (a) inserting the distal end of a rongeur into the container through the single opening of the cap such that the means engages and dislodges cut bone from the cutting area of the distal end of the rongeur whereby the dislodged bone is collected in the container;
- (b) withdrawing the distal end of the rongeur from the cap; and
- (c) detaching the cap with the scrapers from the container for removing the dislodged bone collected in the container.

6. The method of claim 5, wherein the means comprises a scraper.

7. The method of claim 5, wherein the means comprises a brush.

8. The method of claim 5, wherein the means comprises bristles.

9. The method of claim 5, wherein the step of detaching the cap from the container comprises unscrewing the cap.

10. The method of claim 5, further comprising rotating the distal end of the rongeur relative to the container while the distal end of the rongeur is in engagement with the means.

11. The method of claim 5, further comprising viewing of cut bone received within the interior containment space through at least a portion of the container that is transparent.

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12. The method of claim 5, wherein the container comprises a cylindrical body having a length extending between opposite ends that is greater than a diameter of the cylindrical body.

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